

Project Report



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1

2024/2025

SECP3723 – SYSTEM DEVELOPMENT TECHNOLOGY

SECTION 02

TITLE: PROJECT REPORT

LECTURER: DR. AHMAD NAJMI BIN AMERHAIDER NUAR

NAME	MATRIC NUMBER
NURUL IKA SYAFINY BINTI AZHAR	A23CS0164
NURAI SYAH BINTI MOHD ZIKRE	A23CS0160
HARINI A/P SANGARAN	A23CS0081
DAMIYA AINA BINTI BASIR ABD SHAMMAD	A23CS0220

Table of Content

1.0 Company Background	7
2.0 Problem background	9
3.0 Proposed Solution	11
4.0 Development Approach	13
5.0 System Design (ERD, Use Case & Sequence Diagram)	14
5.1 Entities Relationship Diagram (ERD)	14
5.2 Use Case Diagram	15
5.3 Sequence Diagram	16
5.3.1 Module <Member>	16
5.3.1.1 UC17: Use Case View Member Menu	16
5.3.1.2 UC16: Use Case Check Current Saving	17
5.3.1.3 UC23: Use Case Apply Membership Discontinuation	17
5.3.2 Module <Admin/Staff>	18
5.3.2.1 UC10: Use Case View Staff Menu	18
5.3.2.2 UC06: Use Case View Member Application Form	18
5.3.2.3 UC07: Use Case Validate Member Application Form	19
5.3.2.4 UC09: Use Case View Loan Application Form	19
5.3.2.5 UC08: Use Case Validate Loan Application Form	20
5.3.2.6 UC24: Use Case View Monthly Report	20
5.3.2.7 UC25: Use Case View Annual Report	21
5.3.2.8 UC26: Use Case Add New Admin	21
5.3.2.9 US27: Use Case Add New ALK	22
5.3.2.10 UC28: Use Case View Membership Discontinuation Form	22
5.3.2.11 UC29: Use Case Approve Membership Discontinuation	23
5.3.3 Module <ALK>	24
5.3.3.1 UC22: Use Case View ALK Menu	24

5.3.3.2 UC21: Use Case View Validated Member Application Form	24
5.3.3.3 UC18: Use Case Approve Member Application	25
5.3.3.4 UC20: Use Case View Validated Loan Application Form	25
5.3.3.5 UC19: Use Case Approve Loan Application	26
5.3.3.6 UC30: Use Case View Monthly Report	26
5.3.3.7 UC31: Use Case View Annual Report	27
5.3.4 Module <Non-Member>	28
5.3.4.1 UC01: Use Case Apply Membership	28
5.3.4.2 UC03: Use Case View Homepage	28
5.3.4.3 UC02: Use Case Get Member Approval	29
5.3.5 Module <Authentication>	30
5.3.5.1 UC05: Use Case Login	30
5.3.5.2 UC04: Use Case Log Out	30
5.3.6 Module <Loan>	31
5.3.6.1 UC14: Use Case Apply Loan	31
5.3.6.2 UC11: Use Case Check Loan Balance	31
5.3.6.3 UC12: Use Case Check Loan Approval	32
5.3.6.4 UC13: Use Case Find Transaction History	33
6.0 Development Software, Language, Technology, API and Tools	34
7.0 Function/Module Description	35
7.1 Authentication module	35
7.2 Non-member Module	35
7.3 Member Module	36
7.4 Loan Module	36
7.5 Staff Module	37
7.6 ALK Module	38
8.0 System Credentials (Hosting)	39

8.1 Member	39
8.2 Admin	39
8.3 Ahli Lembaga Koperasi (ALK)	39
9.0 Localhost Setup	40
10.0 System Interface for All Users	42
10.1 Non-Member user	42
10.1.1 Homepage Koperasi KADA	42
10.1.2 Apply Membership	42
10.1.3 Check Membership Application Status	44
10.1.4 Approved Membership Status	44
10.1.5 Pending Membership Status	45
10.1.6 Reviewed Membership Status	45
10.1.7 Rejected Membership Status	46
10.2 Member User	47
10.2.1 Login Page	47
10.2.2 Member Login	47
10.2.3 Member Homepage	48
10.2.4 Member Details	48
10.2.5 Update Member Details	49
10.2.6 Apply Loan	50
10.2.7 Check Loan Application Status	51
10.2.8 Check Member Saving	52
10.2.9 Check Loan Balance	53
10.2.10 Check Invoice/Transaction History	53
10.2.11 Apply Membership Termination Form	54
10.2.12 Navigation Bar	55
10.3 Admin User	56

10.3.1 Admin Login	56
10.3.2 Admin Homepage	56
10.3.3 View Membership Application List	57
10.3.4 Successful Membership Application Status Update	57
10.3.5 View Loan Application List	58
10.3.6 Successful Loan Application Status Update	58
10.3.7 View Monthly and Annual Report	59
10.3.8 Add New Admin	60
10.3.9 Add New ALK	60
10.3.10 Successful Add New Admin/ALK	61
10.3.11 View Membership Discontinuation List	61
10.4 ALK User	62
10.4.1 ALK Login	62
10.4.2 ALK Homepage	62
10.4.3 Approve Loan Application	63
10.4.4 Successful Loan Application Status Update	63
10.4.5 Approve Membership Application	64
10.4.6 Successful Membership Application Status	64
10.4.7 Navigation Bar	65
10.3.7 View Monthly and Annual Report	65
10.5 All User	67
10.5.1 Reset Password	67
10.5.2 Successful Reset Password	67
10.5.3 Unsuccessful Reset Password	68
10.5.4 Log Out	68
11.0 Alpha and Beta Testing Reports	69
11. 1 Alpha Testing Report	69

11.1.0 Equivalence Partitioning and Boundary Value Analysis	69
11.1.1 Test Cases Based on Subsystem	75
TC001: Test Member Subsystem: <User Story 1 (US001)>	75
TC002: Test Admin Subsystem: <User Story 2 (US002)>	78
TC003: Test ALK Subsystem: <User Story 3 (US003)>	87
TC004: Test Non-Member Subsystem: <User Story 4 (US004)>	93
TC005: Test Loan Subsystem: <User Story 6 (US006)>	98
11.2 Beta Testing	102
11.2.0 Methods Class	102
11.2.1 Flow Graph	103
11.2.2 Cyclomatic Complexity Calculation	104
11.2.3 Test Cases	105
12.0 UAT Reports	106
13.0 Gantt Chart	107

1.0 Company Background

The Koperasi Kakitangan KADA Berhad is a cooperative organization established to serve the welfare and economic interests of the employees of the Kemubu Agricultural Development Authority (KADA) in Malaysia. Operating from its headquarters at Ibu Pejabat KADA Lundang in Kota Bharu, Kelantan, the cooperative plays a pivotal role in enhancing the financial well-being and professional growth of its members.

Vision and Ambitions:

Koperasi Kakitangan KADA Berhad aspires to be a leading cooperative, committed to uplifting the socio-economic status of its members through diversified financial services and investment opportunities. The cooperative aims to foster a culture of mutual assistance and economic empowerment, ensuring that all members benefit from collective growth and prosperity.

Achievements and Recognition:

Since its inception, the cooperative has been instrumental in providing financial solutions tailored to the needs of KADA employees. Through prudent management and member-centric initiatives, it has established itself as a reliable entity, contributing significantly to the financial stability and advancement of its members.

Collaborations and Initiatives:

Koperasi Kakitangan KADA Berhad collaborates closely with KADA and other financial institutions to offer a range of services, including savings schemes, loan facilities, and investment opportunities. These collaborations are designed to provide members with access to financial resources that support personal development and economic resilience.

Community Impact:

Beyond financial services, the cooperative is dedicated to fostering a sense of community among KADA employees. It organizes various programs and activities aimed at professional development, social engagement, and the overall well-being of its members. By promoting solidarity and mutual support, Koperasi Kakitangan KADA Berhad contributes to a cohesive and empowered workforce within the agricultural sector.

About Koperasi Kakitangan KADA Berhad

As an integral part of the KADA ecosystem, Koperasi Kakitangan KADA Berhad continues to evolve, adapting to the changing needs of its members and the broader economic landscape. Through its unwavering commitment to service and excellence, the cooperative remains a cornerstone of support for KADA employees, driving forward the mission of agricultural development and community enrichment in Malaysia.

2.0 Problem background

Currently Koperasi KADA has been using manual system management for the cooperative for 15 years. To address the issues from the current system, the issues will be broken down into a few points for a better understanding before proposing a new solution to the issues faced.

1. Manual Data Entry Process

As mentioned, Koperasi KADA requires staff to manually enter the application submitted by the applicant into the system. This process is highly inefficient as staff need to manually key in all the details for every application (membership application, loan application & membership termination). This will likely increase the human errors such as typos, incorrect data input and missing information as it can lead to inconsistencies in the records. Moreover, staff will spend some time repetitively keying in the data instead of focusing on higher-value activities. This issues will cause bottlenecks in operation

2. Inconvenient Process of Obtaining and Submitting Physical Form

Applicants who want to become a member of Koperasi KADA need to get a physical form directly in person at the office of Koperasi KADA and return back to the office to submit the form. As this manual process is inconvenient, it will discourage the potential applicant from joining and be part member of Koperasi KADA. Moreover, the form submitted might be misplaced during the submission and applicants need to submit the form again and need to go back and forth to the office in order to get the form and submit the form again. This issue is the same as a member of Koperasi KADA who wants to submit a loan application or membership termination application.

3. Lack of Self-Service Access to Personal Financial Statement

Members of Koperasi KADA need to go to the office in person to access their financial status (savings, loan balance, etc) as they can independently access their financial information. This is inconvenient for those who live far away from the office and for those that have busy schedules where it becomes a burden for them to care about their financial status. As they cannot access their financial status at their own convenience, it will reduce member satisfaction which might lead to membership termination.

4. Inefficient Reporting Mechanism

Due to its reliance on manual data management processes, it is unable to generate an accurate and timely report. These issues will lead to inconsistencies and delay in generating reports such as monthly reports of membership application, loan application, syer etc and even annual reports. The lack of accurate reporting makes it difficult for the cooperative to monitor the financial status or even track member activities. Besides, without timely and accurate report cooperative unable to make a plan strategically for future growth and demands

3.0 Proposed Solution

In this section, we will propose solutions that directly address the issues stated in the previous section

1. Automation of Data Entry Process

To eliminate inefficiencies and reduce human errors, the proposed system will feature automated data entry and validation. When applicants submit their membership, loan, or termination applications online, their details will be directly stored in the database, removing the need for manual keying by staff. The system will include built-in validation checks to ensure all required fields are completed correctly, reducing typos and missing information. This will streamline operations, minimize data inconsistencies, and allow staff to focus on higher-value tasks rather than repetitive data entry.

2. Online Application Submission

To enhance accessibility, applicants will be able to fill out and submit membership, loan, and termination applications online through a dedicated web-based system. This eliminates the need to visit the office in person, reducing inconvenience for both applicants and staff. Additionally, the system will store all application records securely, ensuring that no forms are lost or misplaced. Applicants will also receive real-time status updates via the system, keeping them informed about their application progress.

3. Self-Service Access to Financial Statements

A secure online member portal will allow Koperasi KADA members to view their financial statements, including savings balances, loan statuses, and transaction history. Members will be able to log in anytime, anywhere, eliminating the need for physical visits to the office. This self-service functionality will increase member satisfaction, improve financial transparency, and reduce the workload for staff handling financial inquiries.

4. Automated and Real-Time Reporting

The system will include automated report generation tools that compile real-time financial and membership data into structured reports. Monthly and annual reports for membership applications, loan applications, and syer contributions will be automatically generated and accessible to administrators. The system will also provide data analytics tools for tracking financial trends, member activities, and overall cooperative performance. This will improve decision-making, support strategic planning, and enhance operational efficiency.

4.0 Development Approach

For the development of the KADA Corporation System, we adopted the MVC (Model-View-Controller) architecture as the foundation for coding and structuring the system, along with the Agile development methodology to ensure flexibility and continuous improvement throughout the project. The MVC architecture allowed us to separate the system into three distinct layers: the Model layer, which handled database structures and managed critical data such as member information, transactions, and user roles; the View layer, which focused on designing an interactive and user-friendly interface for different user types, including members, ALK, and administrators; and the Controller layer, which served as the intermediary, processing user requests, executing business logic, and connecting the Model and View components. By following the Agile methodology, we worked in iterative development cycles, allowing us to make continuous enhancements, adapt to changing requirements, and ensure high-quality system performance. This approach enabled our team to collaborate efficiently, identify and resolve issues quickly, and implement new features seamlessly, making the KADA Corporation System well-structured, scalable, and adaptable to future improvements.

5.0 System Design (ERD, Use Case & Sequence Diagram)

5.1 Entities Relationship Diagram (ERD)

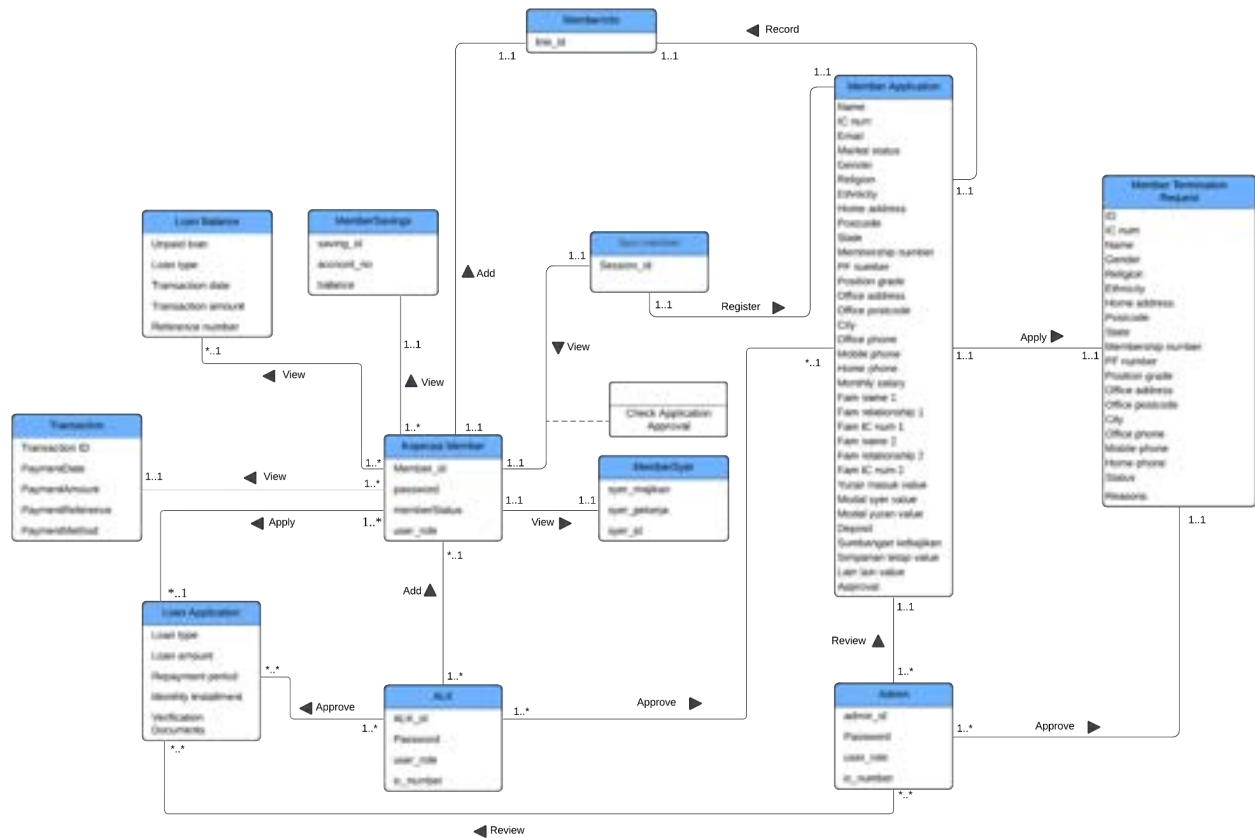


Figure 1: ERD Diagram of Koperasi KADA system

Reference Link:

<https://drive.google.com/file/d/1sRHFm270d1rm9ZB8H4piEvthHacxP6iC/view?usp=sharing>

5.2 Use Case Diagram

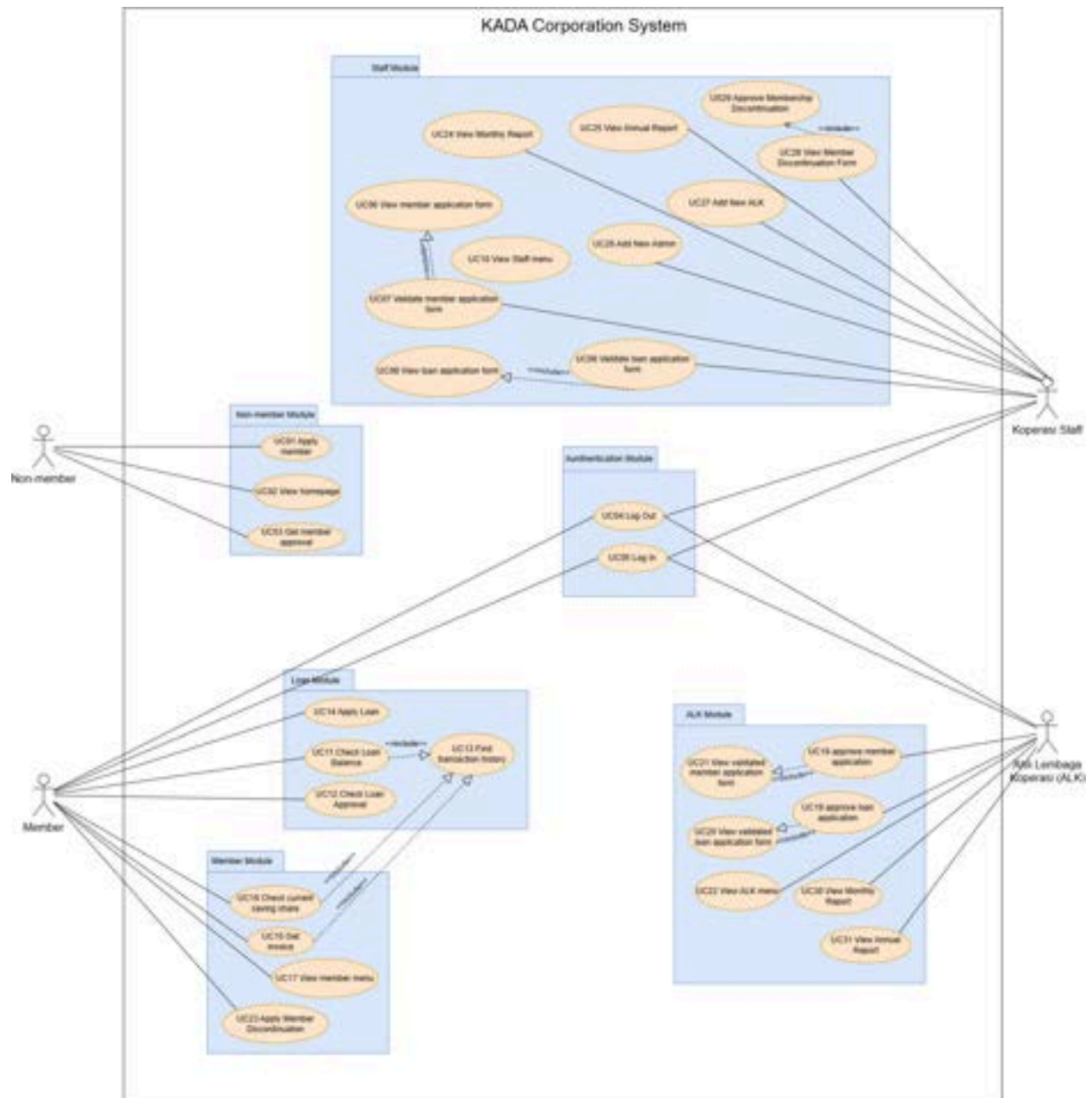


Figure 2: Use Case Diagram for Koperasi KADA system

5.3 Sequence Diagram

5.3.1 Module <Member>

5.3.1.1 UC17: Use Case View Member Menu

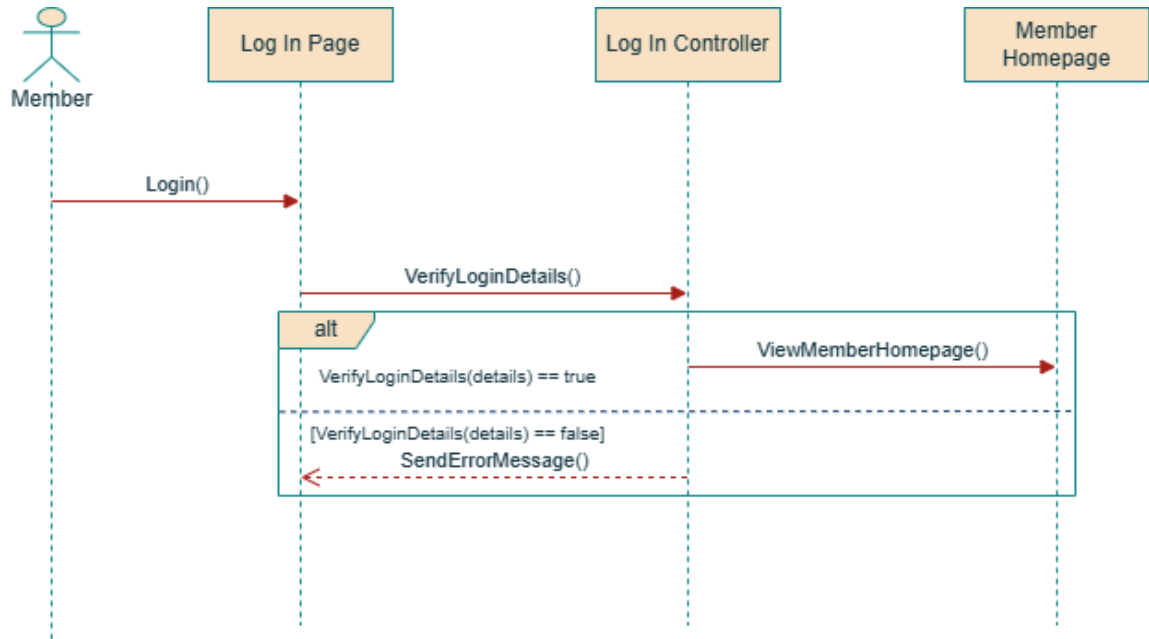


Figure 3: Sequence Diagram for View Member Menu

5.3.1.2 UC16: Use Case Check Current Saving

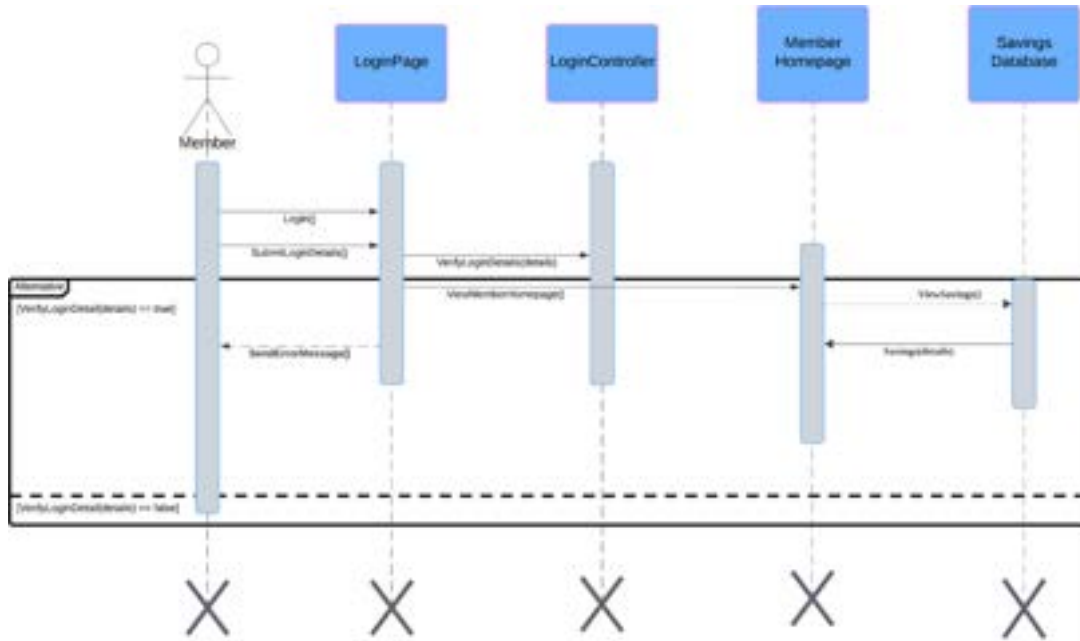


Figure 4: Sequence Diagram for Check Current Saving

5.3.1.3 UC23: Use Case Apply Membership Discontinuation

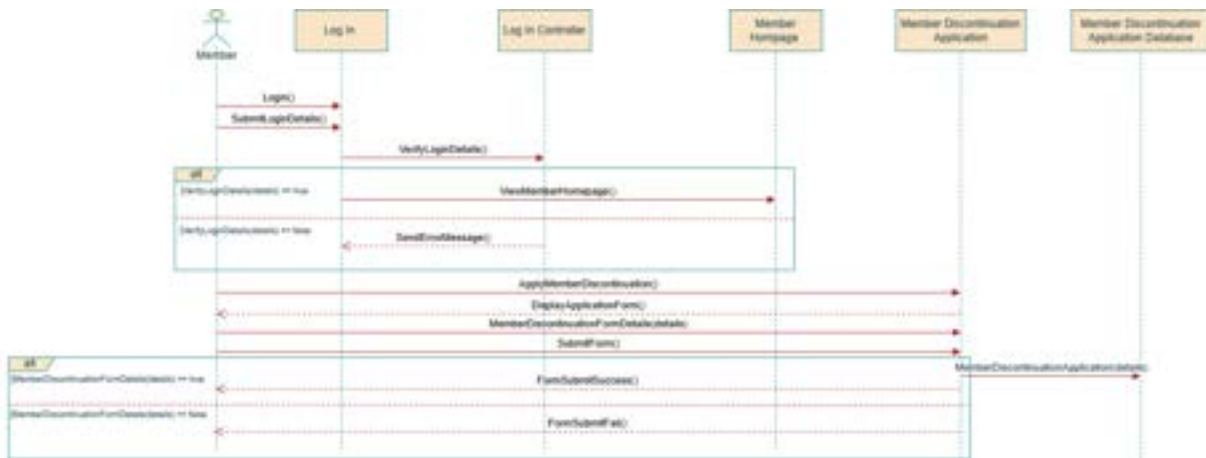


Figure 5: Sequence Diagram for Apply Membership Discontinuation

5.3.2 Module <Admin/Staff>

5.3.2.1 UC10: Use Case View Staff Menu

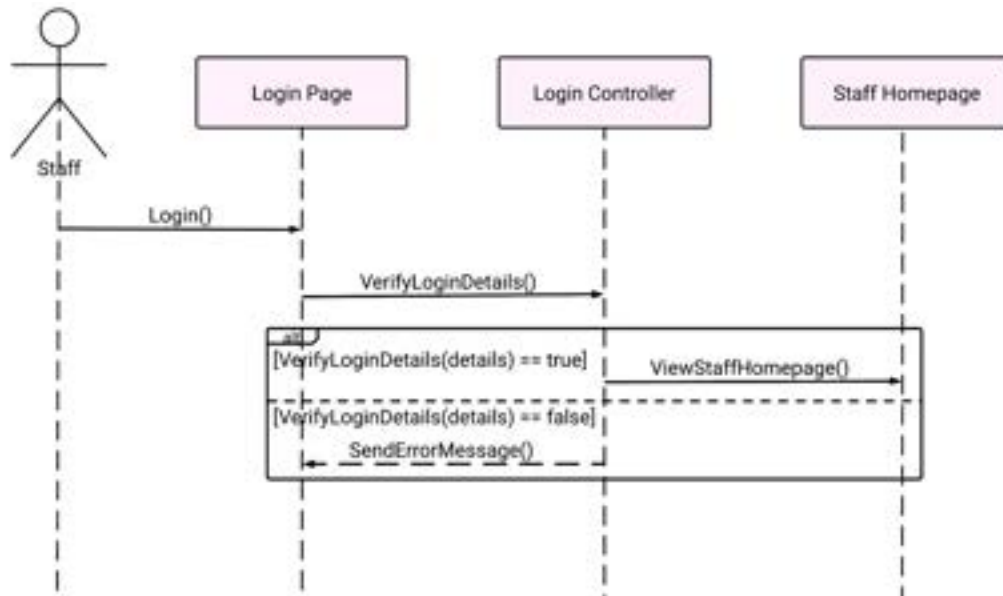


Figure 6: Sequence Diagram for View Staff Menu

5.3.2.2 UC06: Use Case View Member Application Form

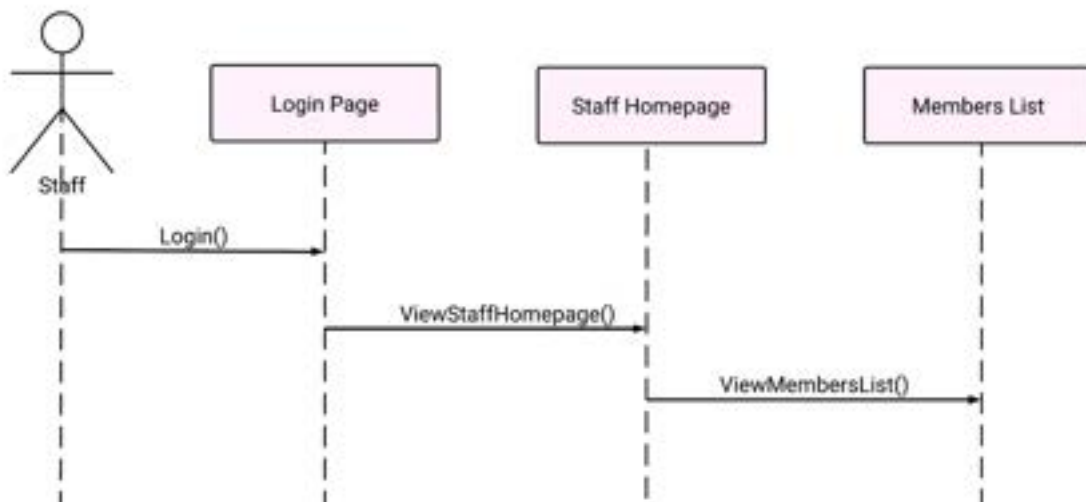


Figure 7: Sequence Diagram for View Member Application Form

5.3.2.3 UC07: Use Case Validate Member Application Form

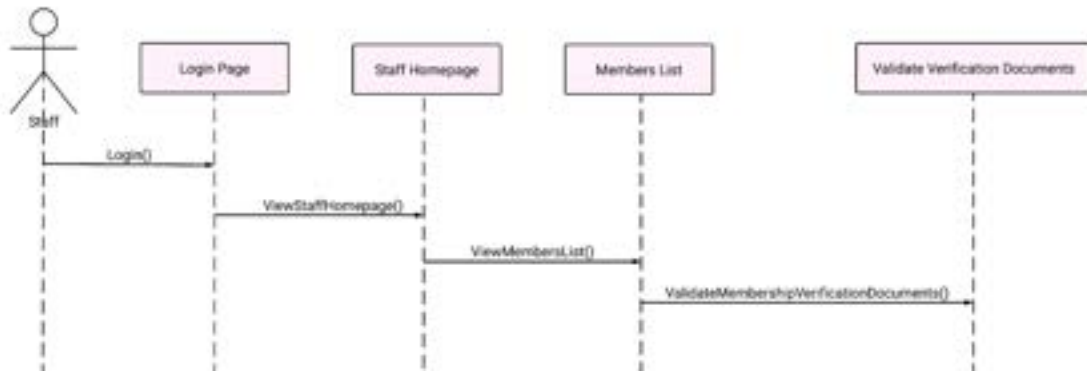


Figure 8: Sequence Diagram for Validate Member Application Form

5.3.2.4 UC09: Use Case View Loan Application Form

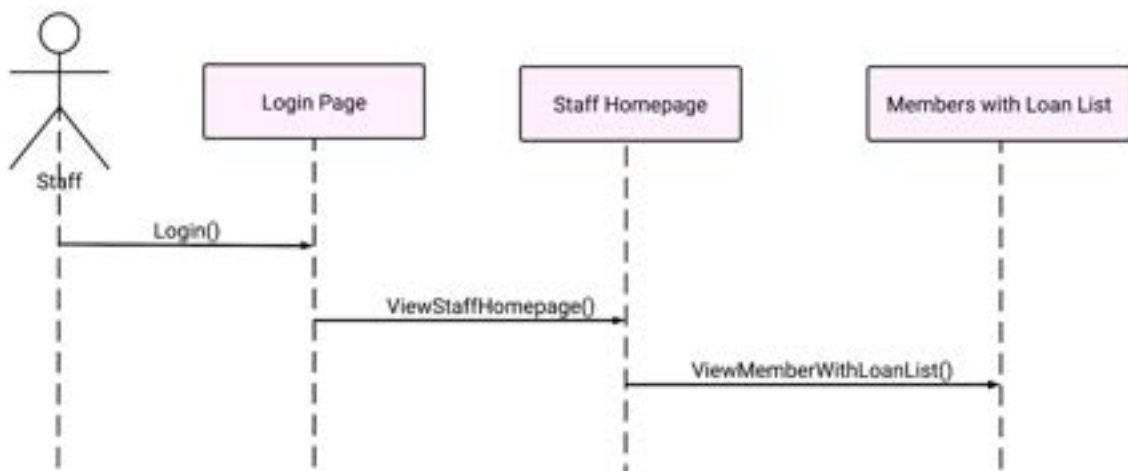


Figure 9: Sequence Diagram for View Loan Application Form

5.3.2.5 UC08: Use Case Validate Loan Application Form

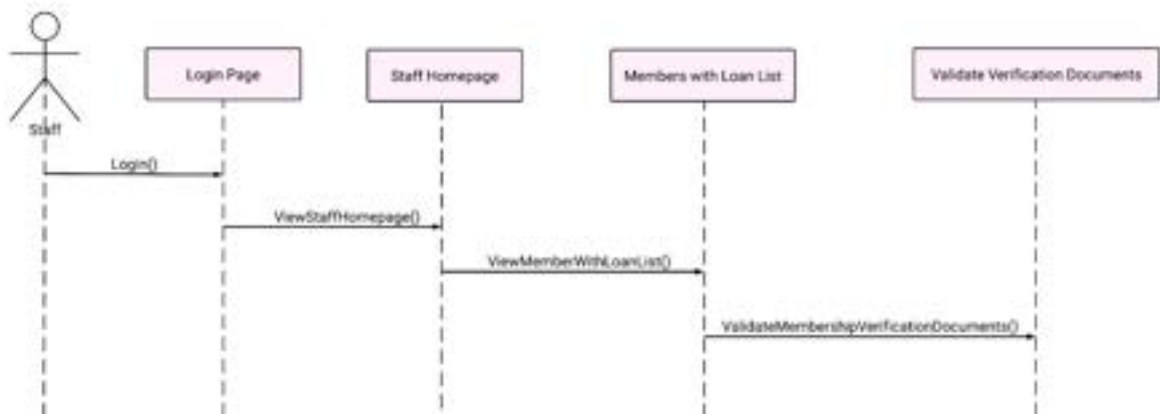


Figure 10: Sequence Diagram for Validate Loan Application Form

5.3.2.6 UC24: Use Case View Monthly Report

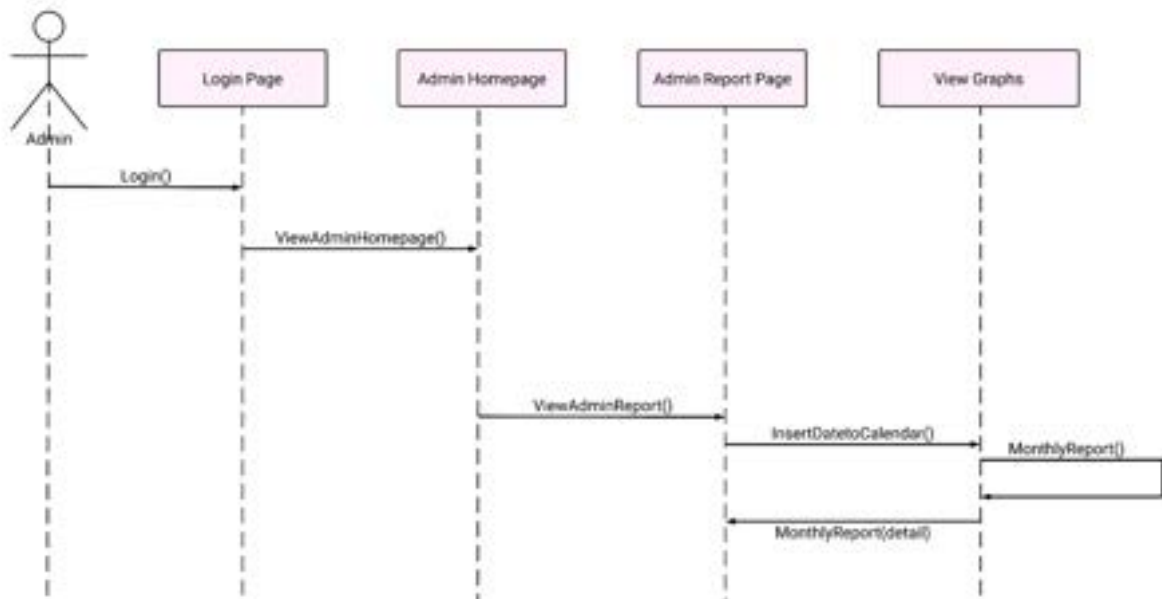


Figure 11: Sequence Diagram for View Monthly

5.3.2.7 UC25: Use Case View Annual Report

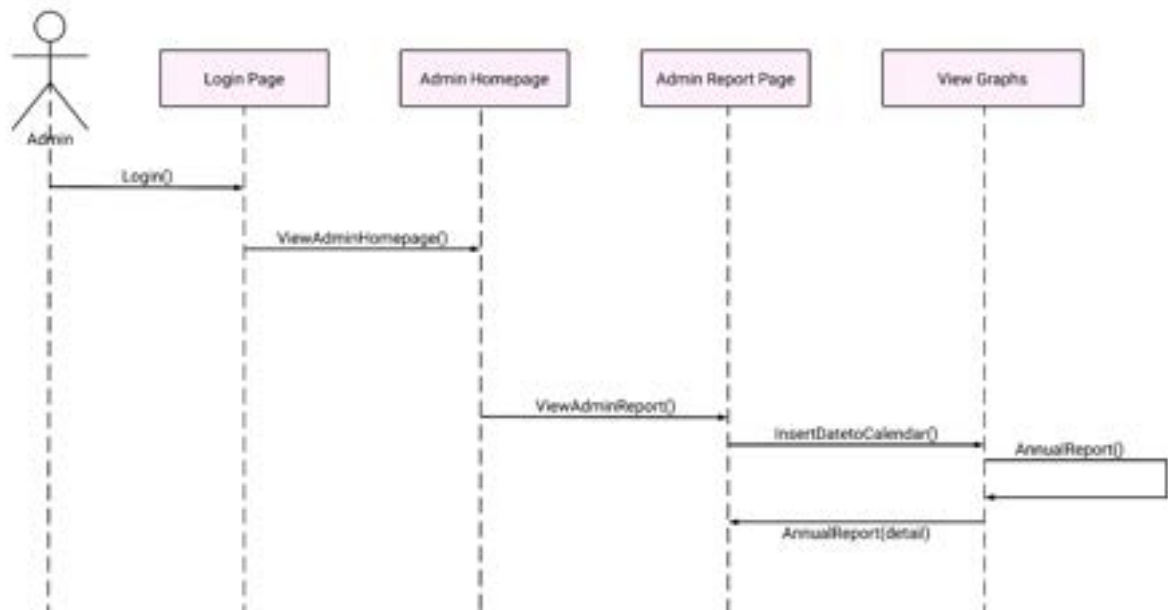


Figure 12: Sequence Diagram for View Annual Report

5.3.2.8 UC26: Use Case Add New Admin

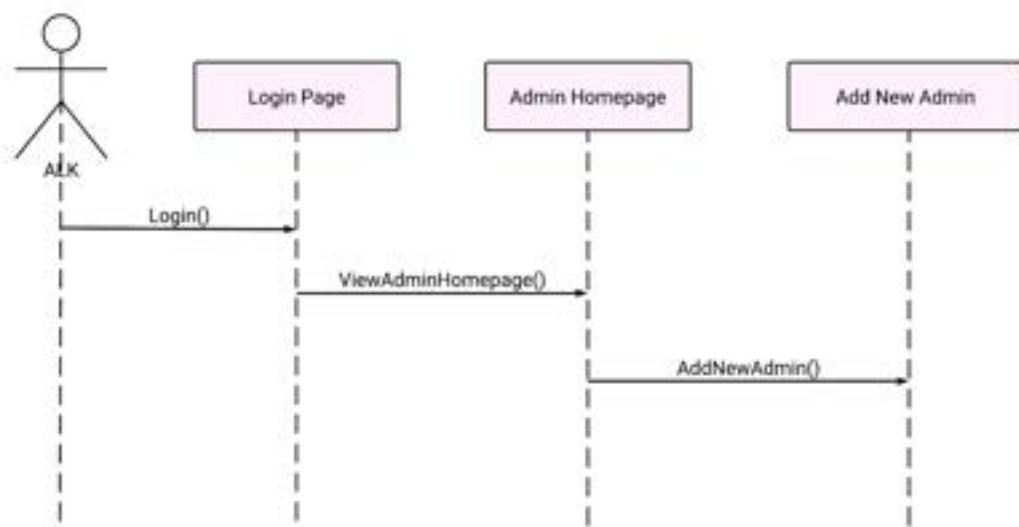


Figure 13: Sequence Diagram for Add New Admin

5.3.2.9 US27: Use Case Add New ALK

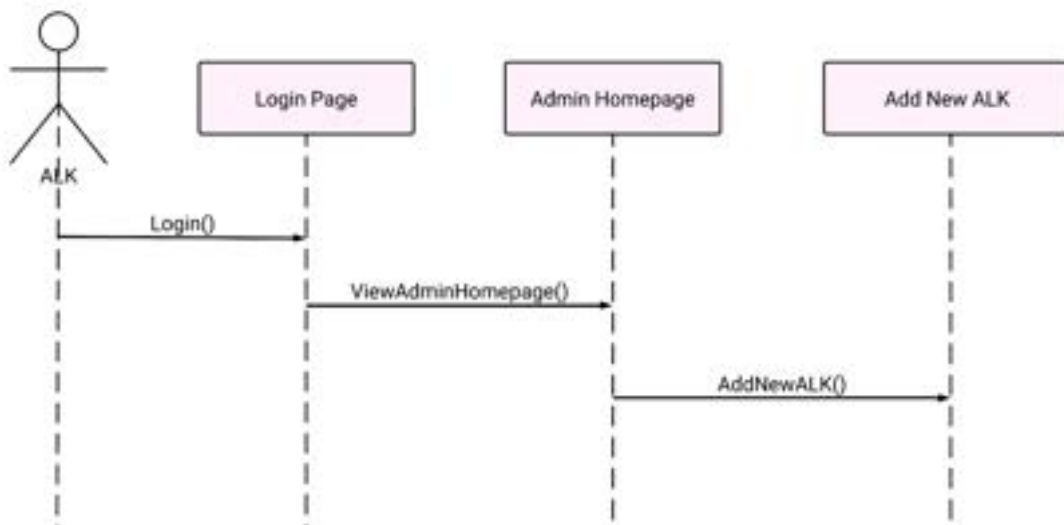


Figure 14: Sequence Diagram for Add New ALK

5.3.2.10 UC28: Use Case View Membership Discontinuation Form

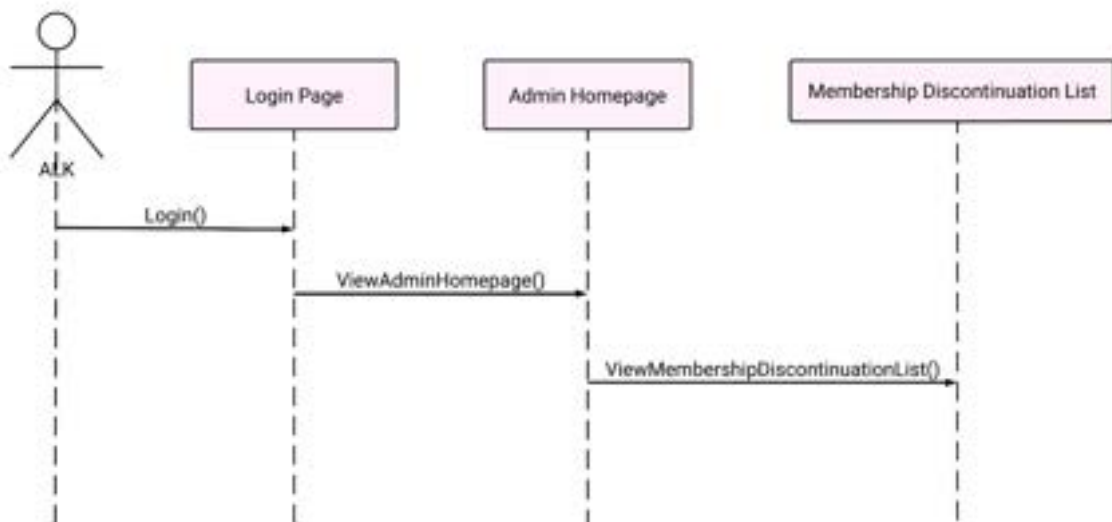


Figure 15: Sequence Diagram for View Membership Discontinuation Form

5.3.2.11 UC29: Use Case Approve Membership Discontinuation

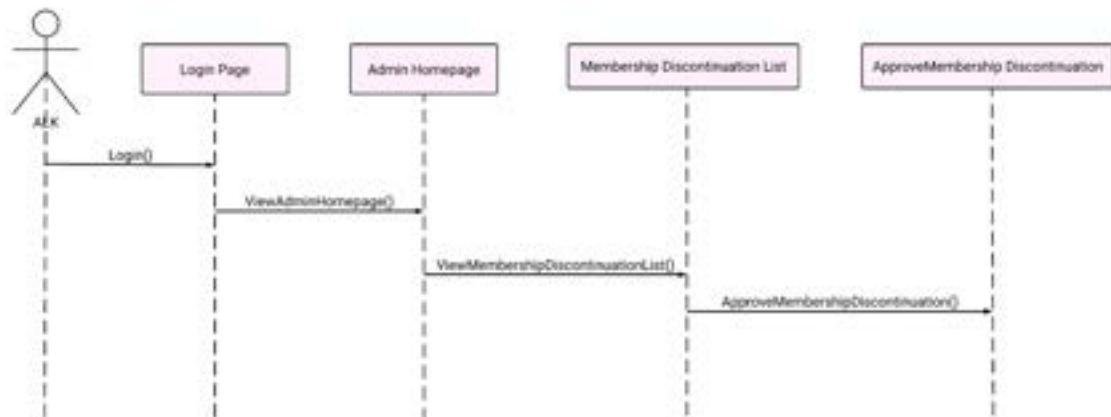


Figure 16: Sequence Diagram for Approve Membership Discontinuation Form

5.3.3 Module <ALK>

5.3.3.1 UC22: Use Case View ALK Menu

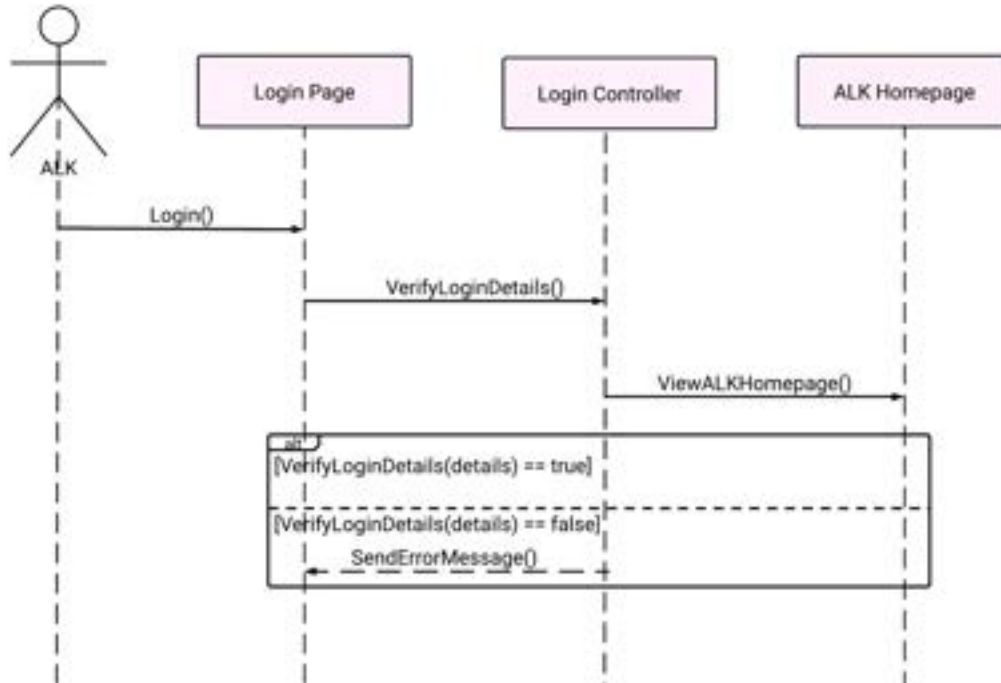


Figure 17: Sequence Diagram for View ALK Menu

5.3.3.2 UC21: Use Case View Validated Member Application Form

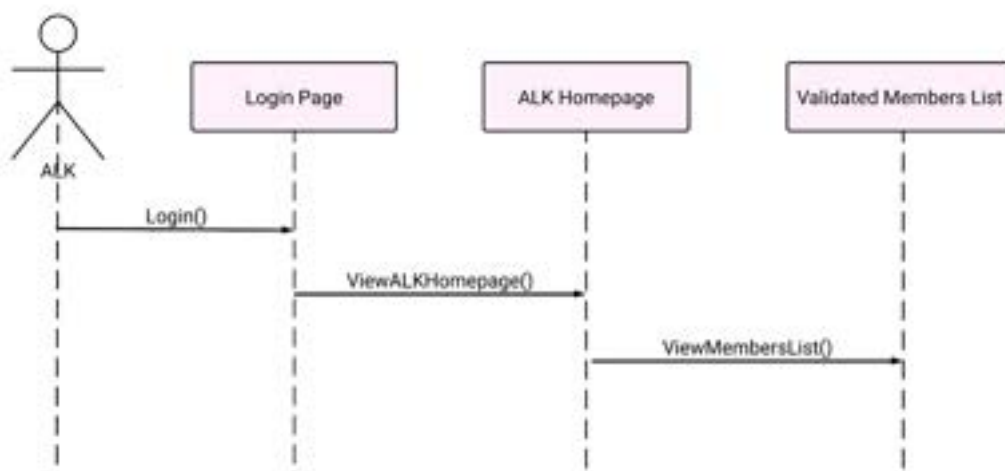


Figure 18: Sequence Diagram for View Validated Member Application Form

5.3.3.3 UC18: Use Case Approve Member Application

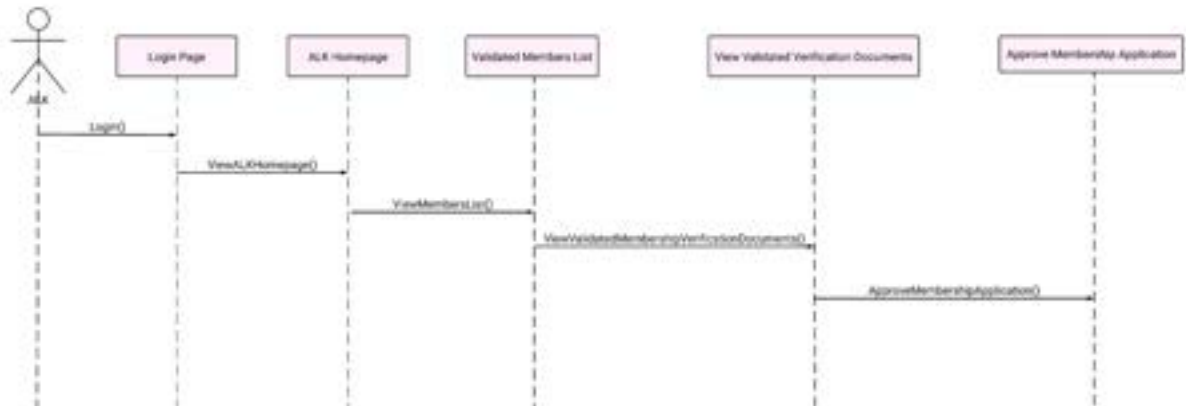


Figure 19: Sequence Diagram for Approve Member Application

5.3.3.4 UC20: Use Case View Validated Loan Application Form

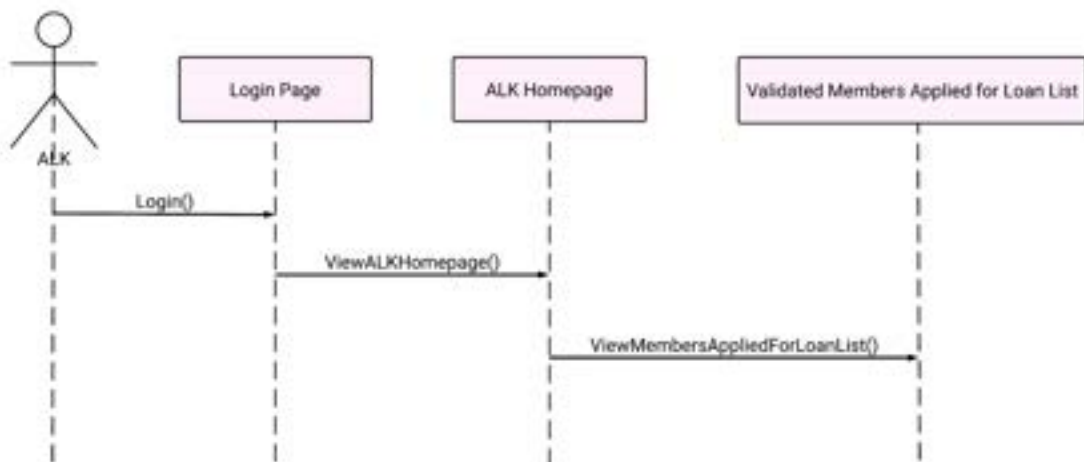


Figure 20: Sequence Diagram for View Validated Loan Application Form

5.3.3.5 UC19: Use Case Approve Loan Application

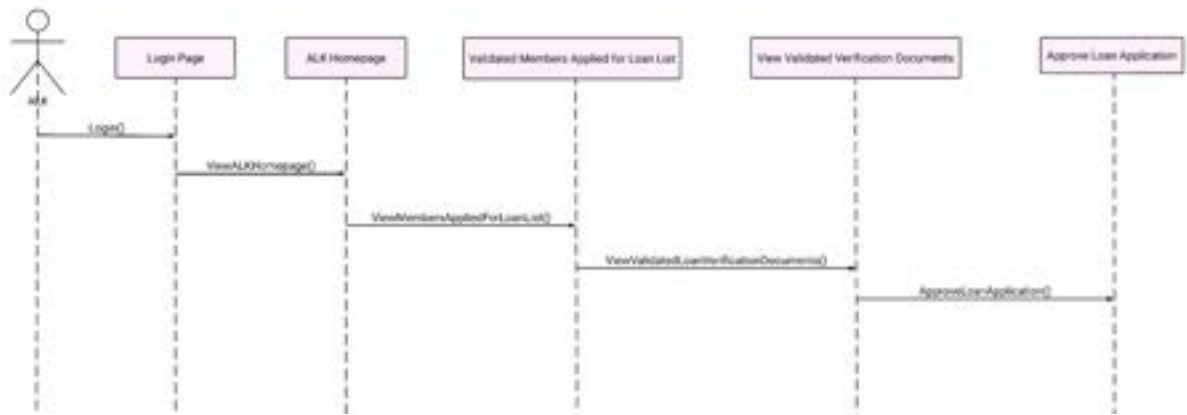


Figure 21: Sequence Diagram for Approve Loan Application

5.3.3.6 UC30: Use Case View Monthly Report

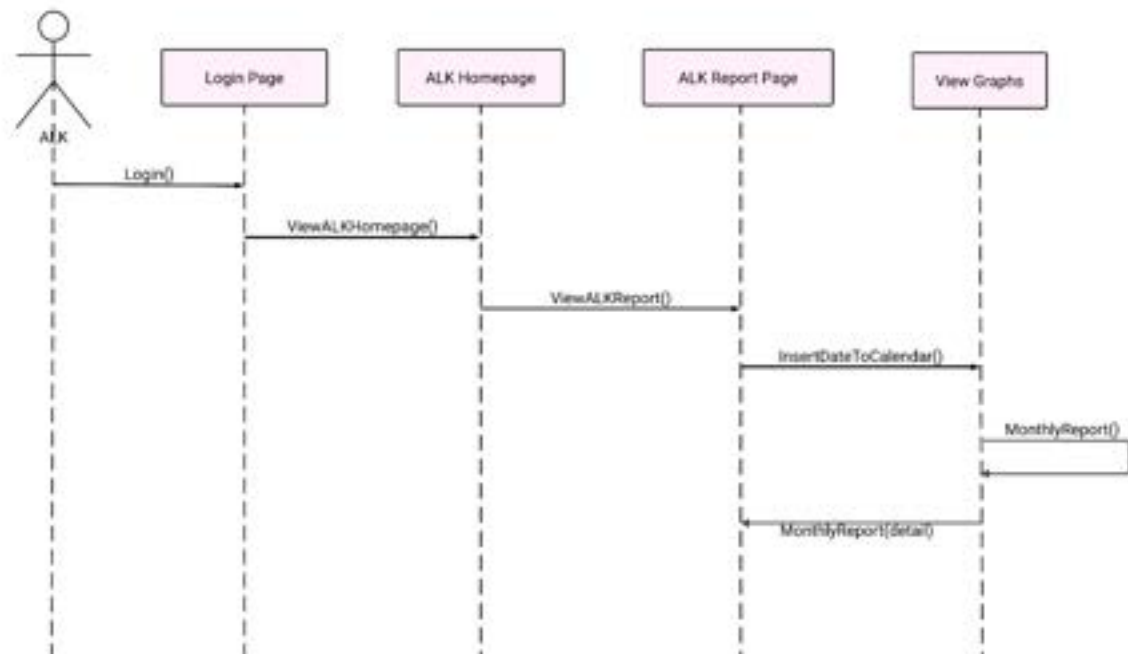


Figure 22: Sequence Diagram for View Monthly Report

5.3.3.7 UC31: Use Case View Annual Report

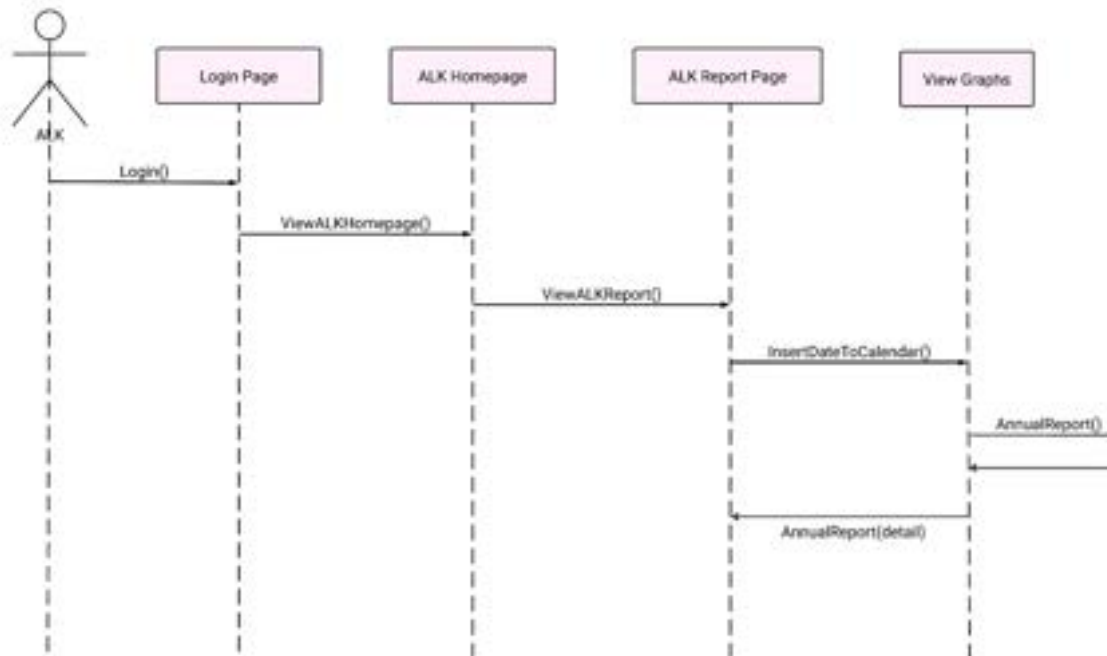


Figure 23: Sequence Diagram for View Annual Report

5.3.4 Module <Non-Member>

5.3.4.1 UC01: Use Case Apply Membership

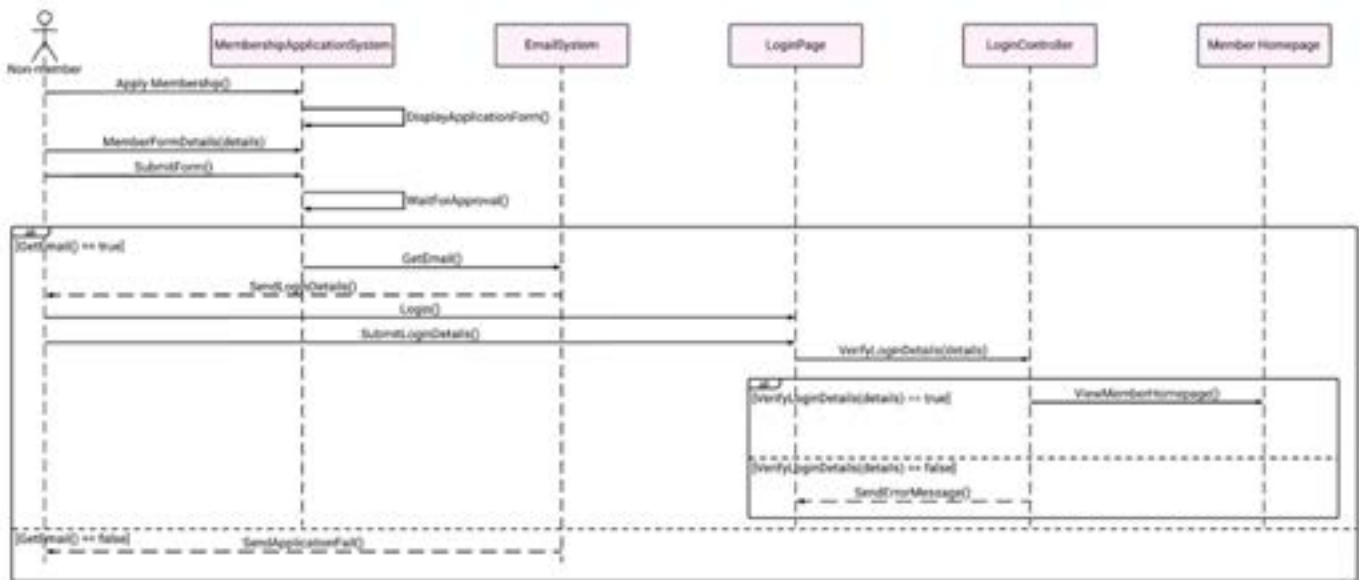


Figure 24: Sequence Diagram for Apply Membership

5.3.4.2 UC03: Use Case View Homepage



Figure 25: Sequence Diagram for View Homepage

5.3.4.3 UC02: Use Case Get Member Approval

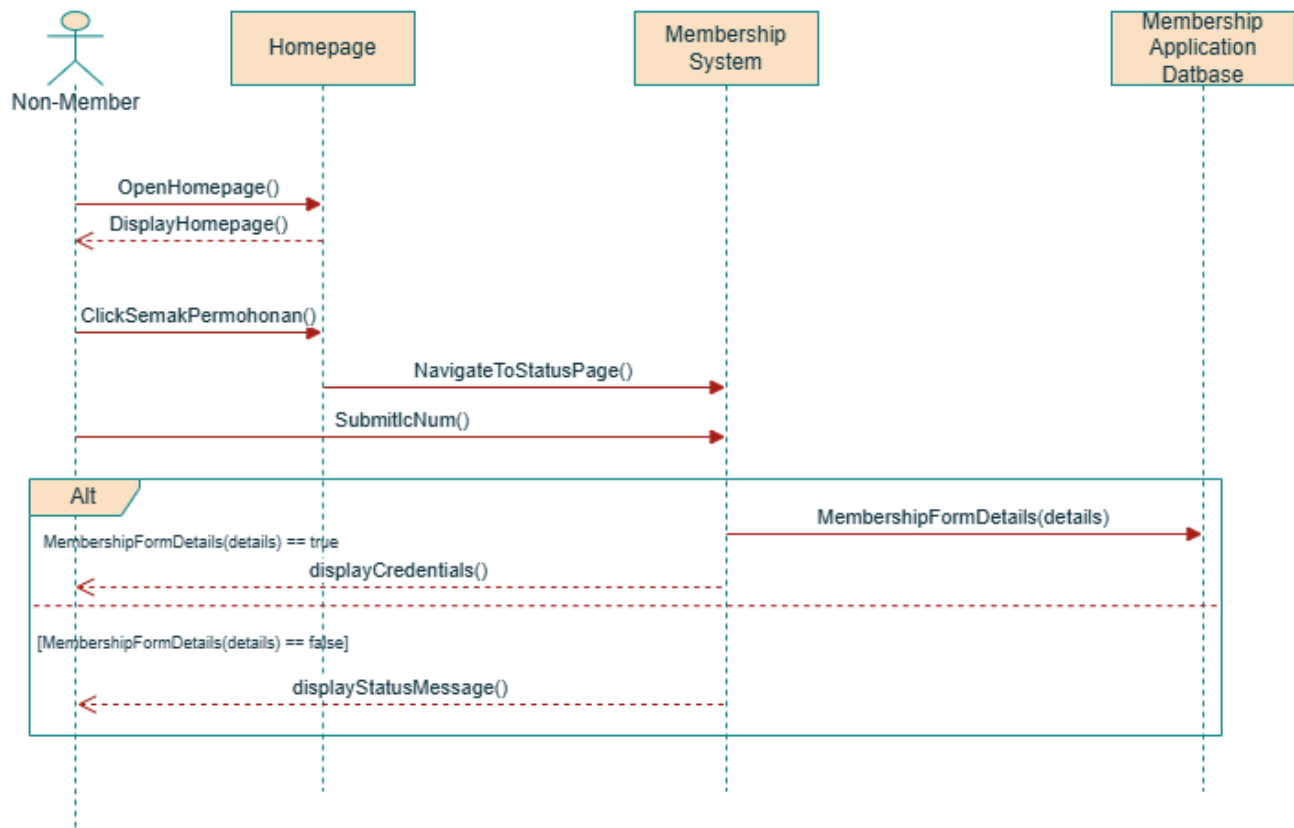


Figure 26: Sequence Diagram for Get Member Approval

5.3.5 Module <Authentication>

5.3.5.1 UC05: Use Case Login

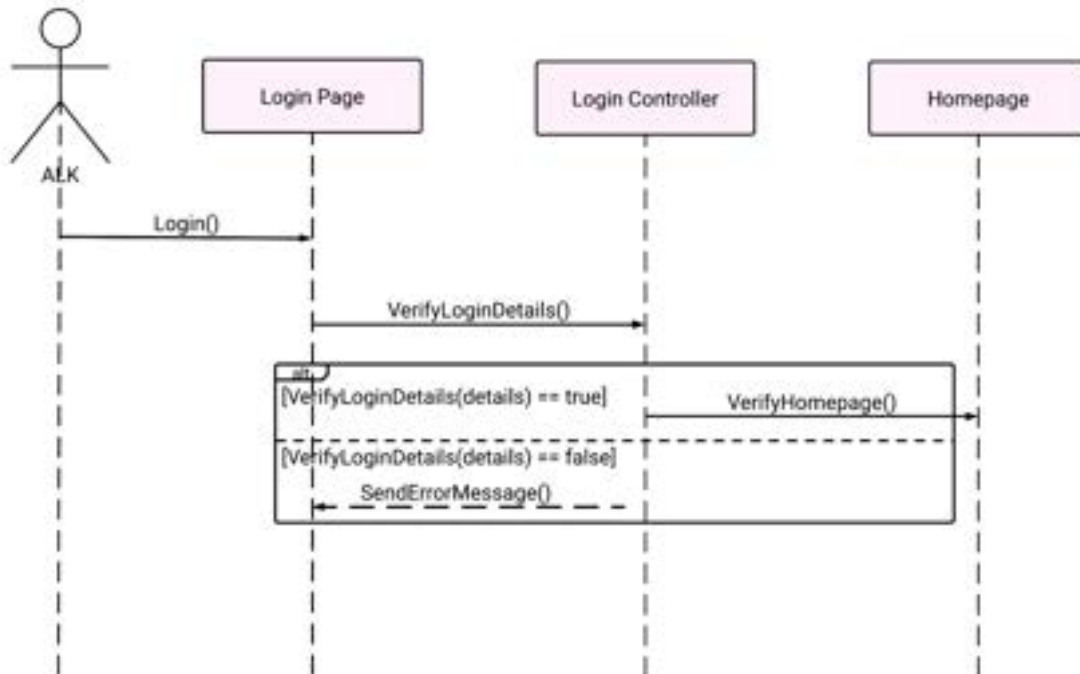


Figure 27: Sequence Diagram for Login

5.3.5.2 UC04: Use Case Log Out

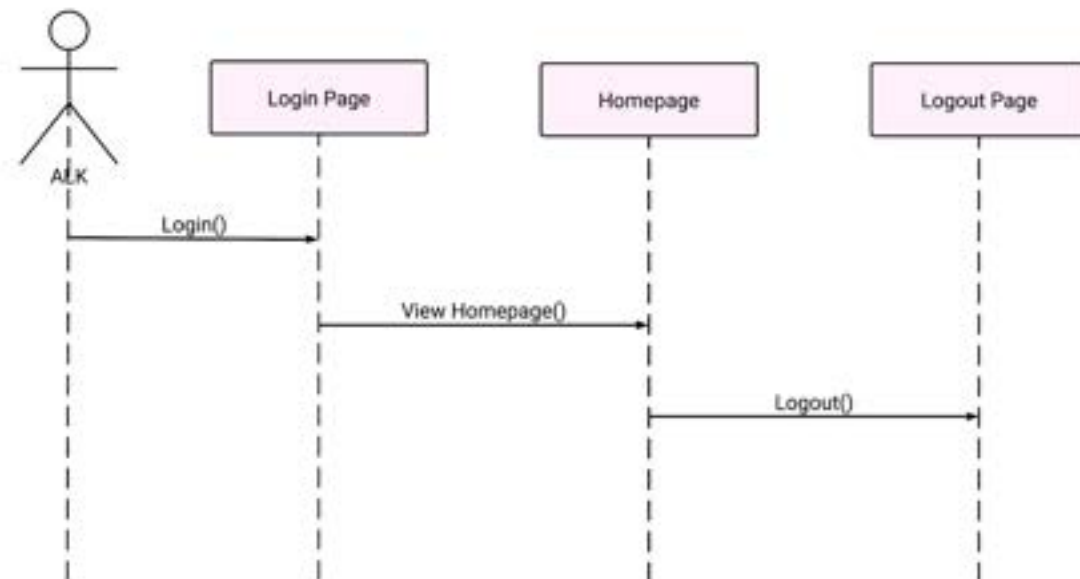


Figure 28: Sequence Diagram for Log Out

5.3.6 Module <Loan>

5.3.6.1 UC14: Use Case Apply Loan

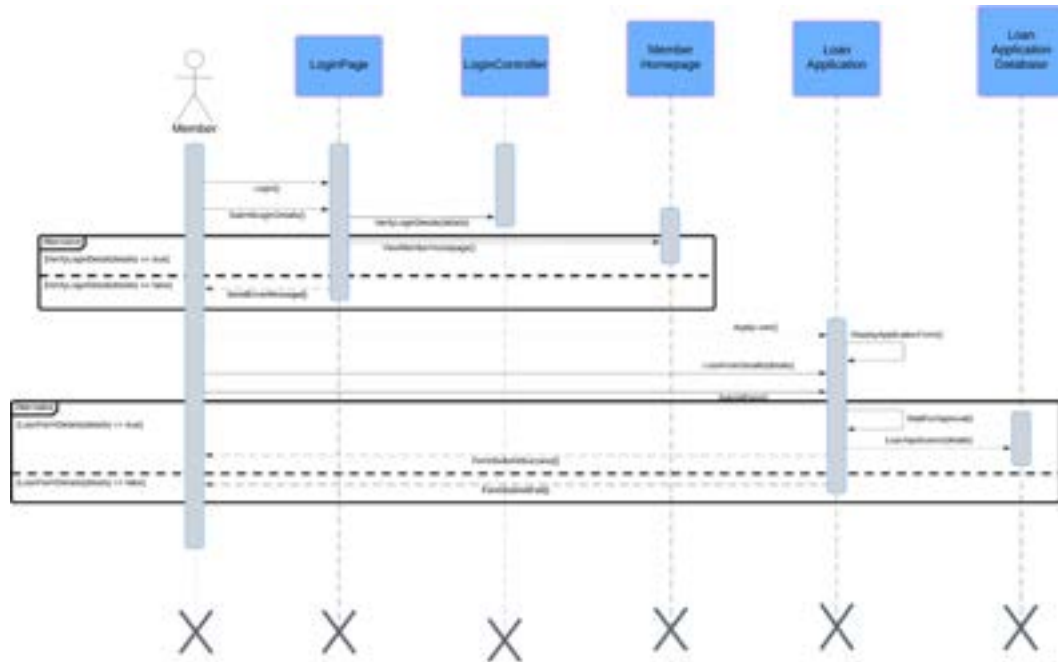


Figure 29: Sequence Diagram for Apply Loan

5.3.6.2 UC11: Use Case Check Loan Balance

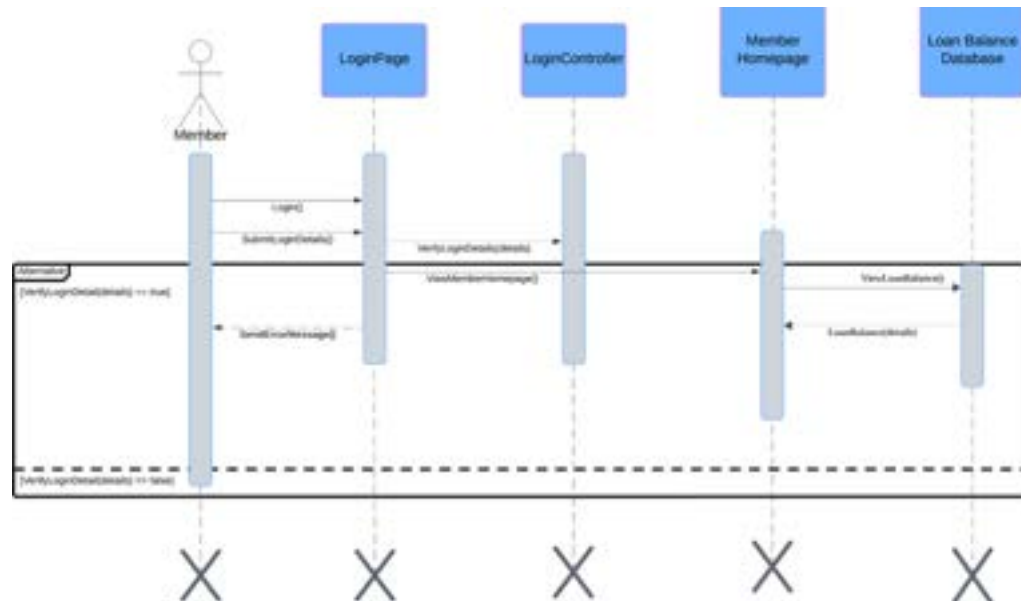


Figure 30: Sequence Diagram for Check Loan Balance

5.3.6.3 UC12: Use Case Check Loan Approval

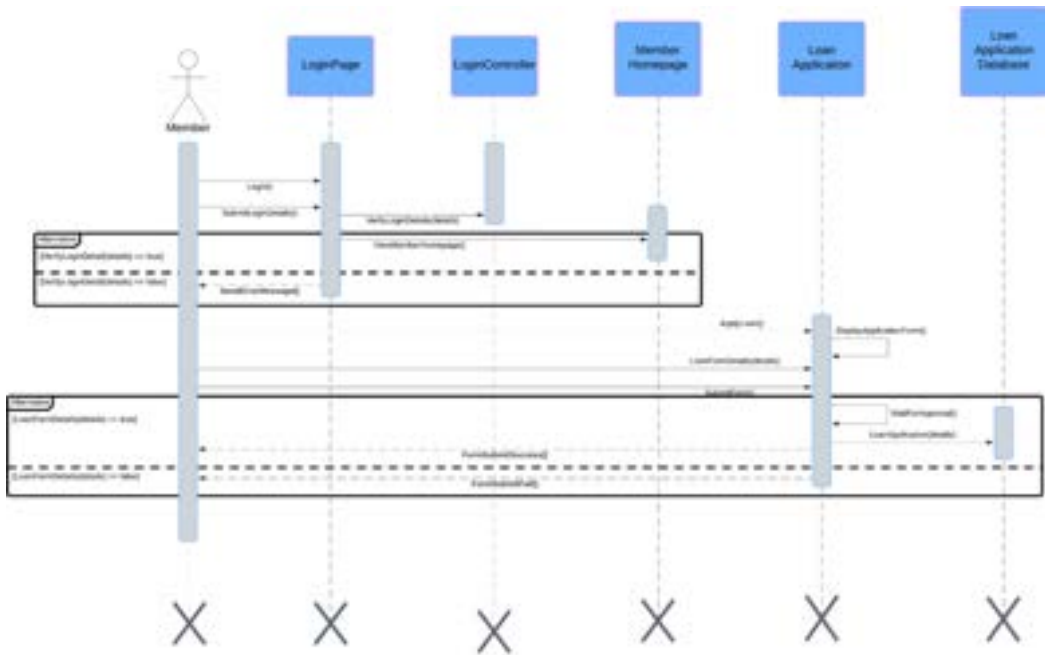


Figure 31: Sequence Diagram for Check Loan Approval

5.3.6.4 UC13: Use Case Find Transaction History

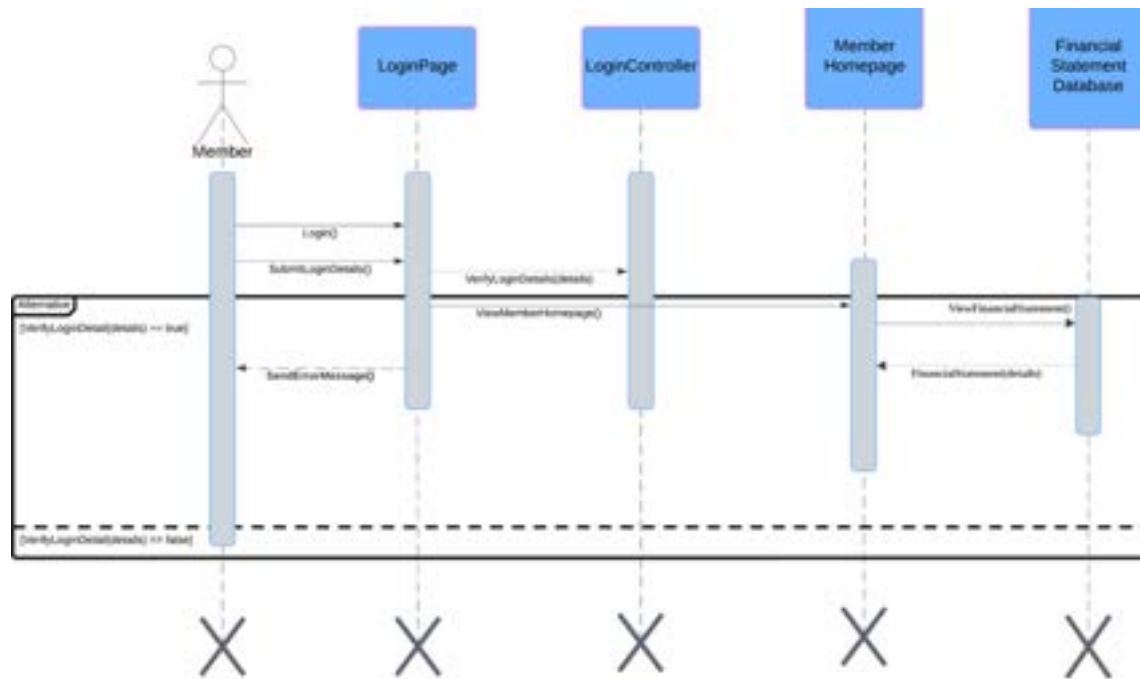


Figure 32: Sequence Diagram for Find Transaction History

6.0 Development Software, Language, Technology, API and Tools

Software

The softwares used while developing Koperasi KADA system are Visual Studio Code, Github, XAMPP Control Panel and phpMyAdmin. We used Visual Studio Code to do the coding part of the system. As we used several different languages in our coding, Visual Studio Code was the optimal software to be used as it supports many types of programming languages. As this was a group project, we linked our Visual Studio Code to Github, so that everyone could have access to the last updated code with ease. After every successful update, each member pushes the code to github and merges it to the existing code. Then, each member is required to pull from Github origin before beginning with their coding to have the updated code. XAMPP Control Panel is the software we used as a web server for the proposed system. XAMPP has multiple modules under its software, such as Apache, MySQL, FileZilla, Mercury and Tomcat. For our proposed system, only Apache and MySQL were used. Apache functions as the web server, enabling the execution of PHP scripts, which are developed using Visual Studio Code. phpMyAdmin was used in the process of the system development to handle the administration of MySQL database servers.

Language

We used multiple programming languages to have a successful functioning system. The programming languages used in the system development are Hypertext Preprocessor (PHP), Hypertext Markup Language (HTML), Cascading Style Sheets (CSS) and Structured Query Language (SQL). The functionalities of the system were developed using PHP language as we used the Model-View-Controller (MVC) architecture. The system was styled mainly using HTML language and CSS language for a few interfaces under the View of MVC architecture. Lastly, SQL language was used for the database declaration and modification of the system.

Technology

The technology we have used in this system development is that we have used Hostinger as our web hosting provider. The system that we developed using Visual Studio Code, Github and phpMyAdmin was pushed to Hostinger to have a website that connected to the database server.

Tools

The tools that we used for the documentation of this system are lucidchart and draw.io. Only the use case diagram of this system was developed using draw.io whereas the rest of the diagrams were developed using lucidchart.

7.0 Function/Module Description

7.1 Authentication module

Person-in-charge: Harini A/P Sangaran

Module Description:

The Authentication Module manages interactions between members, staff and ALK with the login page. The module checks the authenticity of the user ID and password of each member, staff or ALK. If it's authenticated, then the members, staff or ALK can login to the page and interact with the system.

Use Cases:

1. Login (UC05): This use case allows users to access the system.
2. Logout (UC04): This use case allows users to exit the system.

7.2 Non-member Module

Person-in-charge: Nurul Ika Syafiny Binti Azhar

Module Description:

The Non-member Module manages interactions with individuals who are not yet registered members. It facilitates the application process, allows non-members to view the homepage, and provides the necessary steps for membership approval.

Use Cases:

1. Apply for Membership (UC01): Allows a non-member to submit an application to become a member.
2. View Homepage (UC02): Enables non-members to access a public-facing homepage with limited content.
3. Get Membership Approval (UC03): Check status of membership application that has been applied.

7.3 Member Module

Person-in-charge: Nuraisyah Binti Mohd Zikre

Module Description

In this module, it manages individuals who are now a qualified member of Koperasi KADA after applying for membership and they are required to log in into the system in order to gain access to the system. It allows members to view savings, get invoices and view member menus.

Use Cases:

1. Get invoice (UC 15): Allows member to view and get their invoice of every transaction made
2. Check current saving share (UC 16): Allows user to view their current savings and share
3. View member menu (UC 17): Allows member to view the menu where the menu is the function that accessible only for a member of Koperasi KADA
4. Apply membership termination (UC 23): Allows member to terminate the membership with Koperasi KADA by submit the application

7.4 Loan Module

Person-in-charge: Nurul Ika Syafiny Binti Azhar and Nuraisyah Binti Mohd Zikre

Module Description

In this module, it allows the members of Koperasi KADA to manage loan related activities which consists of applying for a loan, checking loan balance, view loan approval (if any) and transaction history. Where they are required to log in into the system in order to get the access.

Use Cases:

1. Check Loan Balance (UC11): It allows member of Koperasi KADA to view outstanding balance of loan they applied
2. Check Loan Approval (UC12): It allows members of Koperasi KADA to be notified with the status of loan application that they applied
3. Find Transaction History (UC13): It allows member of Koperasi KADA to view the transaction record of every transaction they received
4. Apply for Loan (UC14): It allows the member of Koperasi KADA to apply for a loan

7.5 Staff Module

Person-in-charge: Harini A/P Sangaran and Damiya Aina binti Basir Abd Shammad

Module Description

The Staff Module manages interactions with Koperasi KADA staff by providing access to system menus, approving applications, and managing data validation. It supports essential staff operations, ensuring smooth administrative processes.

Use Cases:

1. View staff menu (UC10): Allows staff to see the menu of the system where all their functions are available.
2. Validate member application form (UC07): Validates each submitted membership application form and ensures the details are true before being sent to the ALK.
3. View member application form (UC 06): Allows staff members to view submitted membership application applied by non-member
4. Validate loan application form (UC08): Validates each submitted loan application form and ensures the details are true before being sent to the ALK.
5. View loan application form (UC 09): Allows staff members to view submitted loan application applied by member
6. View member discontinuation form (UC 28): Allows staff members to view submitted member termination application
7. Approve membership discontinuation (UC 29): Staff will approved or disapproved the membership termination request submitted by member of Koperasi KADA
8. Add new admin (UC 26): Add another admin to manage the process in Koperasi KADA
9. Add new ALK (UC 27): Add another ALK to manage the process in Koperasi KADA
10. View monthly report (UC 24): View the monthly generated report
11. View annual report (UC 25): View the annual generated report

7.6 ALK Module

Person-in-charge: Damiya Aina binti Basir Abd Shammad

Module Description

The ALK Module manages tasks and approvals performed by ALK members. It allows viewing and validating member and loan applications, approving or rejecting submissions, and accessing financial transaction data for informed decision-making.

Use Cases:

1. Approve member application (UC18): Allows ALK members to approve or reject membership applications.
2. Approve loan application (UC19): Enables ALK members to approve or reject loan applications.
3. View validated loan application form (UC20): Allows ALK members to view submitted and validated loan application forms.
4. View validated member application form (UC21): Enables ALK members to view submitted and validated membership forms.
5. View ALK menu (UC22): Provides ALK members with access to their menu and related functionalities.
6. View monthly report (UC 30): View the monthly generated report
7. View annual report (UC 31): View the annual generated report

8.0 System Credentials (Hosting)

8.1 Member

- ID number : 0008
- Password : 1234

8.2 Admin

- ID number : 0001
- Password : 1234

8.3 Ahli Lembaga Koperasi (ALK)

- ID number : 0001
- Password : 1234

9.0 Localhost Setup

1. Download and install XAMPP Control Panel from <https://www.apachefriends.org/>.
2. Start Apache and MySQL.

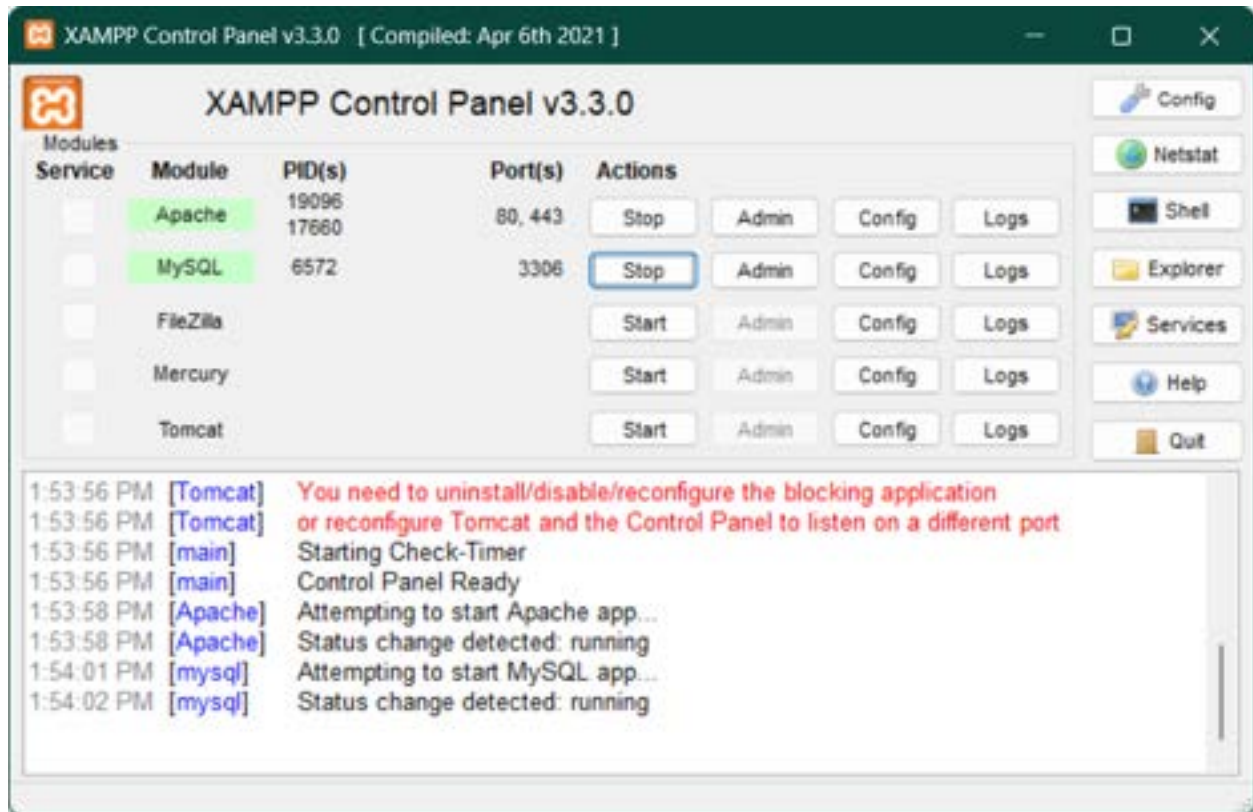


Figure 33: XAMPP Control Panel

3. Click on 'Admin' button under MySQL to go to phpMyAdmin page.
4. Create a database named 'koperasikada'.

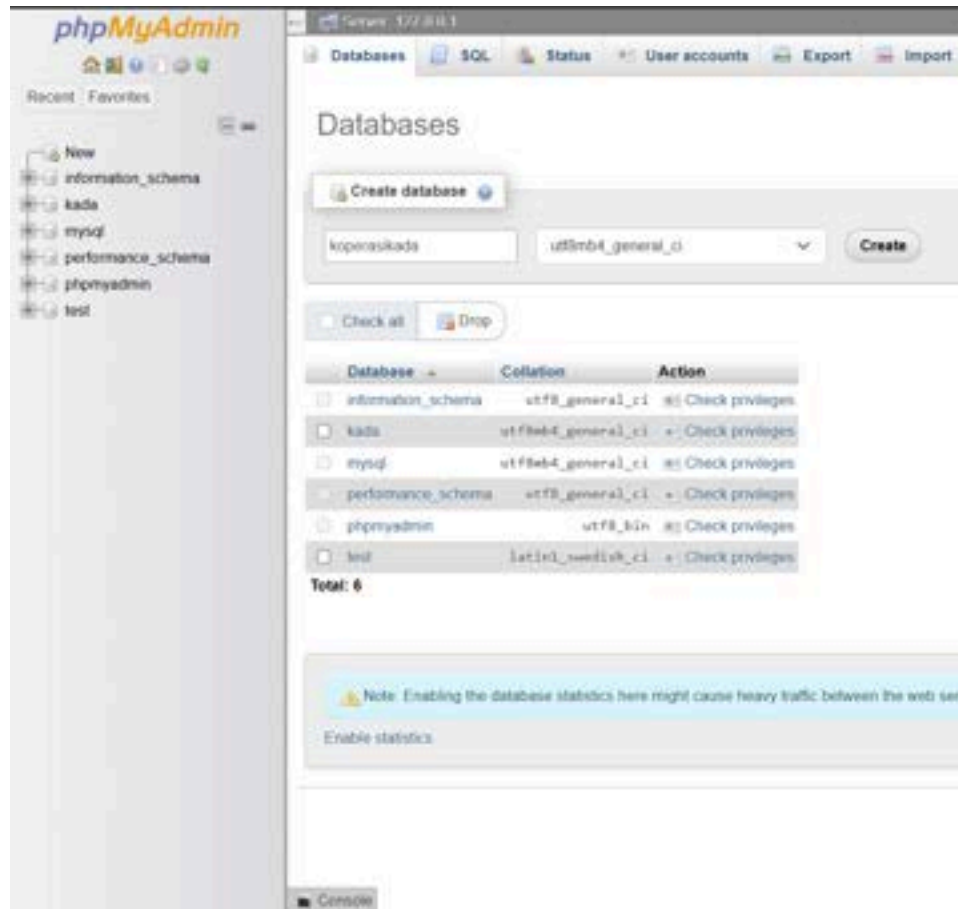


Figure 34: List of Database

5. Enter the SQL for each table under the SQL column of koperasikada database to establish databases.

10.0 System Interface for All Users

10.1 Non-Member user

10.1.1 Homepage Koperasi KADA



Figure 35: Homepage Interface

10.1.2 Apply Membership

The screenshot shows a web form titled 'Maklumat Pemohon' (Applicant Information). The form is divided into two main sections: 'Maklumat Pemohon' and 'Maklumat Pejabat' (Office Information). The 'Maklumat Pemohon' section includes fields for 'Name', 'Nombor K/P', 'Email', 'Taraf Perkahwinan' (Marital Status) with a dropdown menu, 'Jantina' (Gender) with a dropdown menu, 'Agama' (Religion) with a dropdown menu, 'Bangsa' (Race), 'Alamat Rumah' (Home Address), 'Poskod' (Postcode), 'Negari' (Country), and 'Bandar' (City). The 'Maklumat Pejabat' section includes fields for 'No. Anggutan' (Membership No.), 'No. PP' (PP No.), 'Jawatan & Gred' (Position & Grade), 'Alamat Pejabat' (Office Address), 'Poskod' (Postcode), and 'No. Telefon' (Phone No.). The form is displayed on a white background with a light gray border. The Windows taskbar is visible at the bottom.

Perkiraan	RM
Yuran Masuk	
Modal Syer	
Modal Yuran	
Wang Deposit Anggota	
Sumbangan Tabung Kelapangan (Kil Abrai)	
Simpanan Tetap	
Lain-lain	
Simpan Maklumat	

Figure 36: Apply Membership Interface

10.1.3 Check Membership Application Status

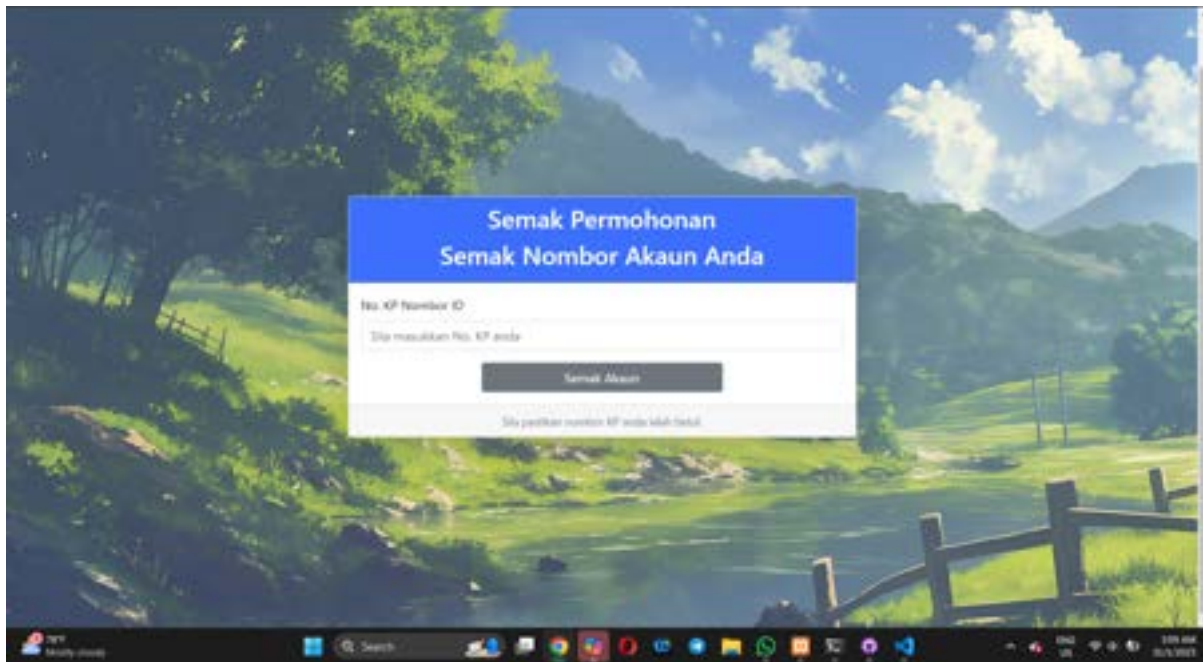


Figure 37: Check Membership Application Interface

10.1.4 Approved Membership Status



Figure 38: Approved Membership Status Interface

10.1.5 Pending Membership Status



Figure 39: Pending Membership Status Interface

10.1.6 Reviewed Membership Status



Figure 40: Reviewed Membership Status Interface

10.1.7 Rejected Membership Status



Figure 41: Rejected Membership Status Interface

10.2 Member User

10.2.1 Login Page

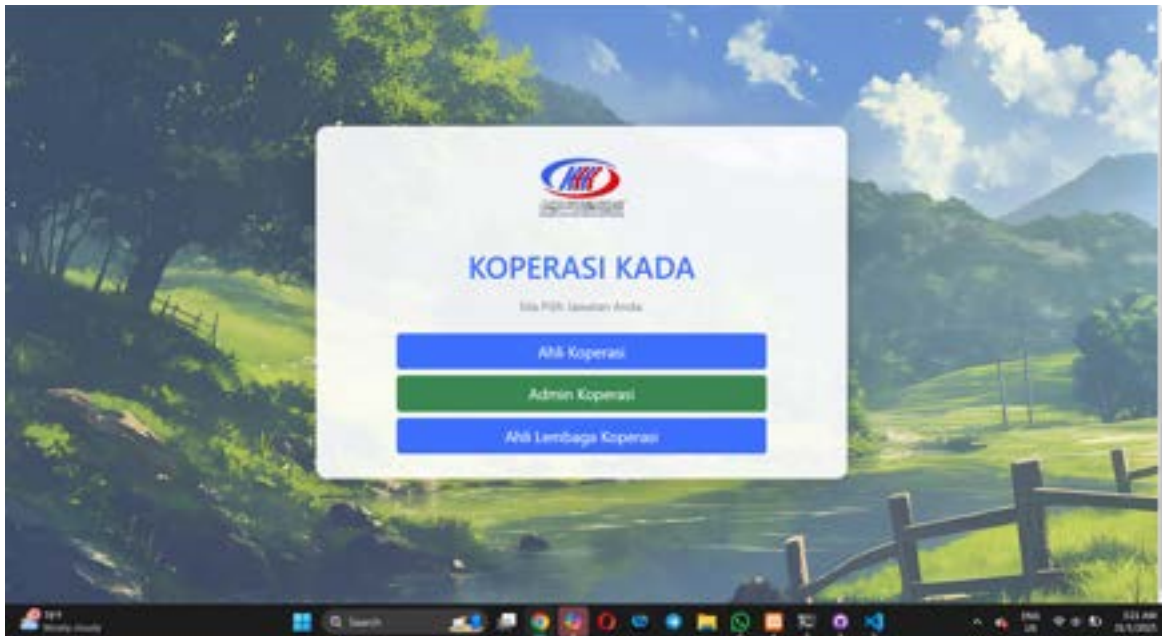


Figure 42: Login Page Interface

10.2.2 Member Login

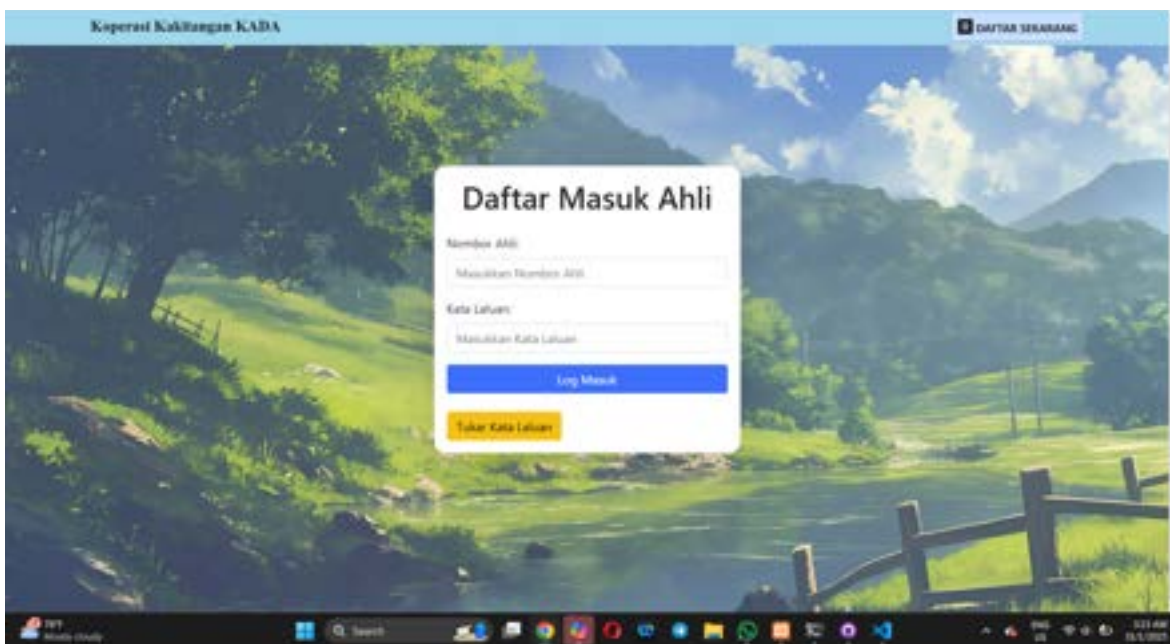


Figure 43: Member Login Interface

10.2.3 Member Homepage



Figure 44: Member Homepage Interface

10.2.4 Member Details

The screenshot shows the 'MAKLUMAT AHLI' (Member Details) form. The form is divided into two sections: 'Maklumat Peribadi' (Personal Information) and 'Maklumat Pejabat' (Office Information). The 'Maklumat Peribadi' section includes fields for Name, Email, Telephone, Gender, Religion, Address, and Postcode. The 'Maklumat Pejabat' section includes fields for No. Anggutan, No. ID, Jawatan & Gred, and Alamat Pejabat. The form is set against a background image of a field at sunset. At the bottom, there's a Windows taskbar with various icons and the system clock showing '1:40 PM 10/1/2021'.

Keperni Kalkulation KADA Pilihan DAFTAR KALAM

Gaji Bulanan

Gaji Bulanan
1000.00

Purani & Sumbangan

Perkara	RM
Yuran Masuk	0.00
Modul Spm	0.00
Modul Tahun	0.00
Wang Deposit Anggota	0.00
Sumbangan Tabung Kalkulation (AI Aliran)	0.00
Simpanan Tabung	0.00
Lain-lain	0.00

[Kemaskini Maklumat](#)

Figure 45: Member Details Interface

10.2.5 Update Member Details

Keperni Kalkulation KADA Pilihan DAFTAR KALAM KEMASKINI MAKLUMAT PEMOHON

Maklumat Pribadi:

Nama: Number K/P:

Email:

Tarikh Perlahutan: Jantina: Agama:

Agama: Alamat Rumah:

Produk: Sejenis: Bandar:

Maklumat Pejabat:

No. Anggota: No. PP:

Jawatan & Grad: Alamat Pejabat:

Koperasi Kakitangan KADA

Pilihan + DAFTAR KESUKSESAN 

No. Kad Pengiraan
75000025000

Gaji Bulanan
5000.00

Yuran & Sumbangan

Pelaksanaan	RM
Yuran Masuk	95.00
Akademi SPM	0.00
Modal Yuran	0.00
Wang Deposit Anggota	0.00
Sumbangan Tabung Kelengkapan (M. Rinal)	0.00
Simpanan Tetap	0.00
Lain-lain	0.00

[SIMPAN & MALLUARI](#)

Figure 46: Update Member Details Interface

10.2.6 Apply Loan

Koperasi Kakitangan KADA

Pilihan + DAFTAR KESUKSESAN 

JADUAL PEMBAYARAN BALIK PEMBIAYAAN

Tempoh (Tahun)	1	2	3	4	5	6
Tempoh (Bulan)	12	24	36	48	60	72
Jumlah Pembiayaan	Jumlah Bulanan					
1,000.00	88.83	45.17	31.28	24.33	20.17	17.39
2,000.00	173.67	90.33	62.56	48.67	40.33	34.78
3,000.00	260.50	135.50	93.83	73.00	60.50	52.17
4,000.00	347.33	180.67	125.11	97.33	80.67	69.54
5,000.00	434.17	225.83	156.39	121.67	100.83	86.94
6,000.00	521.00	271.00	187.67	146.00	121.00	104.33
7,000.00	607.83	316.17	218.94	170.33	141.17	121.72

BORANG PERMOHONAN PINJAMAN

Jenis Pinjaman

[illegible]

Figure 47: Apply Loan Interface

10.2.7 Check Loan Application Status

Figure 48: Check Loan Application Interface

10.2.8 Check Member Saving

Koperasi Kakehungan KADA Pilihan 77 SARANA KUDARA 

SIMPANAN DAN SYER PERIBADI

Nama: Masri A/P Samsoni
No. ID: 041201070100
No. KAD: 0001

Simpanan Peribadi Ahli

Number akaun	Baki Simpanan (RM)	Tarikh Kemaskini
1	100.00	2025-01-08 02:26:54
1	90.00	2024-11-20 02:27:32
2	50.00	2023-12-19 02:26:26
2	130.00	2025-01-14 02:26:47
3	200.00	2024-12-26 02:33:08

Syer Ahli

Syar Majikan	Syar Pekerja	Tarikh
70.00	33.00	2025-01-08 02:49:36
85.00	70.00	2024-11-20 02:30:16
80.00	40.00	2023-12-19 02:58:29
90.00	65.00	2025-01-14 02:58:40
180.00	110.00	2024-12-26 02:50:52

Windows taskbar: 100% zoom, 10:00 AM, 10/10/2024

Figure 49: Check Member Saving Interface

10.2.9 Check Loan Balance



Figure 50: Check Loan Balance Interface

10.2.10 Check Invoice/Transaction History



Figure 51: Check Invoice/Transaction History Interface

10.2.11 Apply Membership Termination Form

Koperasi Kakitangan KADA

Pilihan **DAFTAR KELUAR**

Borang Permohonan Berhenti Keahlian

Nama No KP

Tarikh Lahir Umur

Alamat Rumah

Poskod Bandar Negeri

No Ahli No PF

Pekerjaan Alamat Pejabat

Poskod Pejabat Bandar Pejabat No Tel/Tar

No Tel Berozot Jantina

Agama Nyatakan (Jika Lain Saja)

Bangsa Nyatakan (Jika Lain Saja)

Sebab Berhenti

Hantar Permohonan

Figure 52: Apply Membership Termination Interface

10.2.12 Navigation Bar



Figure 53: Navigation Bar Interface

10.3 Admin User

10.3.1 Admin Login

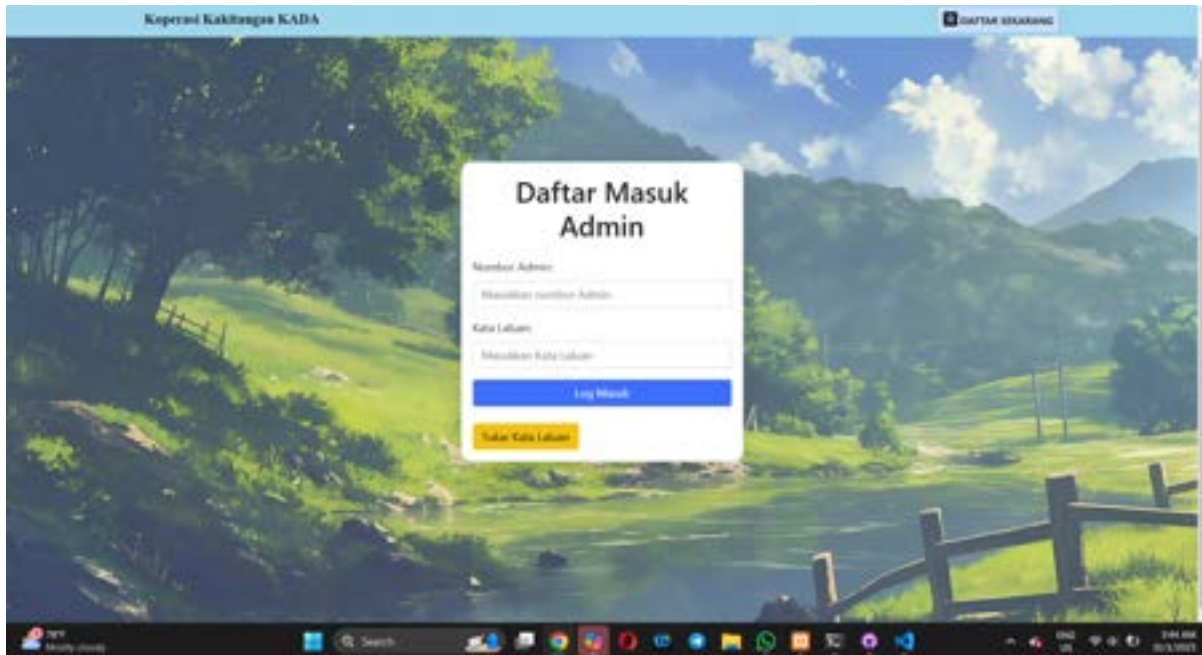


Figure 54: Admin Login Interface

10.3.2 Admin Homepage



Figure 55: Admin Homepage Interface

10.3.3 View Membership Application List

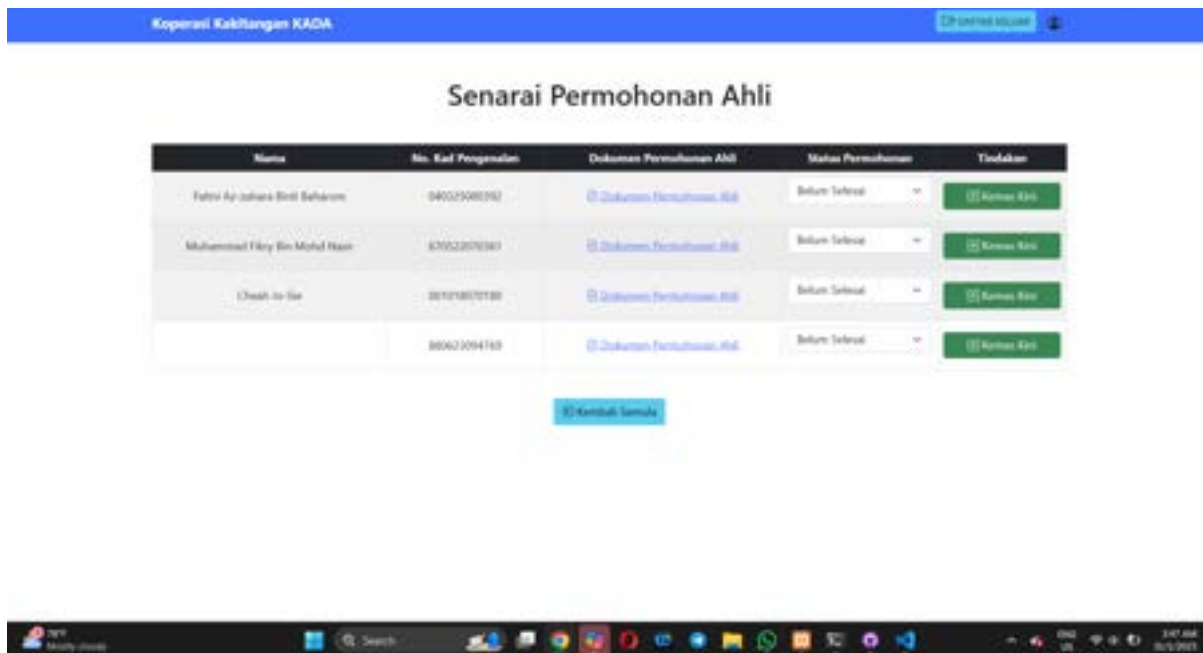


Figure 56: View Membership Application List Interface

10.3.4 Successful Membership Application Status Update

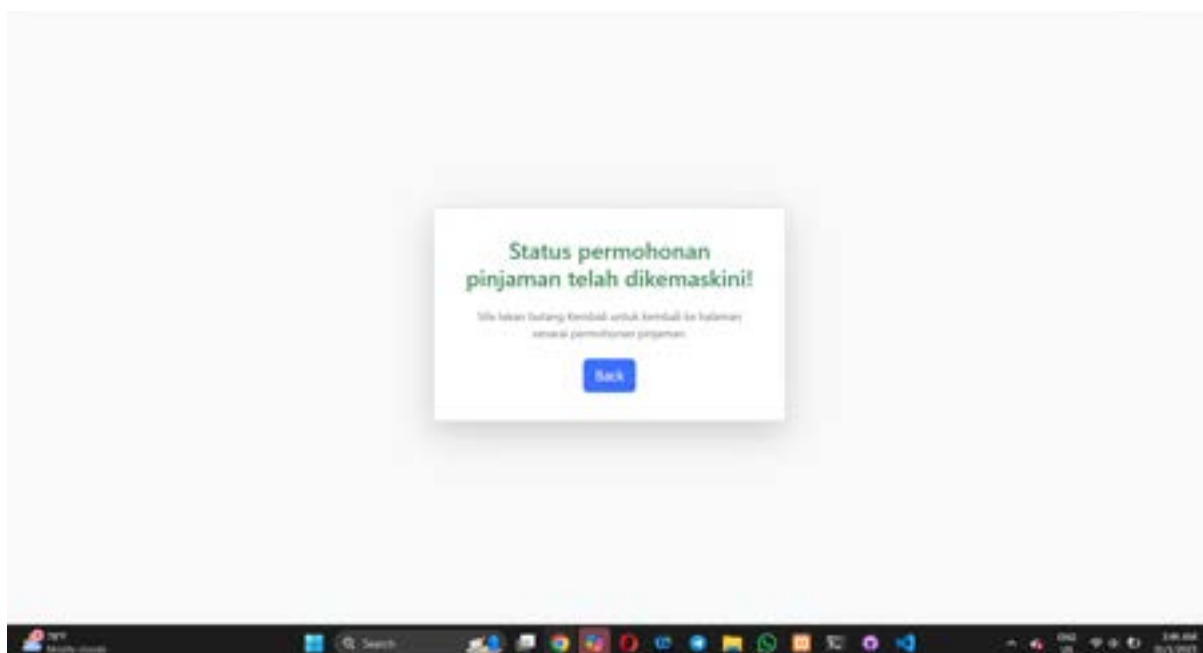


Figure 57: Successful Membership Application Status Update Interface

10.3.5 View Loan Application List



Figure 58: View Loan Application List Interface

10.3.6 Successful Loan Application Status Update

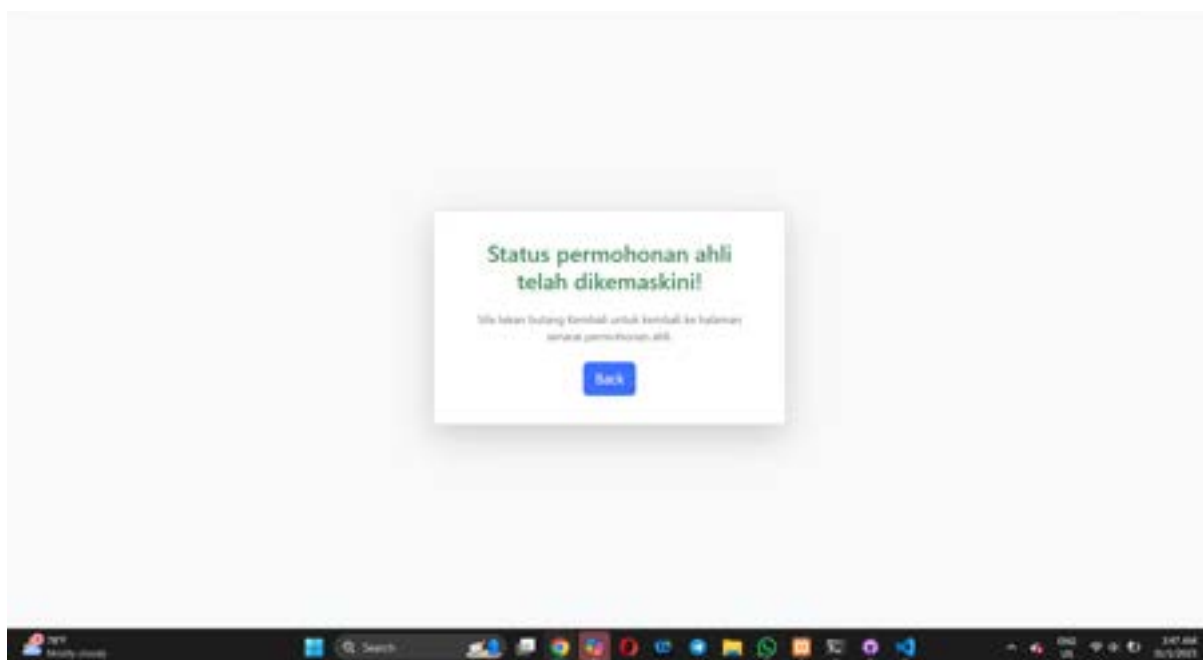


Figure 59: Successful Loan Application Status Update Interface

10.3.7 View Monthly and Annual Report

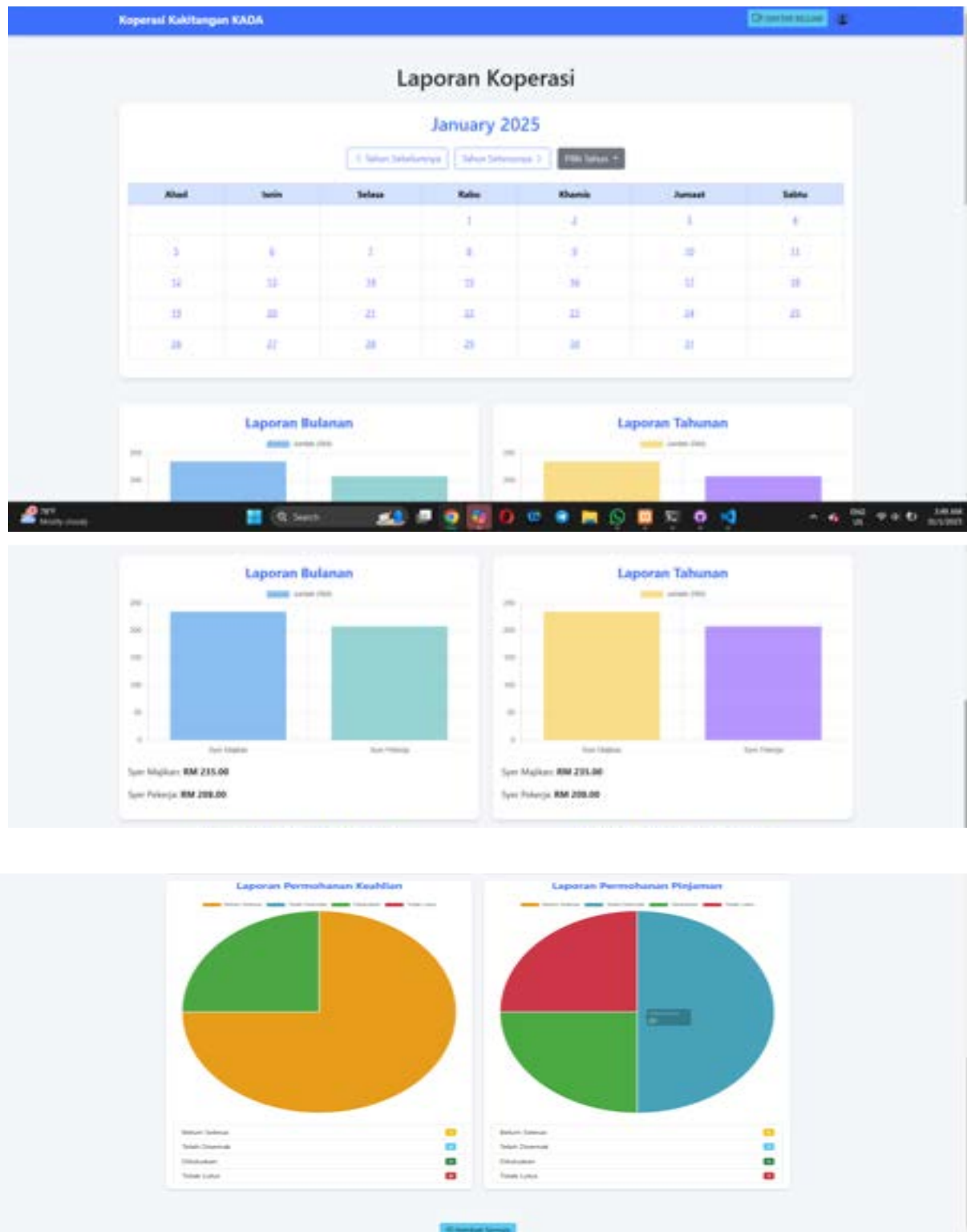


Figure 60: Monthly and Annual Report Interface

10.3.8 Add New Admin

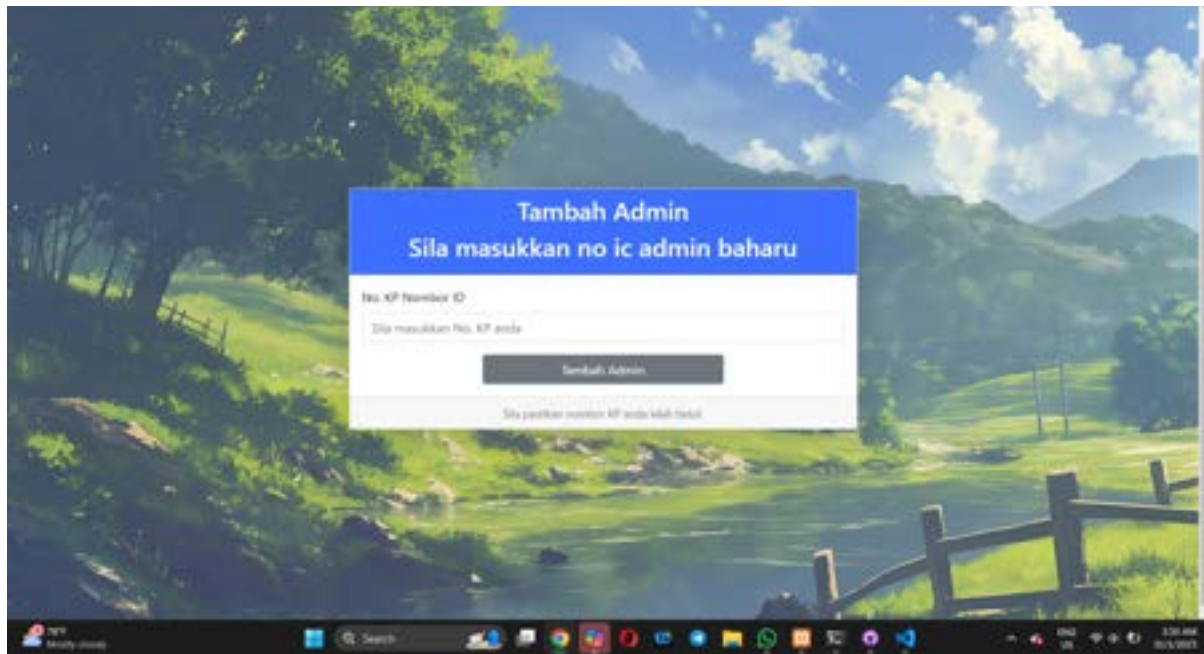


Figure 61: Add New Admin Interface

10.3.9 Add New ALK

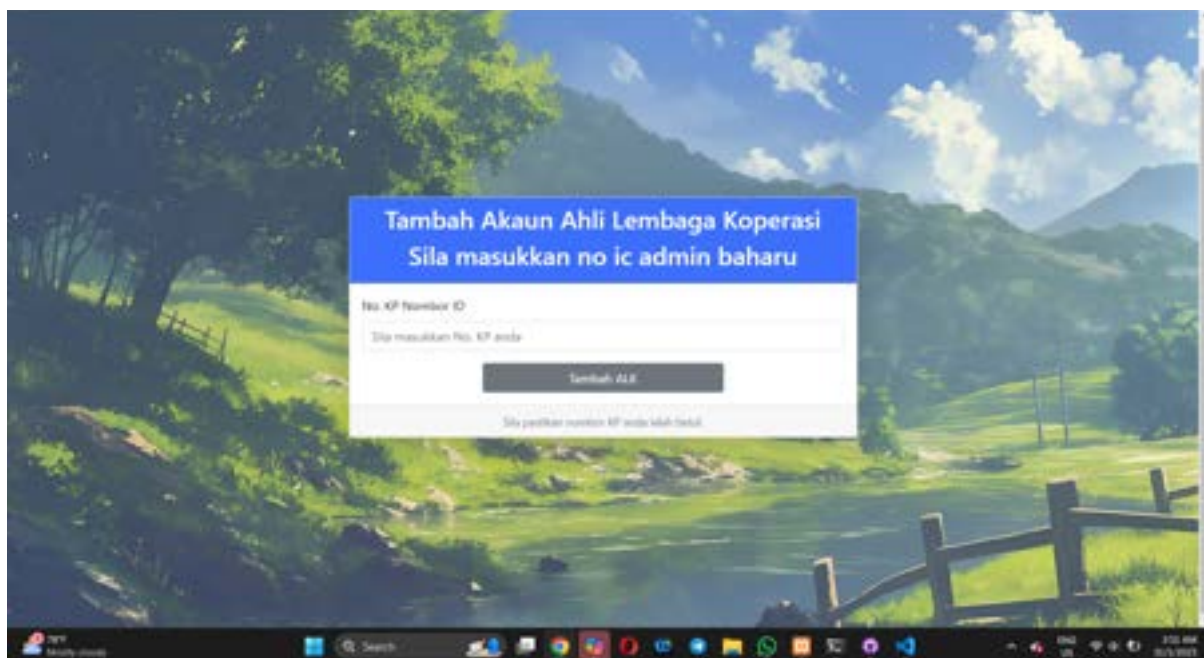


Figure 62: Add New ALK Interface

10.3.10 Successful Add New Admin/ALK

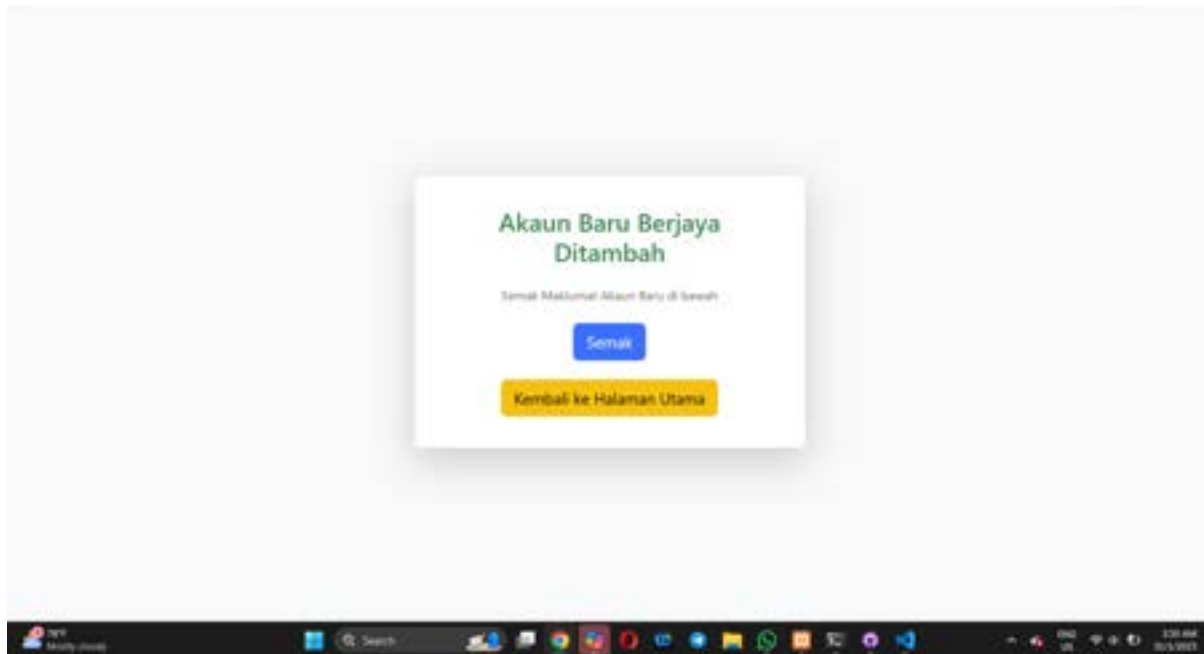


Figure 63: Successful Add New Admin/ALK Interface

10.3.11 View Membership Discontinuation List



Figure 64: View Membership Discontinuation List Interface

10.4 ALK User

10.4.1 ALK Login

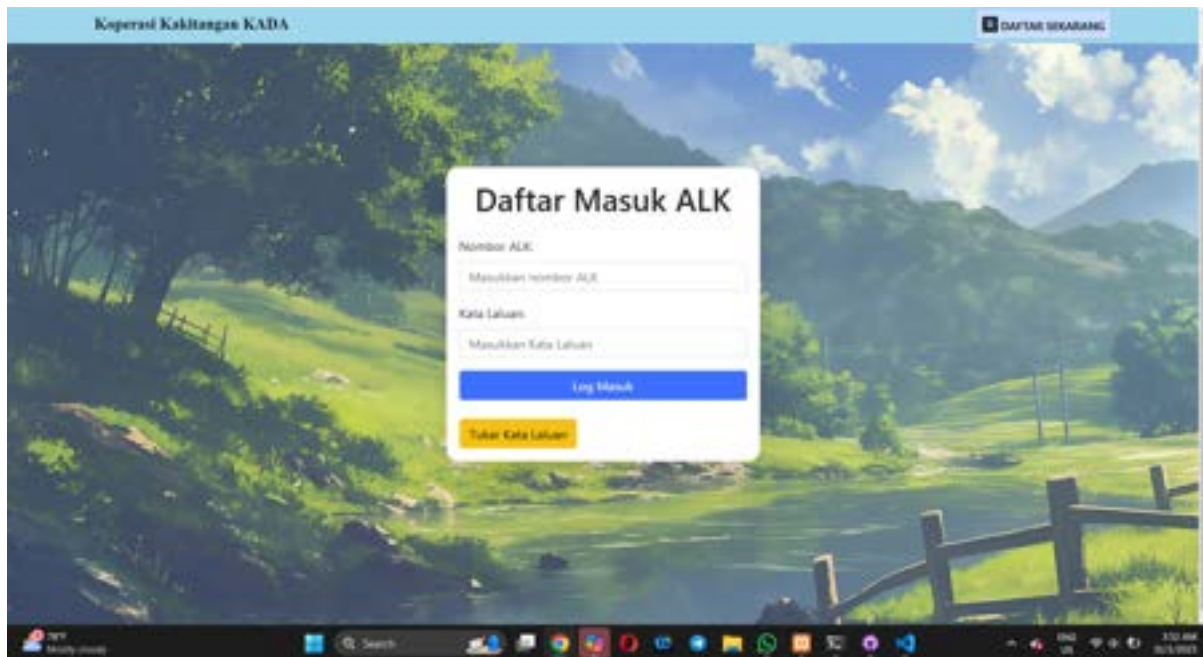


Figure 65: ALK Login Interface

10.4.2 ALK Homepage



Figure 66: ALK Homepage Interface

10.4.3 Approve Loan Application

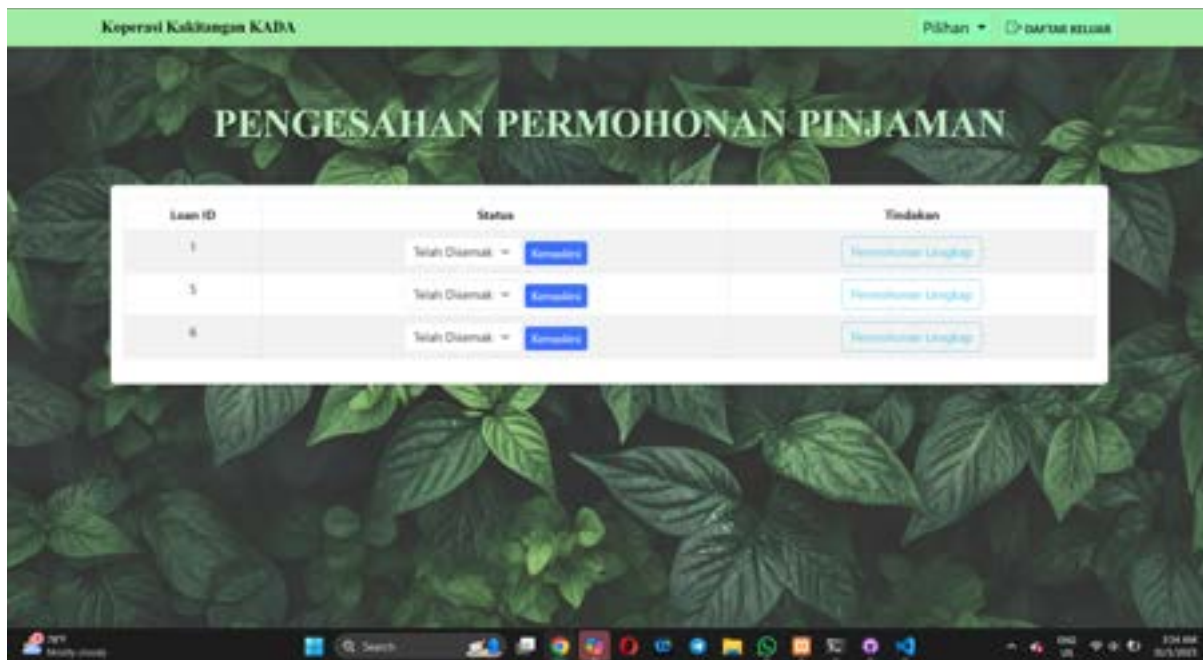


Figure 67: Approve Loan Application Interface

10.4.4 Successful Loan Application Status Update

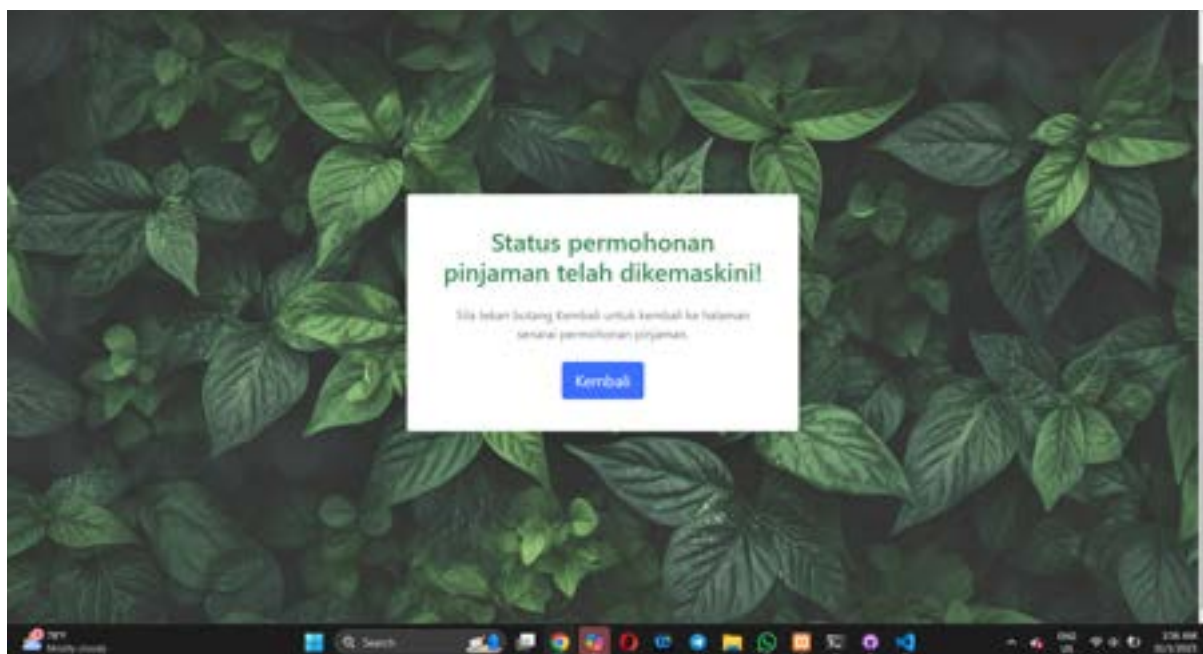


Figure 68: Successful Loan Application Status Update Interface

10.4.5 Approve Membership Application

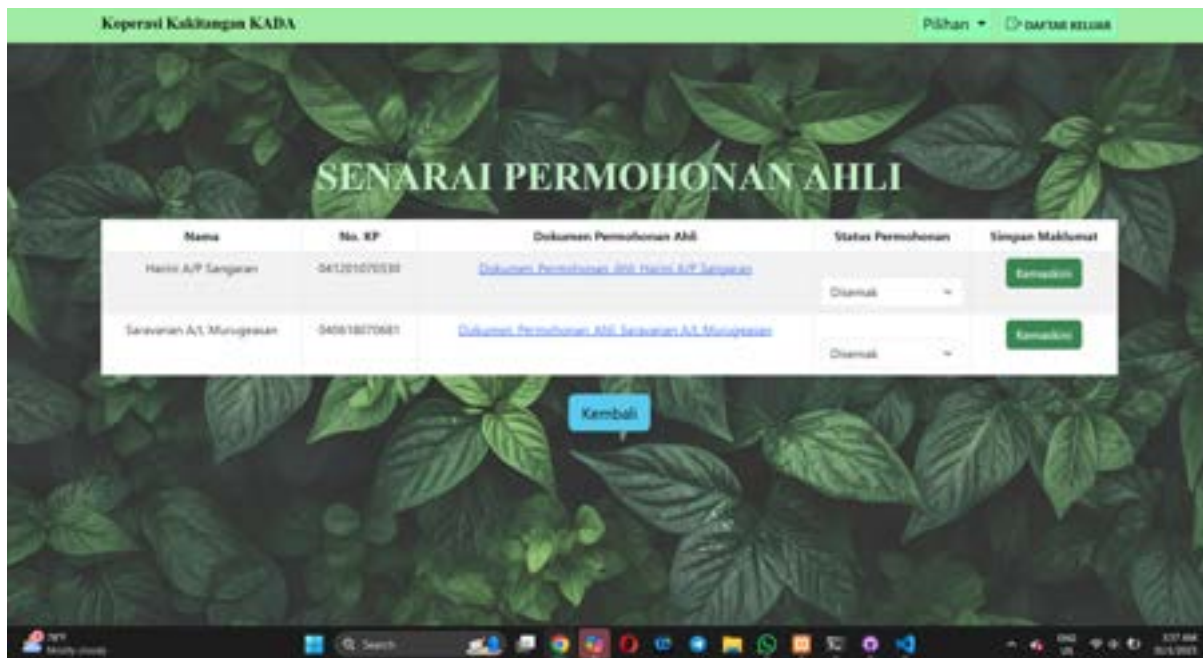


Figure 69: Approved Membership Application Interface

10.4.6 Successful Membership Application Status

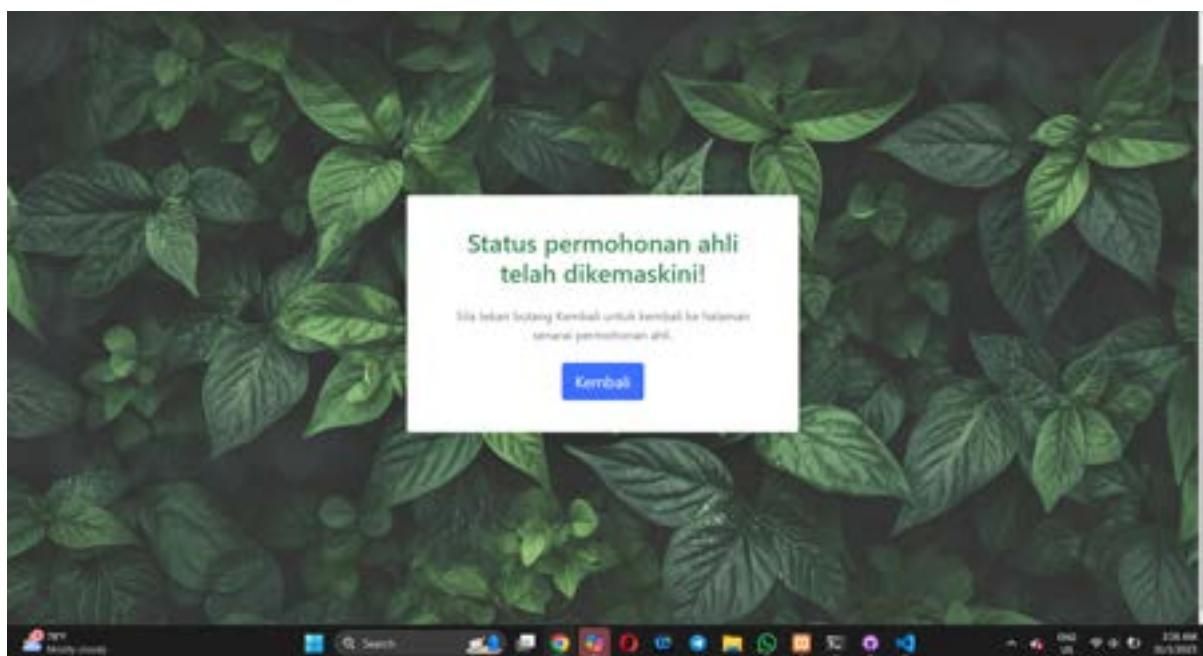


Figure 70: Successful Membership Application Status Update Interface

10.4.7 Navigation Bar

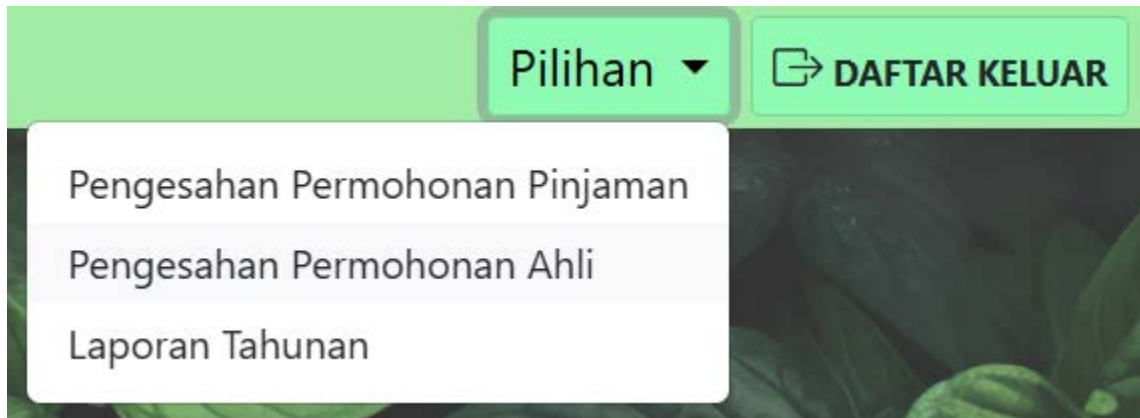
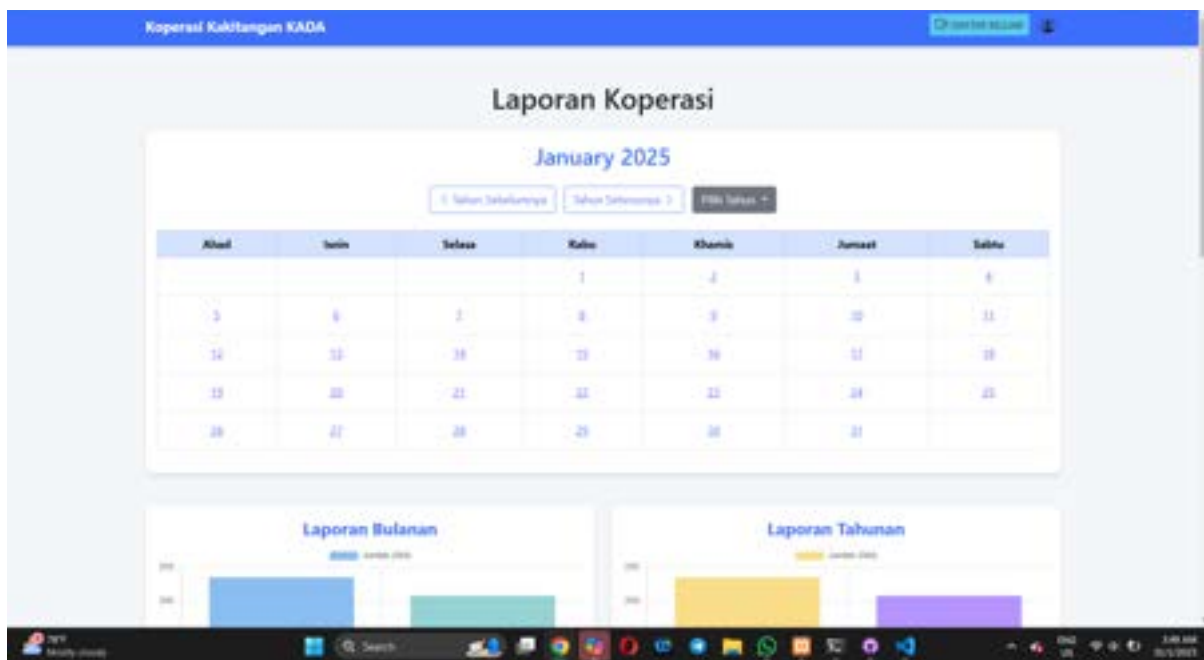


Figure 71: Navigation Bar Interface

10.3.7 View Monthly and Annual Report



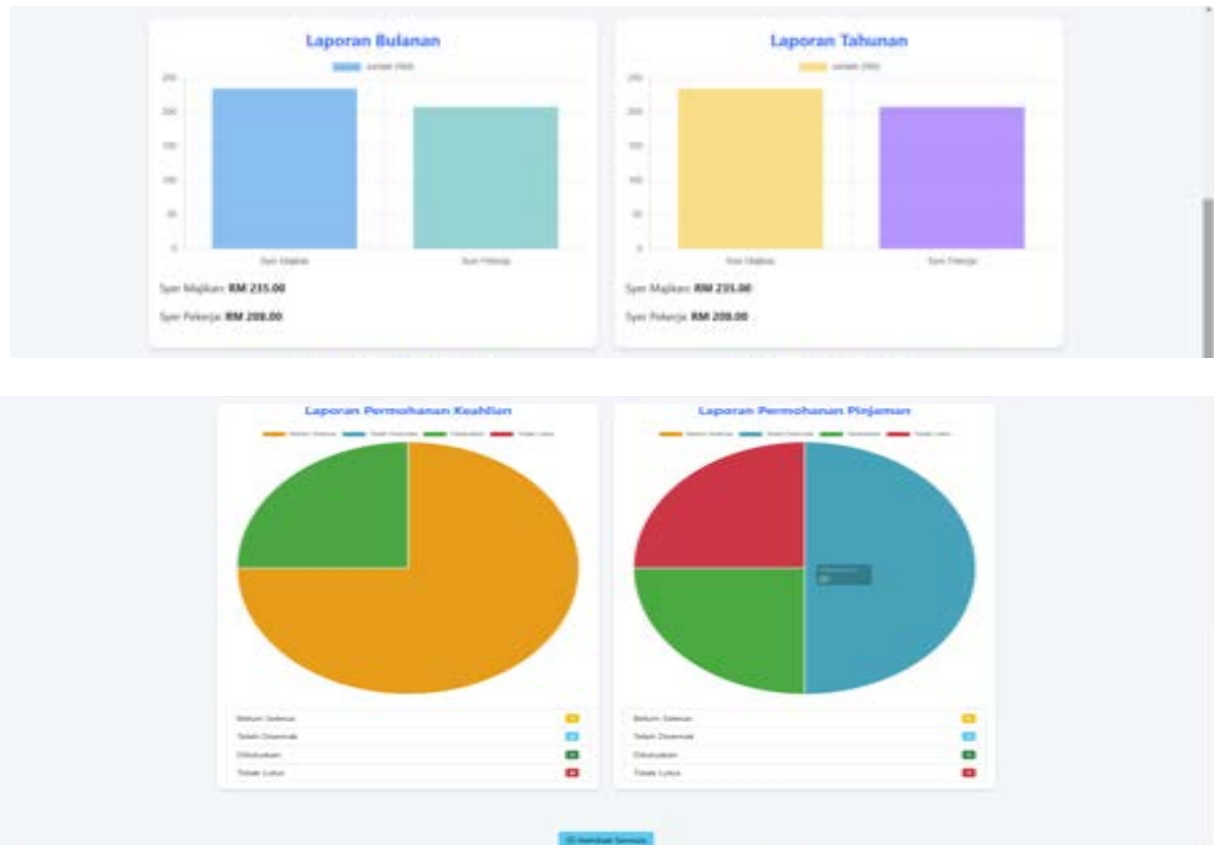


Figure 72: Monthly and Annual Report Interface

10.5 All User

10.5.1 Reset Password



Figure 73: Reset Password Interface

10.5.2 Successful Reset Password



Figure 74: Successful Reset Password Interface

10.5.3 Unsuccessful Reset Password



Figure 75: Unsuccessful Reset Password Interface

10.5.4 Log Out



Figure 76: Log Out Interface

11.0 Alpha and Beta Testing Reports

In this section, we are going to explain the tests that are done to our system.

11.1 Alpha Testing Report

11.1.0 Equivalence Partitioning and Boundary Value Analysis

Object 1: Non-member

Object Class	Attributes	Equivalence Classes	Boundary Values
Member Application	applicant_id	Valid: 1, 100, 1000 Invalid: -1, 1.10, abc	Minimum: 1 Maximum: 2,147,483,647
	name	Valid: "John Doe", "Ali Ahmad" Invalid: "", "J"	Minimum length: 2 characters Maximum length: 255 characters
	ic_number	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
	email	Valid: "user@example.com", "info@domain.net" Invalid: "user123.com", "test@domain"	Minimum length: a@domain.com Maximum length: 255 characters
	marital_status	Valid: "Bujang", "Berkahwin", "Duda", "Janda" Invalid: "Single", "Married"	Enum Values: 4 fixed values
	gender	Valid: "Lelaki", "Perempuan" Invalid: "Male", "Female"	Enum Values: 2 fixed values
	age	Valid: 18, 30, 60 Invalid: 0, -5, 150	Minimum: 18 Maximum: 100
	religion	Valid: "Islam", "Christian", "Hindu" Invalid: "None", "XYZ"	Minimum length: 3 characters Maximum length: 50 characters
	ethnicity	Valid: "Malay", "Chinese",	Minimum length: 3

	"Indian" Invalid: "123", "Unknown"	characters Maximum length: 50 characters
home_address	Valid: "No 10, Jalan ABC, Kuala Lumpur" Invalid: ""	Minimum length: 10 characters Maximum length: 500 characters
postcode	Valid: "50000", "12345" Invalid: "500", "abc12"	Minimum length: 5 characters Maximum length: 10 characters
position_grade	Valid: "Gred 41", "Gred 44" Invalid: "Level 1", "Grade A"	Minimum length: 3 characters Maximum length: 100 characters
phone_mobile	Valid: "012-3456789", "+6012-3456789" Invalid: "12345", "phone123"	Minimum length: 10 characters Maximum length: 15 characters
monthly_salary	Valid: 3000, 10000 Invalid: -500, "abc"	Minimum: 1000 Maximum: 1,000,000
famName1	Valid: "Ahmad Ali", "Siti Nor" Invalid: ""	Minimum length: 2 characters Maximum length: 255 characters
famRelationship1	Valid: "Bapa", "Ibu" Invalid: ""	Minimum length: 2 characters Maximum length: 255 characters
famiC1	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
famName2	Valid: "Ahmad Ali", "Siti Nor" Invalid: ""	Minimum length: 2 characters Maximum length: 255 characters

famRelationship2	Valid: "Bapa", "Ibu" Invalid: ""	Minimum length: 2 characters Maximum length: 255 characters
famiC2	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
yuran_masuk_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
modal_syer_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
modal_yuran_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
wang_deposit_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
sumbangan_kebajikan_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
simpanan_tetap_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
lain_lain_value	Valid: 100.00, 5000.50 Invalid: -100, "abc"	Minimum: 0.01 Maximum: 1,000,000.00
approval	Valid: "Pending", "Reviewed", "Disapproved", "Approved" Invalid: "Yes", "No"	Enum Values: 4 fixed values

Table 1: Equivalence and Boundary Analysis for Membership Application

Object 2: Member:

Object Class	Attributes	Equivalence Classes	Boundary Values
Member Termination Request	id	Valid: 1, 100, 1000 Invalid: -1, 1.10, abc	Minimum: 1 Maximum: 2,147,483,647
	member_id	Valid: 0001, 0012, 1234 Invalid: "abc", -5	Minimum: 0001 Maximum: 9999
	name	Valid: "John Doe", "Ali Ahmad" Invalid: "", "J"	Minimum length: 2 characters Maximum length: 255 characters
	no_kp	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
	tarikh_lahir	Valid: "1990-05-20", "1985-12-01" Invalid: "2025-10-01", "abcd"	Date Format: YYYY-MM-DD
	umur	Valid: 18, 45, 60 Invalid: -5, 150	Minimum: 18 Maximum: 100
	gender	Valid: "Lelaki", "Perempuan" Invalid: "Male", "Female"	Enum Values: 2 fixed values
	age	Valid: 18, 30, 60 Invalid: 0, -5, 150	Minimum: 18 Maximum: 100
	tarikh_permohonan	Valid: "2024-01-10", "2023-12-31" Invalid: "abcd"	Date Format: YYYY-MM-DD
	status	Valid: "Pending", "Approved", "Rejected" Invalid: "Yes", "No"	Enum Values: 3 fixed values

Table 2: Equivalence and Boundary Analysis for Member Termination Request

Object Class	Attributes	Equivalence Classes	Boundary Values
Loan Application	LoanID	Valid: 1, 100, 1000 Invalid: -1, 1.10, abc	Minimum: 1 Maximum: 2,147,483,647
	member_id	Valid: 0001, 0012, 1234 Invalid: "abc", -5	Minimum: 0001 Maximum: 9999
	LoanType	Valid: "Al-Bai", "Al-Inah", "Skim Khas" Invalid: "Personal Loan", "House Loan"	Enum Values: 6 fixed values
	LoanAmount	Valid: 1000.00, 50000.00 Invalid: -500, "abc"	Minimum: 1000.00 Maximum: 1,000,000.00
	RepaymentPeriodMonths	Valid: 12, 24, 60 Invalid: -6, 0, "abc"	Minimum: 6 months Maximum: 360 months
	MonthlyInstallment	Valid: 100.00, 5000.00 Invalid: -10, "xyz"	Minimum: 50.00 Maximum: 50,000.00
	approval	Valid: "Pending", "Reviewed", "Disapproved", "Approved" Invalid: "Yes", "No"	Enum Values: 4 fixed values
	BankName	Valid: "Maybank", "CIMB" Invalid: "", "XYZ Bank 123"	Minimum length: 2 Maximum length: 100
	AccountNumber	Valid: "123456789012" Invalid: "1234", "abcdef"	Minimum length: 10 Maximum length: 20

Table 3: Equivalence and Boundary Analysis for Loan Application

Object 3: Admin

Object Class	Attributes	Equivalence Classes	Boundary Values
Add ALK	ALK_id	Valid: 1, 100, 1000 Invalid: -1, 1.10, abc	Minimum: 1 Maximum: 2,147,483,647
	Password	Valid: "P@ssword123", "MySecurePass99!" Invalid: "", "123"	Minimum length: 8 Maximum length: 255
	no_kp	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
	user_role	Valid: "ALK" Invalid: "Admin", "User", "123"	Enum Values: Fixed value "ALK"

Table 4: Equivalence and Boundary Analysis for Add ALK

Object Class	Attributes	Equivalence Classes	Boundary Values
Add Admin	Admin_id	Valid: 1, 100, 1000 Invalid: -1, 1.10, abc	Minimum: 1 Maximum: 2,147,483,647
	Password	Valid: "P@ssword123", "MySecurePass99!" Invalid: "", "123"	Minimum length: 8 Maximum length: 255
	no_kp	Valid: "900101-14-5678" Invalid: "1234", "abc"	Minimum length: 12 characters Maximum length: 20 characters
	user_role	Valid: "ADMIN" Invalid: "ALK", "User", "123"	Enum Values: Fixed value "ADMIN"

Table 5: Equivalence and Boundary Analysis for Add Admin

11.1.1 Test Cases Based on Subsystem

TC001: Test Member Subsystem: <User Story 1 (US001)>

This test contains the following test cases:

- (a) TC001_01: Test <View Member Menu (SD024)>
- (b) TC001_02: Test <Check Current Saving (SD003)>
- (c) TC001_03: Test <Apply Membership Termination (SD006)>

DESCRIPTION	VERSION	TEST DATE
View Member Menu	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the members are able to view the appropriate homepage once they have successfully logged in.	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Harini A/P Sangaran	-
2	View Member homepage.	-	Display the features for member to do in the system	As Expected	Pass	Harini A/P Sangaran	

Table 6: TC001_01: Test <View Member Menu (SD024)>

DESCRIPTION	VERSION	TEST DATE
Check Current Saving	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
Ensure that the system correctly displays the member's savings including the personal and member syer	Nuraisyah binti Mohd Zikre	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Damiya Aina binti Basir Abd Shammd	-
2	Navigate to the Semak Simpanan section from the Member homepage.	-	Successfully load the member saving section, displaying all the information of the member saving including the personal saving and member syer	As Expected	Pass	Damiya Aina binti Basir Abd Shammd	
3	view the table saving and member syer	-	table saving and member syer is successfully appear	As Expected	Pass	Damiya Aina binti Basir Abd Shammd	

Table 7: TC001_02: Test <Check Current Saving (SD003)>

DESCRIPTION	VERSION	TEST DATE
Applying the Membership Termination Application	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
allows non-members to apply for membership termination by completing the form and giving the reasonable reason of quitting	Nurul Ika Syafiny binti Azhar	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Navigate to Mohon Berhenti Ahli section from the member homepage	-	Successfully load the membership termination section, displaying all the form that need to be filled in	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
3	Fill in the membership termination application form fields (e.g: nama, no.kp, etc)	nama: Ali bin Abu no.kp: 780506030201	Form is completed	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
4	click Hantar Permohonan button after completing the form	-	successfully submit the membership termination application form	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	

Table 8: TC001_03: Test <Apply Membership Termination (SD006)>

TC002: Test Admin Subsystem: <User Story 2 (US002)>

This test contains the following test cases:

- (a) TC002_01: Test <View Staff Menu (SD025)>
- (b) TC002_02: Test <View Member Application Form (SD007)>
- (c) TC002_03: Test <View Loan Application Form (SD008)>
- (d) TC002_04: Test <View Membership Termination Form (SD009)>
- (e) TC002_05: Test <Validate Member Application Form (SD010)>
- (f) TC002_06: Test <Validate Loan Application Form (SD011)>
- (g) TC002_07: Test <Approve Membership Termination (SD012)>
- (f) TC002_08: Test <View Monthly Report (SD013)>
- (g) TC002_09: Test <View Annual Report (SD014)>
- (h) TC002_10: Test <Add New Admin (SD015)>
- (i) TC002_11: Test <Add New ALK (SD016)>

DESCRIPTION	VERSION	TEST DATE
View Staff Menu	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view the appropriate homepage once they have successfully logged in.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Harini A/P Sangaran	-
2	View admin homepage.	-	Display the features for admin to do in the system	As Expected	Pass	Harini A/P Sangaran	-

Table 9: TC002_01: Test <View Staff Menu (SD017)>

DESCRIPTION	VERSION	TEST DATE
View Member Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view the list of pending member application forms.	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View admin homepage.	-	Display the features for admin to do in the system	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-
2	Click on the 'Senarai Permohonan Ahli' button.	-	Display list of pending member application forms.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-

Table 10: TC002_02: Test <View Member Application Form (SD007)>

DESCRIPTION	VERSION	TEST DATE
View Loan Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view the list of pending loan application forms.	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View admin homepage.	-	Display the features for admin to do in the system	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Click on the 'Senarai Permohonan Pinjaman' button.	-	Display list of pending loan application forms.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 11: TC002_03: Test <View Loan Application Form (SD008)>

DESCRIPTION	VERSION	TEST DATE
View Membership Termination Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view the list of pending membership termination forms.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View admin homepage.	-	Display the features for admin to do in the system	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-
2	Click on the 'Senarai Permohonan Berhenti Ahli' button.	-	Display list of pending membership termination forms.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-

Table 12: TC002_04: Test <View Membership Termination Form (SD009)>

DESCRIPTION	VERSION	TEST DATE
Validate Member Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view applicant's verification documents and update the application status.	Nuraisyah binti Mohd Zikre	Harini A/P Sangaran

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Senarai Permohonan Ahli' button.	-	Display list of pending member application forms.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Click on the 'Dokumen Permohonan Ahli' hyperlink.	-	Display applicant's application form.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
3	Click on the drop down button in the 'Status Permohonan' column.	-	Display the current application status by default and display all the possible statuses when clicked on the drop down button.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
4	Click on the 'Kemas Kini' button.	-	Update applicant's application status.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-

Table 13: TC002_05: Test <Validate Member Application Form (SD010)>

DESCRIPTION	VERSION	TEST DATE
Validate Loan Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view member's verification documents and update the loan application status.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Senarai Permohonan Pinjaman' button.	-	Display list of pending loan application forms.	As Expected	Pass	Harini A/P Sangaran	-
2	Click on the 'Dokumen Permohonan Pinjaman' hyperlink.	-	Display member's loan application form.	As Expected	Pass	Harini A/P Sangaran	-
3	Click on the 'Muat Turun Dokumen' button.	-	Download member's verification documents for the loan application	As Expected	Pass	Harini A/P Sangaran	-
4	Click on the drop down button in the 'Status Permohonan' column.	-	Display the current application status by default and display all the possible statuses when clicked on the drop down button.	As Expected	Pass	Harini A/P Sangaran	-
5	Click on the 'Kemas Kini' button.	-	Update member's loan application status.	As Expected	Pass	Harini A/P Sangaran	-

Table 14: TC002_05: Test <Validate Loan Application Form (SD011)>

DESCRIPTION	VERSION	TEST DATE
Approve Membership Termination Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admins are able to view membership termination form and update the membership status.	Nuraisyah binti Mohd Zikre	Harini A/P Sangaran

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Senarai Permohonan Berhenti Ahli' button.	-	Display list of pending membership termination forms.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Click on the 'Dokumen Permohonan Berhenti Keahlian' hyperlink.	-	Display applicant's membership termination form.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
3	Click on the drop down button in the 'Status Permohonan' column.	-	Display the current application status by default and display all the possible statuses when clicked on the drop down button.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
4	Click on the 'Kemas Kini' button.	-	Update applicant's application status.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-

Table 15: TC002_07: Test <Approve Membership Termination Form (SD012)>

DESCRIPTION	VERSION	TEST DATE
View Monthly Report	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the monthly report displays the correct data for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications in a graphical format for a selected month.	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Laporan Koperasi' button.	-	Display overall report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Choose a specific month and year from the dropdown or calendar.	Selected month & year: January 2024	Graph diisplays report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications for the selected month.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 16: TC002_08: Test <View Monthly Report (SD013)>

DESCRIPTION	VERSION	TEST DATE
View Annual Report	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the annual report displays the correct data for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications in a graphical format for a selected year.	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Laporan Koperasi' button.	-	Display overall report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-
2	Choose a specific year from the dropdown or calendar.	Selected year: 2024	Graph displays report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications for the selected year.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-

Table 17: TC002_09: Test <View Annual Report (SD014)>

DESCRIPTION	VERSION	TEST DATE
Add New Admin	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admin can add a new admin to the system.	Nuraisyah binti Mohd Zikre	Harini A/P Sangaran

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Tambah Admin' button.	-	Display a column to enter an identity card number for the new admin.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Enter the Identity Card (IC) number of the new admin and click on 'Tambah Admin' button.	IC number : 04120107 0330	Displays appropriate message for successful admission of the new admin.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-

Table 18: TC002_10: Test <Add New Admin (SD015)>

DESCRIPTION	VERSION	TEST DATE
Add New ALK	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the admin can add a new ALK to the system.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Tambah ALK' button.	-	Display a column to enter an identity card number for the new ALK.	As Expected	Pass	Harini A/P Sangaran	-
2	Enter the Identity Card (IC) number of the new ALK and click on 'Tambah ALK' button.	IC number : 04120107 0330	Displays appropriate message for successful admission of the new ALK.	As Expected	Pass	Harini A/P Sangaran	-

Table 19: TC002_11: Test <Add New ALK (SD016)>

TC003: Test ALK Subsystem: <User Story 3 (US003)>

This test contains the following test cases:

- (a) TC003_01: Test <View ALK Menu (SD026)>
- (b) TC003_02: Test <View Validated Member Application Form (SD017)>
- (c) TC003_03: Test <View Validated Loan Application Form (SD018)>
- (d) TC003_04: Test <Approve Member Application (SD019)>
- (e) TC003_05: Test <Approve Loan Application (SD020)>
- (f) TC002_06: Test <View Monthly Report (SD021)>
- (g) TC002_07: Test <View Annual Report (SD022)>

DESCRIPTION	VERSION	TEST DATE
View ALK Menu	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the ALKs are able to view the appropriate homepage once they have successfully logged in.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	ALK ID: 0001 Password : 1234	Credential can be entered.	As Expected	Pass	Harini A/P Sangaran	-
2	View ALK homepage.	-	Display the features for ALK to do in the system.	As Expected	Pass	Harini A/P Sangaran	-

Table 20: TC003_01: Test <View ALK Menu (SD026)>

DESCRIPTION	VERSION	TEST DATE
View Validated Member Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the ALKs are able to view the list of reviewed member application forms.	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View ALK homepage.	-	Display the features for ALK to do in the system.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-
2	Click on the 'Pengesahan Permohonan Ahli' button.	-	Display list of reviewed member application forms.	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-

Table 21: TC003_02: Test <View Validated Member Application Form (SD017)>

DESCRIPTION	VERSION	TEST DATE
View Validated Loan Application Form	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the ALKs are able to view the list of reviewed loan application forms.	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View ALK homepage.	-	Display the features for ALK to do in the system.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Click on the 'Pengesahani Permohonan Pinjaman' button.	-	Display list of reviewed loan application forms.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 22: TC003_03: Test <View Validated Loan Application Form (SD018)>

DESCRIPTION	VERSION	TEST DATE
Approve Member Application	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the ALKs are able to view applicant's verification documents and update the application status.	Nuraisyah binti Mohd Zikre	Harini A/P Sangaran

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Pengesahan Permohonan Ahli' button.	-	Display list of pending member application forms.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Click on the 'Dokumen Permohonan Ahli' hyperlink.	-	Display applicant's application form.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
3	Click on the drop down button in the 'Status Permohonan' column.	-	Display the current application status by default and display all the possible	As Expected	Pass	Damiya Aina binti	-

			statuses when clicked on the drop down button.			Basir Abd Shammad	
4	Click on the 'Kemas Kini' button.	-	Update applicant's application status.	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-

Table 23: TC003_04: Test <Approve Member Application (SD019)>

DESCRIPTION	VERSION	TEST DATE
Approve Loan Application	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the ALKs are able to view member's verification documents and update the loan application status.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Pengesahan Permohonan Pinjaman' button.	-	Display list of reviewed loan application forms.	As Expected	Pass	Harini A/P Sangaran	-
2	Click on the 'Dokumen Permohonan Pinjaman' hyperlink.	-	Display member's loan application form.	As Expected	Pass	Harini A/P Sangaran	-
3	Click on the 'Muat Turun Dokumen' button.	-	Download member's verification documents for the loan application	As Expected	Pass	Harini A/P Sangaran	-
4	Click on the drop down button in the	-	Display the current application status by default and display	As Expected	Pass	Harini A/P Sangaran	-

	'Status Permohonan' column.		all the possible statuses when clicked on the drop down button.				
5	Click on the 'Kemas Kini' button.	-	Update member's loan application status.	As Expected	Pass	Harini A/P Sangaran	-

Table 24: TC003_05: Test <Approve Loan Application (SD020)>

DESCRIPTION	VERSION	TEST DATE
View Monthly Report	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the monthly report displays the correct data for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications in a graphical format for a selected month.	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Laporan Tahunan' button.	-	Display overall report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Choose a specific month and year from the dropdown or calendar.	Selected month & year: January 2024	Graph displays report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications for the selected month.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 25: TC003_06: Test <View Monthly Report (SD021)>

DESCRIPTION	VERSION	TEST DATE
View Annual Report	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the annual report displays the correct data for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications in a graphical format for a selected year.	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Click on the 'Laporan Tahunan' button.	-	Display overall report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Choose a specific year from the dropdown or calendar.	Selected year: 2024	Graph displays report for <i>syer majikan</i> , <i>syer pekerja</i> , total membership applications and total loan applications for the selected year.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 26: TC003_07: Test <View Annual Report (SD034)>

TC004: Test Non-Member Subsystem: <User Story 4 (US004)>

This test contains the following test cases:

- (a) TC004_01: Test <Apply Membership (SD023)>
- (b) TC004_02: Test <View Homepage (SD027)>
- (c) TC004_03: Test <Get Member Approval (SD028)>

DESCRIPTION	VERSION	TEST DATE
Apply Membership	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
allows non-members to apply for membership by providing all necessary information and documents and that the application is successfully recorded in the system for staff and ALK review.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Register into the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Log in back to the Koperasi KADA system with a valid id and password	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
3	Click on the “Daftar Sekarang” to apply for the membership	-	Successfully load the membership form section, displaying all the membership application form	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
4	Fill in the membership	nama: Ali bin Abu	Form is completed	As Expected	Pass	Damiya Aina binti	

	application form fields (e.g: nama, no. k/p, etc)	no.k/p: 76041303 7761				Basir Abd Shammad	
5	Click the "Simpan Maklumat" button after completing the form.	-	successfully submit the membership application form	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	

Table 27: TC004_01: Test <Apply Membership (SD023)>

DESCRIPTION	VERSION	TEST DATE
View Homepage	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
Ensure that the homepage is accessible to all users before registration or login and displays relevant information along with clear options for registration and login without exposing any member-specific details.	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View homepage	-	Display the information of the Koperasi KADA	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-

Table 28: TC004_02: Test <View Homepage (SD027)>

DESCRIPTION	VERSION	TEST DATE
Get Member Approval	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To ensure that the system correctly retrieves and displays the membership application status for a non-member. If the application is approved, the system should provide login credentials. If the application is pending, rejected, or under review, the system should display an appropriate status message.	Harini A/P Sangaran	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View homepage	-	Display the information of the Koperasi KADA	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	-
2	Click on "Semak Permohonan" button	-	Membership status page is loaded	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
3	Enter IC number into the form	IC num: 770605030402	System accepts the IC number input	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
4	Click "Semak Akaun"	-	System processes the request	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
5	If approved, display credentials	id no: 0001 password: 1234	ID and password are displayed	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	

6	If pending, display "Pending" message	IC num: 770605030402(pending)	"Permohonan anda sedang diproses" message is displayed	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
7	If rejected, display "Rejected" message	IC num: 770605030402(rejected)	"Permohonan anda ditolak" message is displayed	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	
8	If in reviewed, display "in reviewed" message	IC num: 770605030402(in reviewed)	"Permohonan anda sedang dinilai" message is displayed	As Expected	Pass	Damiya Aina binti Basir Abd Shammad	

Table 29: TC004_03: Test <Get Member Approval (SD028)

DESCRIPTION	VERSION	TEST DATE
Log in	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the users can log in to the system by entering valid identity number and password.	Nurul Ika Syafiny binti Azhar	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password for any user.	Chosen user : admin Admin ID: 0001 Password : 1234	Credentials can be entered.	As Expected	Pass	Harini A/P Sangaran	-
2	View admin homepage.	-	Display the features for admin to do in the system	As Expected	Pass	Harini A/P Sangaran	-

Table 29: TC005_01: Test <Log in (SD029)>

DESCRIPTION	VERSION	TEST DATE
Log out	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the users can log in to the system by entering valid identity number and password.	Harini A/P Sangaran	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	View homepage.	-	Display the features for user to do in the system.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Click on 'Daftar Keluar' button.	-	Log out of the system and display general homepage.	As Expected	Pass	Nuraisyah binti Mohd Zikre	-

Table 30: TC005_02: Test <Log out (SD030)>

TC005: Test Loan Subsystem: <User Story 6 (US006)>

This test contains the following test cases:

- (a) TC006_01: Test <Apply Loan (SD001)>
- (b) TC006_02: Test <Check Loan Approval (SD002)>
- (c) TC006_03: Test <Find Transaction History (SD004)>
- (d) TC006_04: Test <Check Loan Balance (SD005)>

DESCRIPTION	VERSION	TEST DATE
Apply Loan	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
to verify that a member can successfully apply for a loan after logging in	Damiya Aina binti Basir Abd Shammad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Harini A/P Sangaran	-
2	Navigate to the Mohon Pinjaman section from the Member homepage.	-	Successfully load the loan form section, displaying all the loan application form	As Expected	Pass	Harini A/P Sangaran	
3	Fill in the loan application form fields (e.g: tempoh bayaran balik, jumlah pinjaman, etc)	Tempoh Bayaran Balik: 36 bulan Jumlah Pinjaman: RM10000	Form is complete	As Expected	Pass	Harini A/P Sangaran	

4	Upload all required documents (e.g., salary slips, IC copies, etc).	Salary Slip: uploaded IC copies: uploaded	Successfully upload the file required and can submit the form	As Expected	Pass	Harini A/P Sangaran	
5	Click the "Submit" button after completing the form.	-	successfully submit the loan application form	As Expected	Pass	Harini A/P Sangaran	

Table 31: TC006_01: Test <Apply Loan (SD001)>

DESCRIPTION	VERSION	TEST DATE
Check Loan Approval	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
to verify that loan application status is displayed correctly	Harini A/P Sangaran	Nuraisyah binti Mohd Zikre

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Nuraisyah binti Mohd Zikre	-
2	Navigate to the Semak Kelulusan Pinjaman section from the Member homepage.	-	Successfully load the loan application status section, displaying all the information of the member with the loan status	As Expected	Pass	Nuraisyah binti Mohd Zikre	
3	Check the status of loan apply	-	Loan Status is appeared successfully	As Expected	Pass	Nuraisyah binti Mohd Zikre	

Table 32: TC006_02: Test <Check Loan Approval (SD002)>

DESCRIPTION	VERSION	TEST DATE
Find Transaction History	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
Ensure that members can access a detailed, accurate transaction history with relevant filters to provide clarity on past transactions	Harini A/P Sangaran	Damiya Aina binti Basir Abd Shammad

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Nurul Ika Syafiny binti Azhar	-
2	Navigate to the Semak Penyata section from the Member homepage.	-	Successfully load the member invoice section, displaying all the information of the member financial history	As Expected	Pass	Nurul Ika Syafiny binti Azhar	
3	view the table of member's finance	-	table of member's finance is successfully appear	As Expected	Pass	Nurul Ika Syafiny binti Azhar	

Table 33: TC006_03: Test <Find Transaction History (SD004)>

DESCRIPTION	VERSION	TEST DATE
Check Loan Balance	1.0	23-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
Ensure that the system accurately displays the member's current loan balance, repayment history, and any outstanding amounts to help members keep track of their loan obligations.	Damiya Aina binti Basir Abd Shamad	Nurul Ika Syafiny binti Azhar

Test Case ID (TC001_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS /FAIL	Tested By	Tester Comments
1	Log in to the Koperasi KADA system with a valid id and password.	Admin ID: 0001 Password : 1234	Credential can be entered	As Expected	Pass	Harini A/P Sangaran	-
2	Navigate to the Baki Pinjaman section from the Member homepage.	-	Successfully load the loan balance section, displaying the loan balance amount and also the repayment history	As Expected	Pass	Harini A/P Sangaran	
3	view the table of loan repayment history and also the loan balance amount	-	table of repayment history and loan balance amount is successfully appear	As Expected	Pass	Harini A/P Sangaranr	

Table 34: TC006_04: Test <Check Loan Balance (SD005)>

11.2 Beta Testing

11.2.0 Methods Class

Entity Name	NonMember
Method Name	applyForMembership
Input	Non-member details (name, ID number, email, marital status, gender, religion, ethnicity, home address, postcode, state, membership number, office details, phone contacts, salary, and other financial details)
Output	Membership application submitted and stored in the database
Algorithm	1. Start 2. Non-member accesses the membership application form page via website. 3. Non-member fills in the required information 4. System validates input fields 5. If the information is valid, the form is submitted 6. Else, an error message is displayed, asking for corrections 7. System stores applications in the database with 'Pending' status 8. Admin reviews the application and either approves or rejects it 9. If approved, applicants can check for their login details via the website. 10. If rejected, the system notifies the applicant when they check their application via the website 11. End

Table 35: Method Class of Beta Testing: Membership Application

11.2.1 Flow Graph

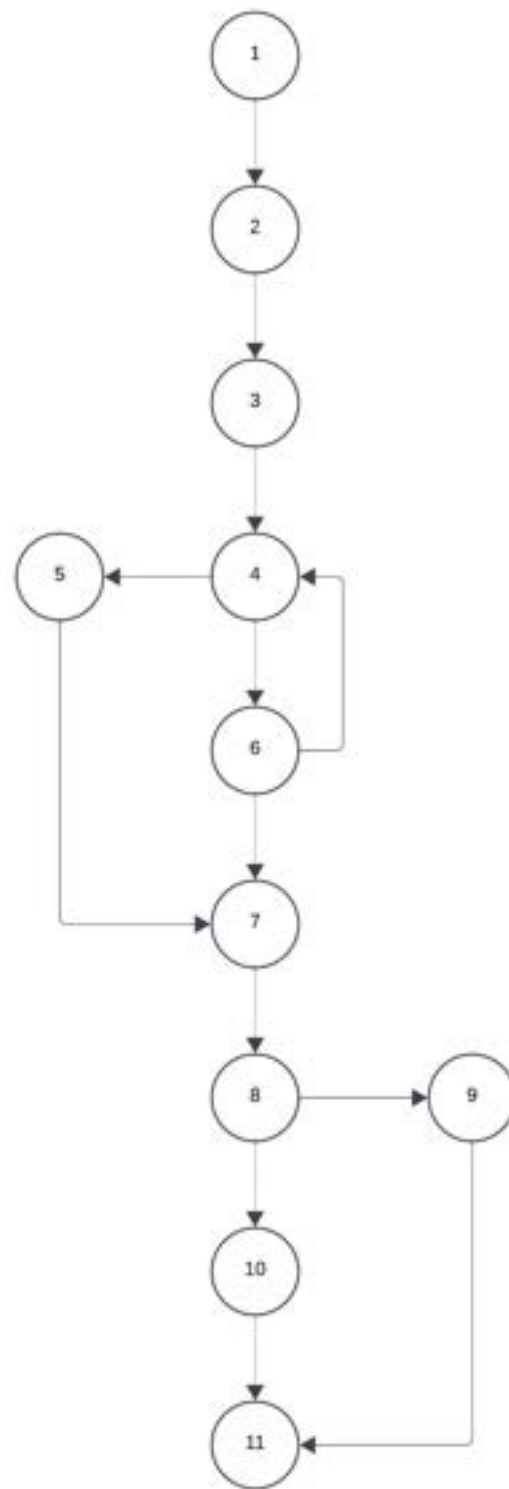


Figure 77: Flow Chart of Beta Testing: Membership Application

11.2.2 Cyclomatic Complexity Calculation

Using different formulas:

1. Formula 1: $V(G) = E - N + 2$
 - Edges (E) = 13, Nodes (N) = 11
 - $V(G) = 13 - 11 + 2 = 3$
2. Formula 2: $V(G) = P + 1$
 - Decision points (P) = 2
 - $V(G) = 2 + 1 = 3$
3. Formula 3: Number of regions in the flow graph
 - $V(G) = 3$

Thus, the Cyclomatic Complexity is 3, meaning there are three independent paths in the program.

11.2.3 Test Cases

Test Case No.	Independent Path	Test Data	Expected Result
TC001	1 → 2 → 3 → 4 → 5 → 7 → 8 → 9 → 11	Non-member fills the form correctly and submits	Membership application stored in the database and approved
TC002	1 → 2 → 3 → 4 → 6 → 4 (loop) → 5 → 7 → 8 → 10 → 11	Non-member enters invalid input and submits	System rejects form and prompts for corrections before resubmission
TC003	1 → 2 → 3 → 4 → 5 → 7 → 8 → 10 → 11	Form submission disapproved by ALK or Admin due to missing fields or invalid information.	System displays fail application when applicator check their applications

Table 36: Test Cases of Beta Testing: Membership Application

12.0 UAT Reports

Date: 23/1/2025

Tested by: En. Ahmad Rohailan bin Hani, representative of Koperasi KADA Sdn. Bhd.

Feedback:

- System process is accurate to the current existing business process.
- Need to add Member Termination Application as well into the system.
- Include the report for the quantity of inactive and active members throughout the month.

13.0 Gantt Chart

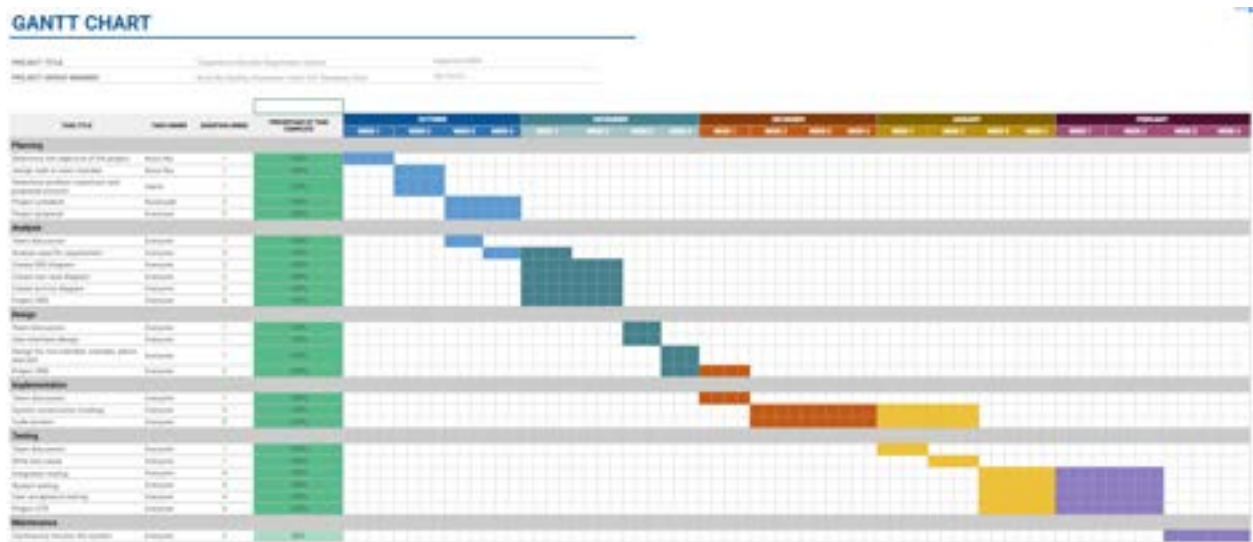


Figure 78: Gantt Chart

Reference Link:

https://docs.google.com/spreadsheets/d/1MncYaFx7WbLbXWMqy9_3xP6GLY3oKiqbybpXH_Mi-r4/edit?usp=sharing

User Manual



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1

2024/2025

SECP3723 – SYSTEM DEVELOPMENT TECHNOLOGY

SECTION 02

TITLE: USER MANUAL

LECTURER: DR. AHMAD NAJMI BIN AMERHAIDER NUAR

NAME	MATRIC NUMBER
NURUL IKA SYAFINY BINTI AZHAR	A23CS0164
NURAI SYAH BINTI MOHD ZIKRE	A23CS0160
HARINI A/P SANGARAN	A23CS0081
DAMIYA AINA BINTI BASIR ABD SHAMMAD	A23CS0220

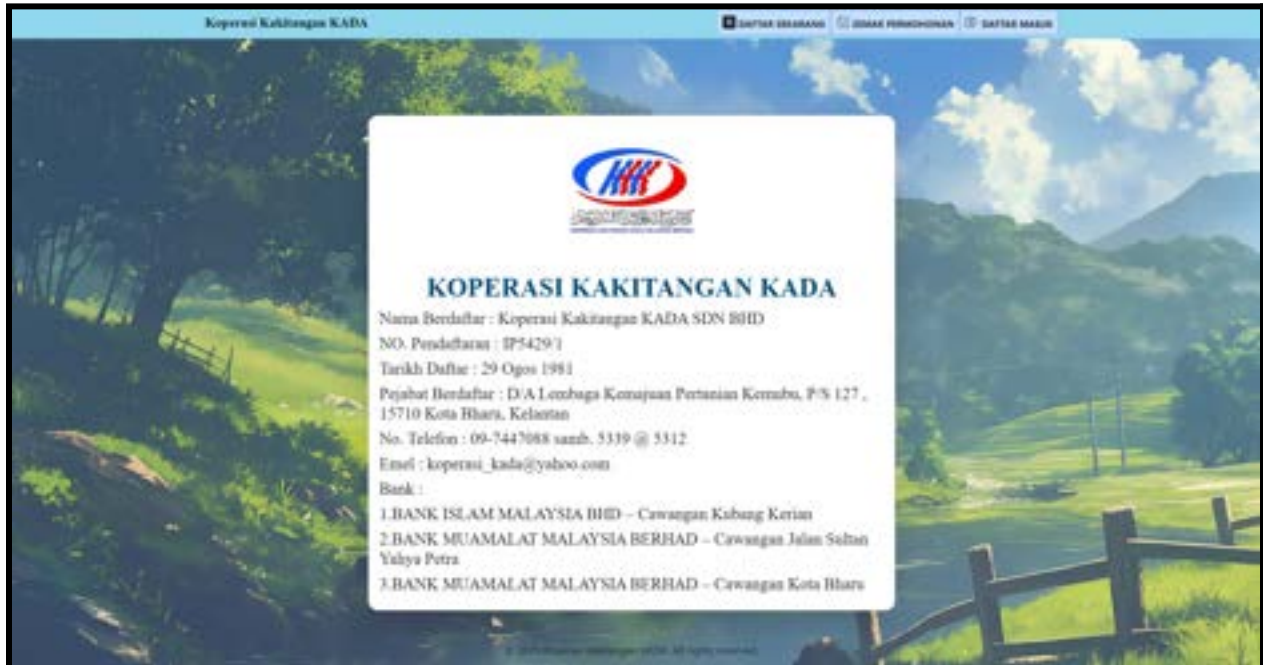
Table of Content

1.0 All users	3
1.1 Non-member Homepage	3
1.2 Apply for membership	4
1.3 Check membership application status	5
1.4 Reset Password	6
1.5 Successful Reset Password	6
1.6 Unsuccessful Reset Password	7
2.0 Member	8
2.1 Common Login Page	8
2.2 Member Login Page	9
2.3 Member Menu Page	10
2.4 Navigation Bar	11
2.5 View Personal Information	12
2.6 Editing Personal Information	13
2.7 Apply for a Loan	14
2.8 Check Loan Application Status	15
2.9 Check Member Savings	17
2.10 Check Loan Balance	18
2.11 Check Invoice/Transaction History	19
2.12 Apply for Membership Termination	20
3.0 Admin User	21
3.1 Admin Login Page	21
3.2 View Loan Application and send it to ALK for approval	22
3.3 View Member Application and send it to ALK for approval	24
3.4 View Member Termination Application and approve it	26
3.5 View System Report for Admin	28

3.6 Add New Admin and Ahli Lembaga Koperasi (ALK) to the system	30
3.7 Log Out	31
4.0 ALK User	32
4.1 ALK Log In Page	32
4.2 ALK Homepage	33
4.3 Approve Membership Application	34
4.4 Successful Membership Application Status Update	34
4.5 Navigation Bar	36
4.6 Approve Loan Application	37
4.7 Successful Loan Application List	37
4.8 Annual and Monthly Report	38
4.9 Log Out ALK System	39

1.0 All users

1.1 Non-member Homepage



1. Non-member of Koperasi KADA will be greet with this homepage as they do not have full access to log in into the system
2. Non-member can only apply for membership and check the status of membership application applied provided at the navigation bar of the homepage

1.2 Apply for membership

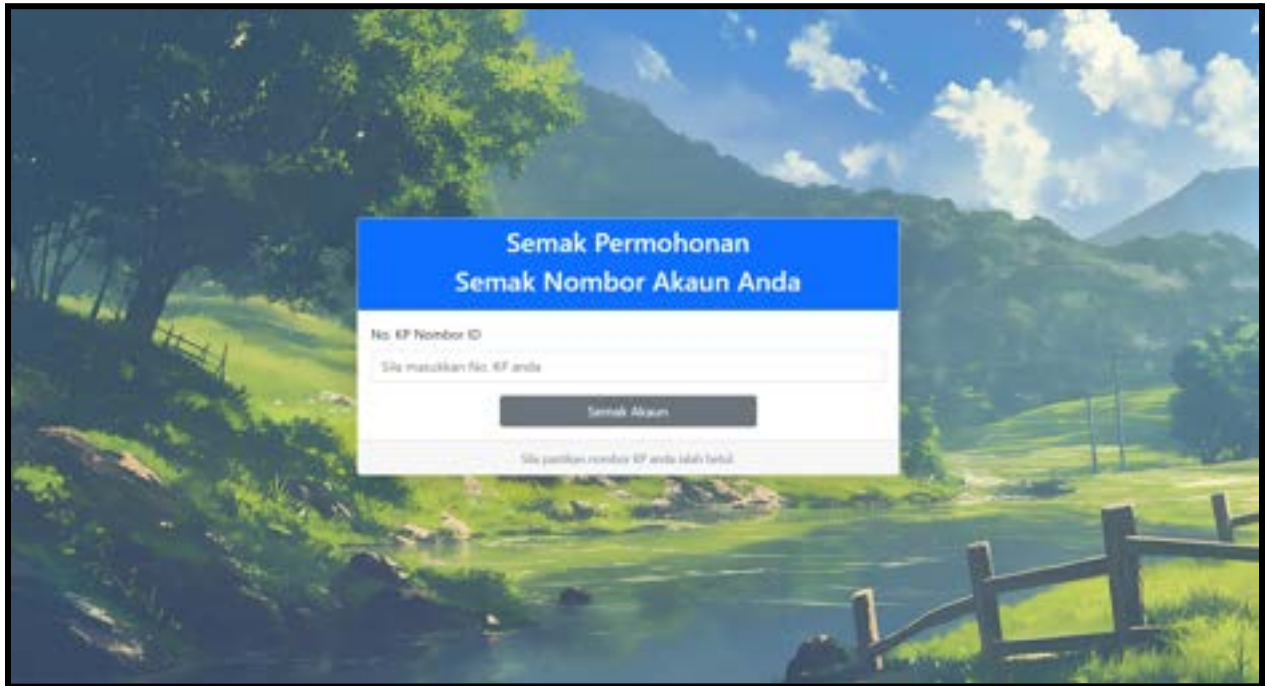
The screenshot displays a web form titled "Permohonan Menjadi Anggota" (Membership Application). The form is organized into several sections with labels in Indonesian:

- Personal Information:** Includes fields for Name, Gender, Date of Birth, and Address.
- Registration Information:** Includes fields for Registration Number, Registration Date, and a signature field.
- Contract Details:** Includes fields for Contract Type, Contract Duration, and Contract Amount.
- Membership Fee:** Includes fields for Membership Fee, Fee Type, and Fee Amount.
- Payment Information:** Includes fields for Payment Method, Payment Date, and Payment Amount.

At the bottom of the form, there is a blue button labeled "Daftar Sekarang" (Register Now).

1. By clicking on “Daftar Sekarang” the membership application will appear as shown below.
2. Non-member needs to fill in the blank box as required before submit the form.
3. After successfully submitting the membership application form, non-member need to wait for 30 days for the application to be approved.

1.3 Check membership application status

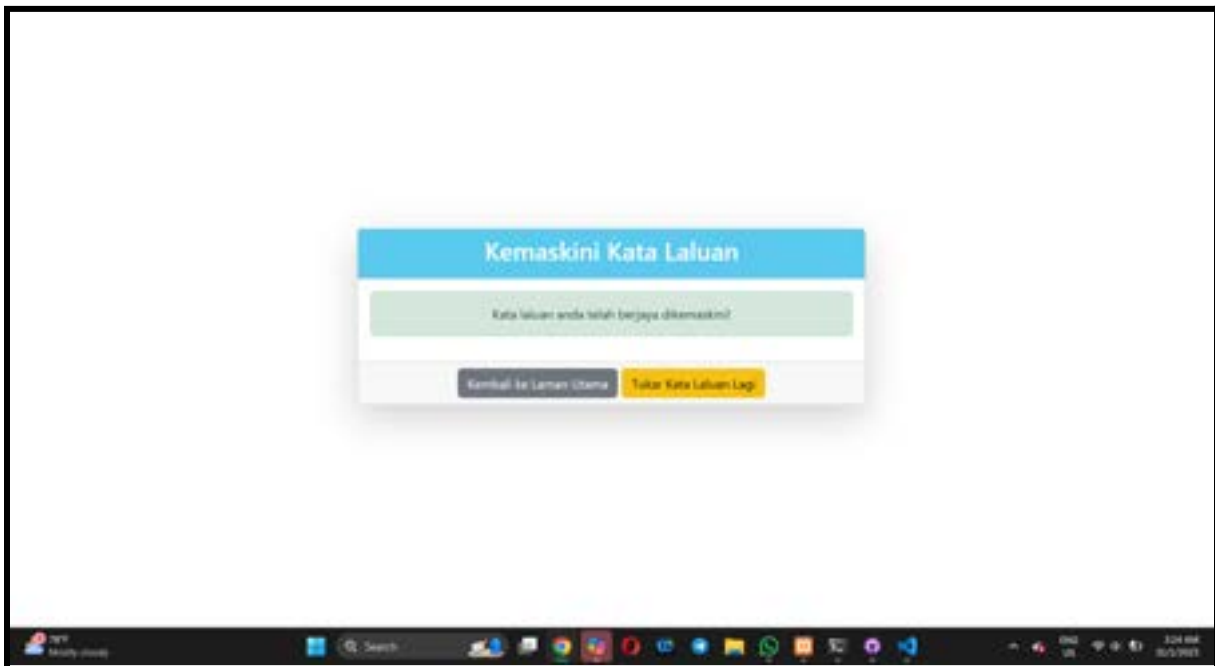


1. By clicking on “Semak Permohonan” non-member can check the status of membership application by filling in the IC number to search for the status of application.
2. If the IC number is in the database, the status of application will appear where it will show either the application is approved, reviewed or disapproved.
3. If the application status is approved by Koperasi KADA, the details of the member account will be displayed

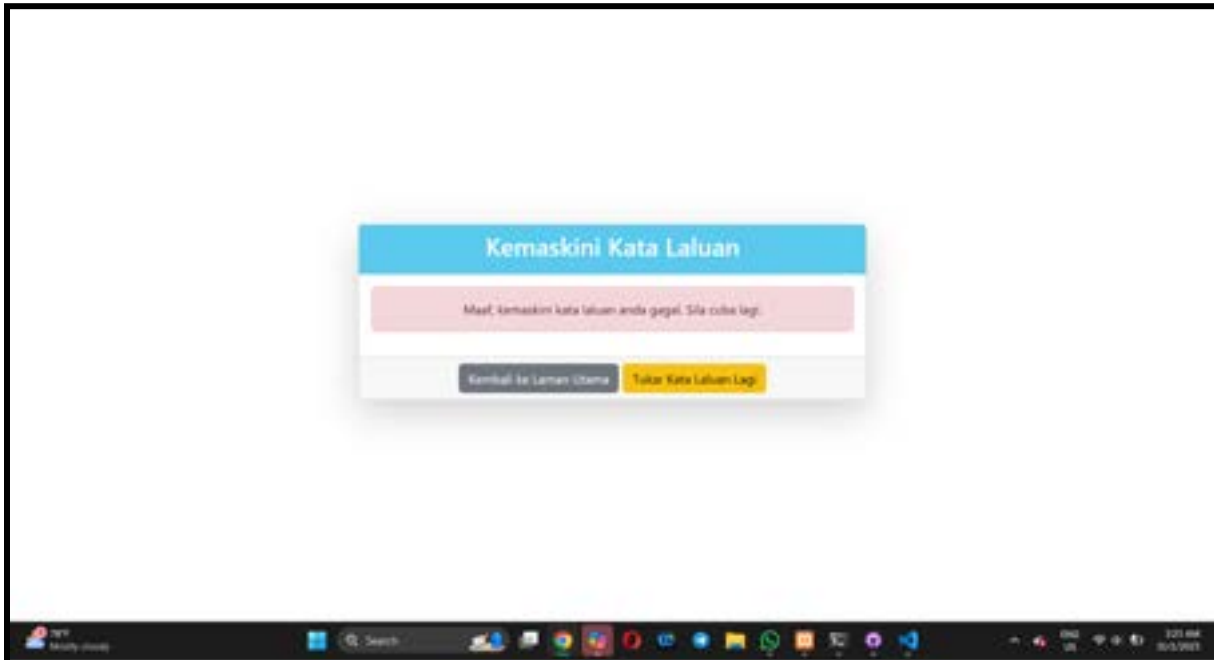
1.4 Reset Password



1.5 Successful Reset Password



1.6 Unsuccessful Reset Password

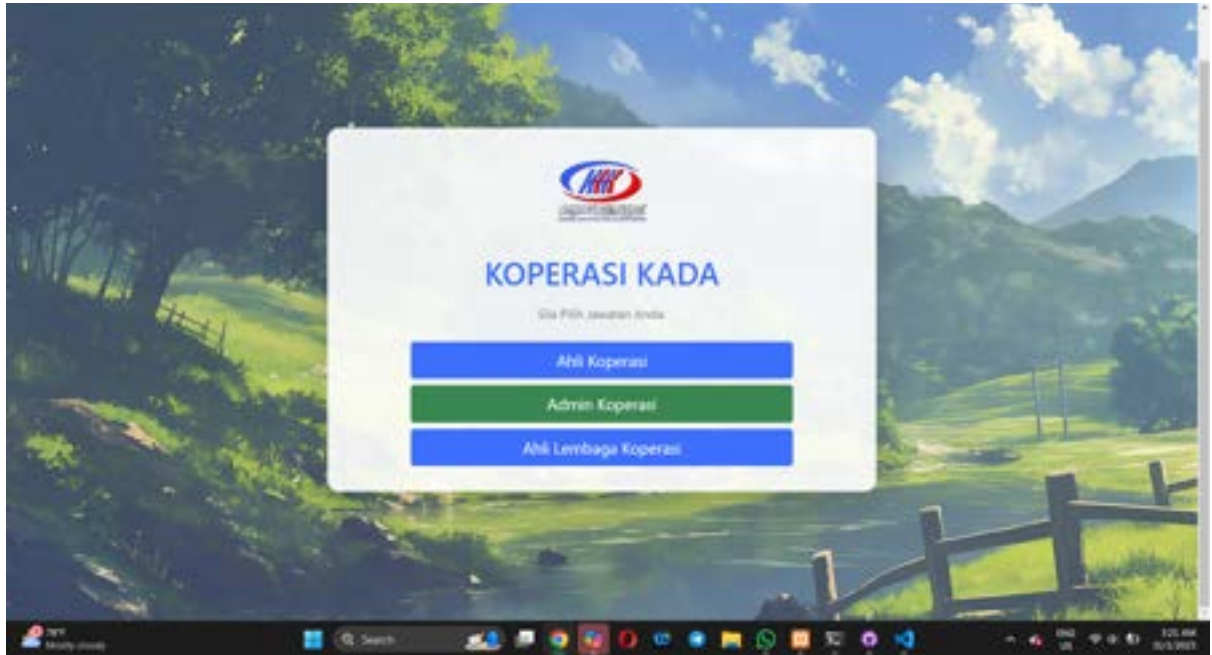


Resetting Password:

1. If Members, Admins and ALKs wish to change their password, they can click on the 'Tukar Kata Laluan' button to change their password
2. They will have to enter their identification card number and enter their new password
3. They have to click on the 'Simpan Maklumat' button to successfully update their password.
4. Appropriate messages will be displayed based on successful or unsuccessful password reset.

2.0 Member

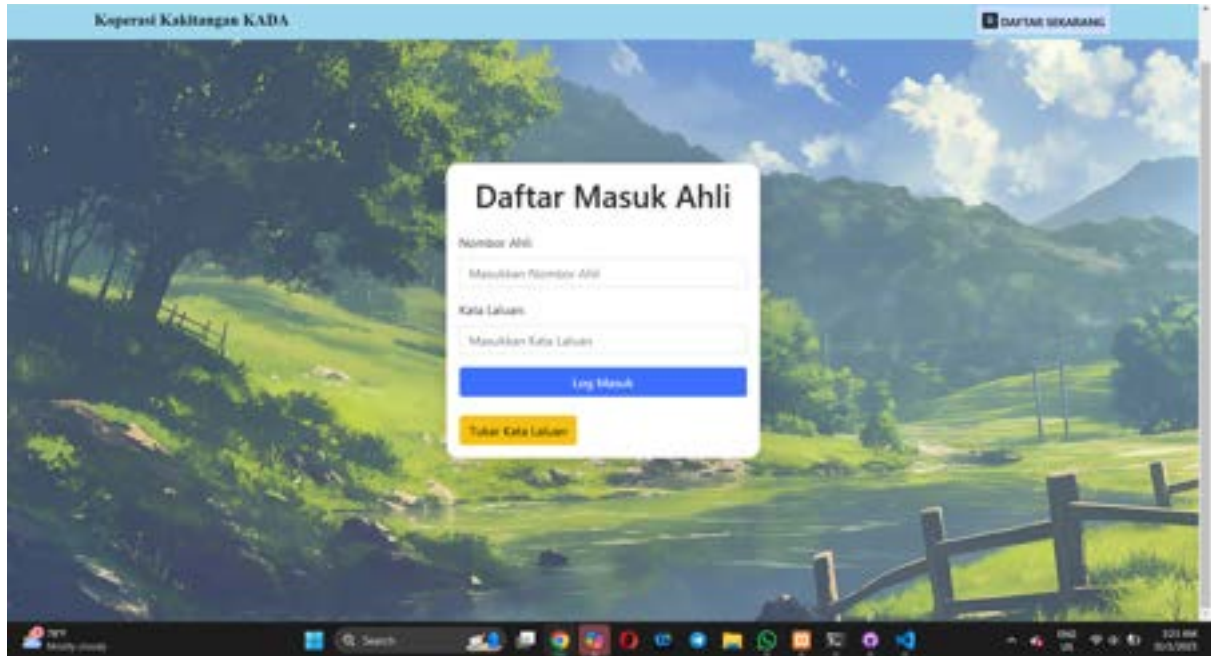
2.1 Common Login Page



Logging In:

1. This page will be displayed when members click on 'Daftar Masuk' button on the navigation bar of the homepage.
2. This page displays the three user roles of the system which are members as 'Ahli Koperasi', admins as 'Admin Koperasi' and ALK as 'Ahli Lembaga Koperasi'.
3. Members have to click on 'Ahli Koperasi' button to login.

2.2 Member Login Page



Logging In:

1. Once members have clicked on the 'Ahli Koperasi' button, this page will be displayed as the member login page.
2. Members have to enter their identification number and password and click on 'Log Masuk' to log in to the system.

2.3 Member Menu Page



Menu Page :

1. This is the menu page members will be seeing once they have successfully logged in.
2. There are six features for members to do which is to apply for a loan shown as 'Mohon Pinjaman' button, check loan application status shown as 'Semak Kelulusan Pinjaman' button, check savings shown as 'Semak Simpanan' button, check loan balance shown as 'Baki Pinjaman' button, check invoice shown as 'Semak Penyata' button and apply for membership termination shown as 'Mohon Berhenti Ahli' button.
3. Members can also view and update their personal information if they click on the user icon on the navigation bar.

2.4 Navigation Bar



Navigation bar features :

1. On the navigation bar, there is a dropdown button named 'Pilihan' which has all the features of the members to do in it. Members can also click on these to go to the allocated pages or features.
2. Then, there is a 'Daftar Keluar' button to log out of the system.
3. If the user icon on the navigation bar is clicked, the member's information will be displayed.

2.5 View Personal Information

Koperasi Kakitangan KADA Pilihan Daftar Koperasi

MAKLUMAT AHLI

Maklumat Pribadi

Nama	Husni A/P Sangari			Nombor K/P	04101070330
Email	husni.sangari@gmail.com				
Taraf Perkhidmatan	Jantina	Agama			
Begang	Pemangsaan	Hindu			
Rangsa	Alamat Rumah				
India	27, Lorong Impian Indah 2, Taman Impian Indah				
Poskod	Negeri	Bandar			
14000	Pulau Pinang	Bukit Mertajam			

Maklumat Pejabat

No. Anggota	No. PF
0001	5678
Jawatan & Grade	Alamat Pejabat
A	MNO KTD

Koperasi Kakitangan KADA Pilihan Daftar Koperasi

Gaji Bulanan

Gaji Bulanan
1000.00

Purata & Sumbangan

Perkara	RM
Purata Masuki	11.00
Modal Syer	0.00
Modal Yuran	0.00
Wang Deposit Anggota	0.00
Sumbangan Tabung Kelengkapan (AI Aliran)	0.00
Sumbangan Tabung	0.00
Lain-lain	0.00

Kemaskini Maklumat

Viewing personal information :

1. All the personal details of the member such as name, email, pf number, etc. will be displayed when members click on the user icon on the navigation bar to view their information.

2.6 Editing Personal Information

KEMASKINI MAKLUMAT PEMOHON

Maklumat Peribadi

Nama: Nombor K/P:

Email:

Tarikh Perkahwinan: Jantina: Agama:

Agama: Alamat Rumah:

Produk: Segen: Bendera:

Maklumat Pejabat

Nis. Anggot: Nis. PR:

Jawatan & Grad: Alamat Pejabat:

Kemaskini Maklumat

No. Kad Pengiraan:

Gaji Bulanan:

Yuran & Sumbangan

Perkara	RM
Yuran Masuk	0.00
Modul 1 yer	0.00
Modul 2 yer	0.00
Modul 3 yer	0.00
Wang Deposit Anggot	0.00
Sumbangan Tabung Kelengkapan (M-Rawat)	0.00
Sumbangan Tabung	0.00
Lain-lain	0.00

Editing personal information :

1. If members click on the 'Kemaskini Maklumat' button on the personal details profile, the system will redirect to these interfaces.
2. The existing information about the member is shown and members can make changes to the information.

- ## 2.7 Apply for a Loan

[illegible]

Applying for loans :

1. If members click on the 'Mohon Pinjaman' button, the system will redirect to these interfaces.
2. The system will display a table with monthly installment value for each loan amount according to the repayment period.
3. Then, members can scroll down the page to view the loan application form and fill in the details accordingly.
4. Members can refer to the table given above for the monthly installment value column according to their preferred loan amount and repayment period.

2.8 Check Loan Application Status



Checking loan application status :

1. If members click on the 'Semak Kelulusan Pinjaman' button, the system will redirect to this interface.

2. The system will display loan information if the loan application status is approved or will display appropriate messages if the loan is pending, being reviewed or is disapproved.

2.9 Check Member Savings

The screenshot displays a web application interface for 'Koperasi Kakitangan KADA'. The main heading is 'SIMPANAN DAN SYER PERIBADI'. Below this, a yellow box contains member details: Name (Hafizi A/P Langgani), No. IC (041201070100), and No. KAD (0001). The interface is divided into two main sections: 'Simpanan Peribadi Ahli' and 'Syer Ahli'. The 'Simpanan Peribadi Ahli' section features a table with columns for 'Nombor akaun', 'Baki Simpanan (RM)', and 'Tarikh Kemasuk'. The 'Syer Ahli' section features a table with columns for 'Syer Majikan', 'Syer Pekerja', and 'Tarikh'. The background of the interface shows a sunset over a field.

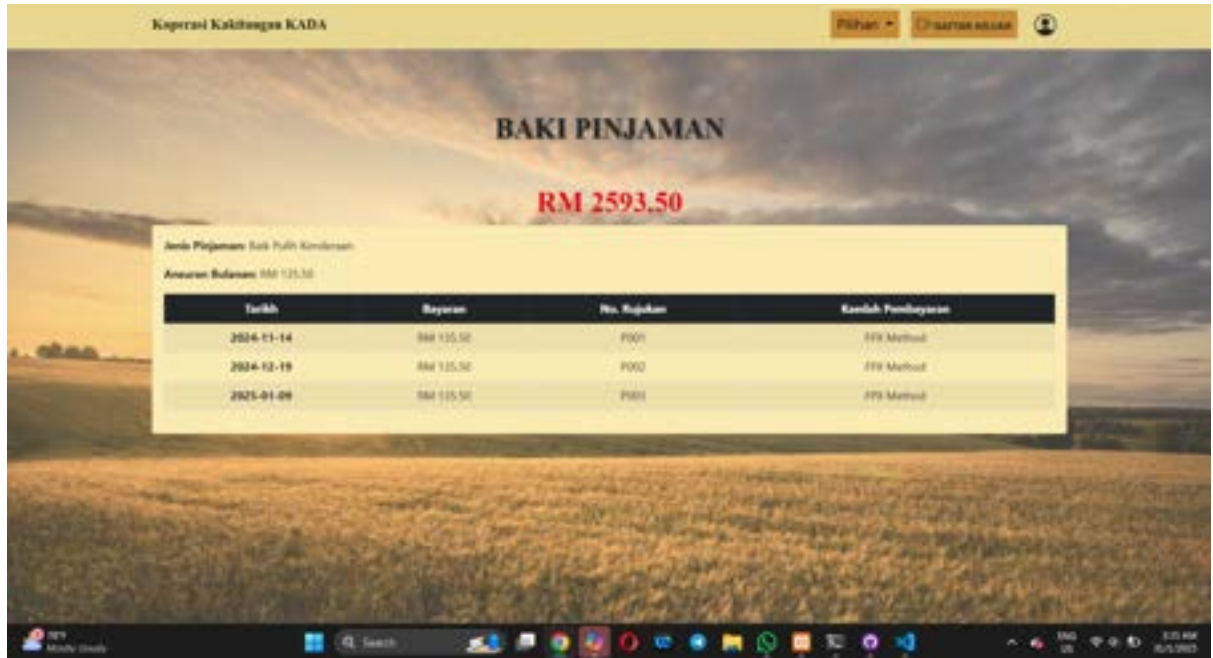
Nombor akaun	Baki Simpanan (RM)	Tarikh Kemasuk
0	100.00	2023-01-08 02:26:54
0	90.00	2024-11-20 02:27:32
0	50.00	2023-12-19 02:26:05
0	130.00	2023-01-14 02:26:47
0	230.00	2024-12-26 02:33:06

Syer Majikan	Syer Pekerja	Tarikh
70.00	55.00	2023-01-08 02:40:36
85.00	70.00	2024-11-20 02:30:16
60.00	40.00	2023-12-19 02:30:29
90.00	65.00	2023-01-14 02:30:40
180.00	110.00	2024-12-26 02:30:52

Checking member savings :

1. If the members click on the 'Semak Simpanan' button, the system will redirect to this interface.
2. The system will display the member's name, Identification Card (IC) number, and member ID at the top.
3. The system also displays the savings amount of the member in each account of the member after every transaction or update in a tabular form.
4. Beneath the personal savings table, the interface displays the employee and employer shares after every transaction in a tabular form.

2.10 Check Loan Balance



Checking loan balance :

1. If the members click on the 'Baki Pinjaman' button, the system will redirect to this interface.
2. The balance loan amount to be paid is displayed in bold and red.
3. Beneath it, the loan type and monthly installment amount is displayed.
4. Then, the previous transactions made to pay off the loan are displayed along with the date, amount, reference number and payment method in a tabular form.

2.11 Check Invoice/Transaction History



Transaksi	Tarikh	Jumlah (RM)	Catatan	Cara Bayaran
1	2024-11-14	125.00	P001	DPS Method
2	2024-12-14	125.00	P002	DPS Method
3	2025-01-08	125.50	P003	DPS Method

Checking invoice :

1. If the members click on the 'Semak Penyata' button, the system will redirect to this interface.
2. The system will display the member's name, Identification Card (IC) number, and member ID at the top.
3. The system also displays every transaction made along with the date, amount, reference and payment method in a tabular form.

2.12 Apply for Membership Termination

Koperasi Kakitangan KADA

Pilihan **Daftar Ahli**

Borang Permohonan Berhenti Keahlian

Nama No KP

Tarikh Lahir Jenis

Alamat Rumah

Poskod Bandar Negeri

No Akaun No PF

Alamat Pejabat

Poskod Pejabat Bandar Pejabat No Tel/Fax

No Tel Bimbit Jantina

Agama Mistaken (Jika Lain Islam)

Bangsa Mistaken (Jika Lain Melayu)

Sebab Berhenti

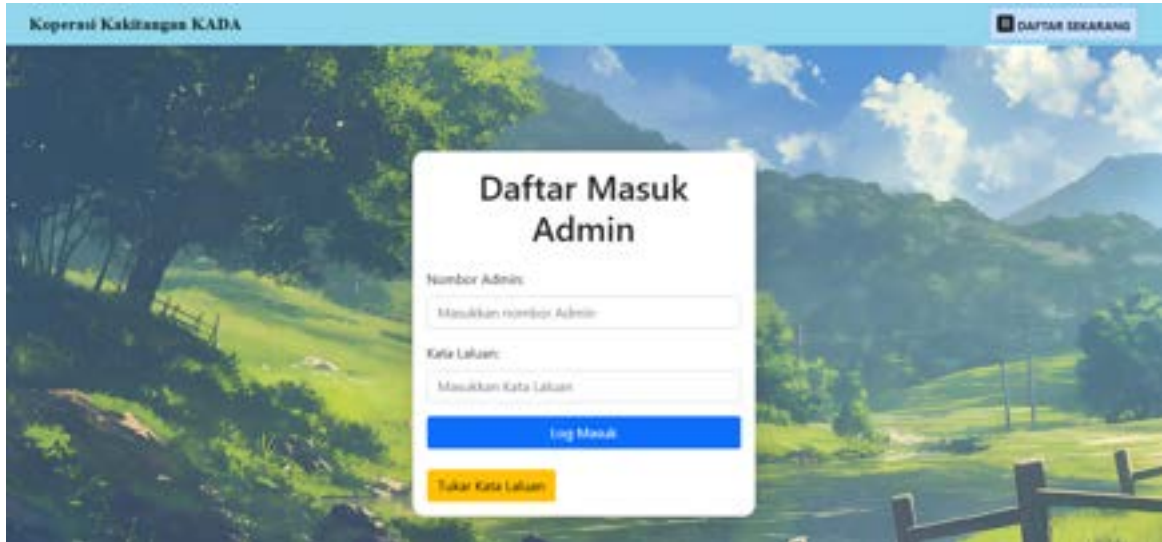
Hantar Permohonan

Applying for membership termination:

1. If the members click on the 'Mohon Berhenti Ahli' button, the system will redirect to this interface.
2. The system will display a membership termination form.
3. Members are required to fill in their details in the form and click on 'Hantar Permohonan' button to submit the form.

3.0 Admin User

3.1 Admin Login Page



Koperasi Kakitangan KADA

DAFTAR SEKARANG

Daftar Masuk Admin

Number Admin:

Kata Laman:

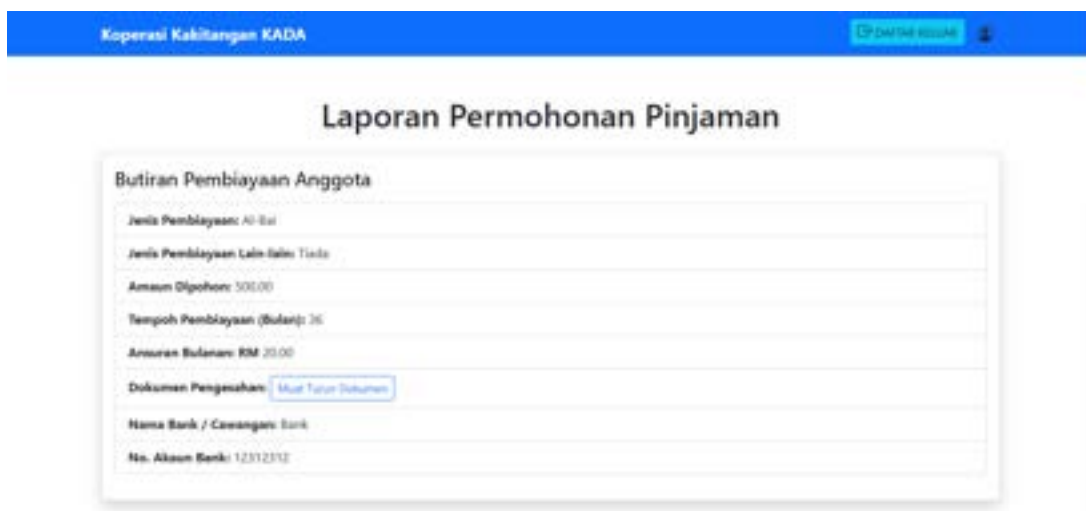
log Masuk

Tukar Kata Laman

Logging In:

4. Admin have to click on the 'Daftar Masuk' button on the homepage to login to the system
5. A button page with three user views will be displayed, which are 'Ahli Koperasi', 'Admin Koperasi' and 'Ahli Lembaga Koperasi'. Admins have to click on the 'Ahli Lembaga Koperasi' button
6. Admin key-in their Admin identification number and password to login to the system

3.2 View Loan Application and send it to ALK for approval



View Loan Form for each application:

1. From the admin homepage, choose 'Senarai Permohonan Pinjaman' to see the list of loan applications that are waiting to be checked before being sent to ALK.

2. Click the hyperlink 'Dokumen Permohonan Pinjaman' to view the complete form of loan application for each application.

Koperasi Kakitangan KADA DAFTAR KELUAR

Senarai Permohonan Pinjaman

ID Pinjaman	Dokumen Permohonan Pinjaman	Status Permohonan	Tindakan
10	Dokumen Permohonan Pinjaman	Belum selesai	Kemas Kini
12	Dokumen Permohonan Pinjaman	Belum selesai Telah Disemak Tidak Lulus	Kemas Kini

[Kembali Semula](#)

Approve Loan Application to next level or decline it:

1. From the loan application list page, press on the drop down button in the 'Status Permohonan' column and choose 'Telah Disemak' to approve the loan application or 'Tidak Lulus' to decline the application.
2. Press the green button with 'Kemas Kini' to update the new status of the application.

3.3 View Member Application and send it to ALK for approval



Koperasi Kakitangan KADA

DAFTAR KELUAR

Laporan Maklumat Pemohon

Maklumat Peribadi

Nama: AHMAD ALI

No. Kad Pengenalan: 102345

Email: ahmad04@gmail.com

Taraf Perkahwinan: Bujang

Jantina: Lelaki

Umur: 30

Agama: Islam

Bangsa: Melayu

Alamat Rumah: Somewhere

Bilik-bilik: 07070

View Member Application Form for each application:

1. From the admin homepage, choose 'Senarai Permohonan Ahli' to see the list of member applications that are waiting to be checked before being sent to ALK.
2. Click the hyperlink 'Dokumen Permohonan Ahli' to view the complete form of member application for each application.



Approve Member Application to next level or decline it:

1. From the member application list page, press on the drop down button in the 'Status Permohonan' column and choose 'Telah Disemak' to approve the loan application or 'Tidak Lulus' to decline the application.
2. Press the green button with 'Kemas Kini' to update the new status of the application.

3.4 View Member Termination Application and approve it



ID Pinjaman	Nama	No IC	Dokumen Permohonan Pinjaman	Status Permohonan	Tindakan
5	Rahman	0302992	Dokumen Permohonan Berhenti Ahli	Belum selesai	<button>Kemas Kini</button>
6	Rahman	929292	Dokumen Permohonan Berhenti Ahli	Belum selesai	<button>Kemas Kini</button>

Kembali Semula



Borang Permohonan Berhenti Keahlian

Nama: Rahman No KP: 0302992

Tarikh Lahir: 12/02/1988 Umur: 40

Alamat Rumah: Somewhere

No AHS: 0083 No PF: 2529

Jantina: Lelaki Agama: Islam

View Member Termination Application Form for each application:

1. From the admin homepage, choose 'Senarai Permohonan Berhenti Ahli' to see the list of member termination applications that are waiting to be checked before being sent to ALK.
2. Click the hyperlink 'Dokumen Permohonan Berhenti Ahli' to view the complete form of member termination application for each application.

Senarai Permohonan Berhenti Ahli

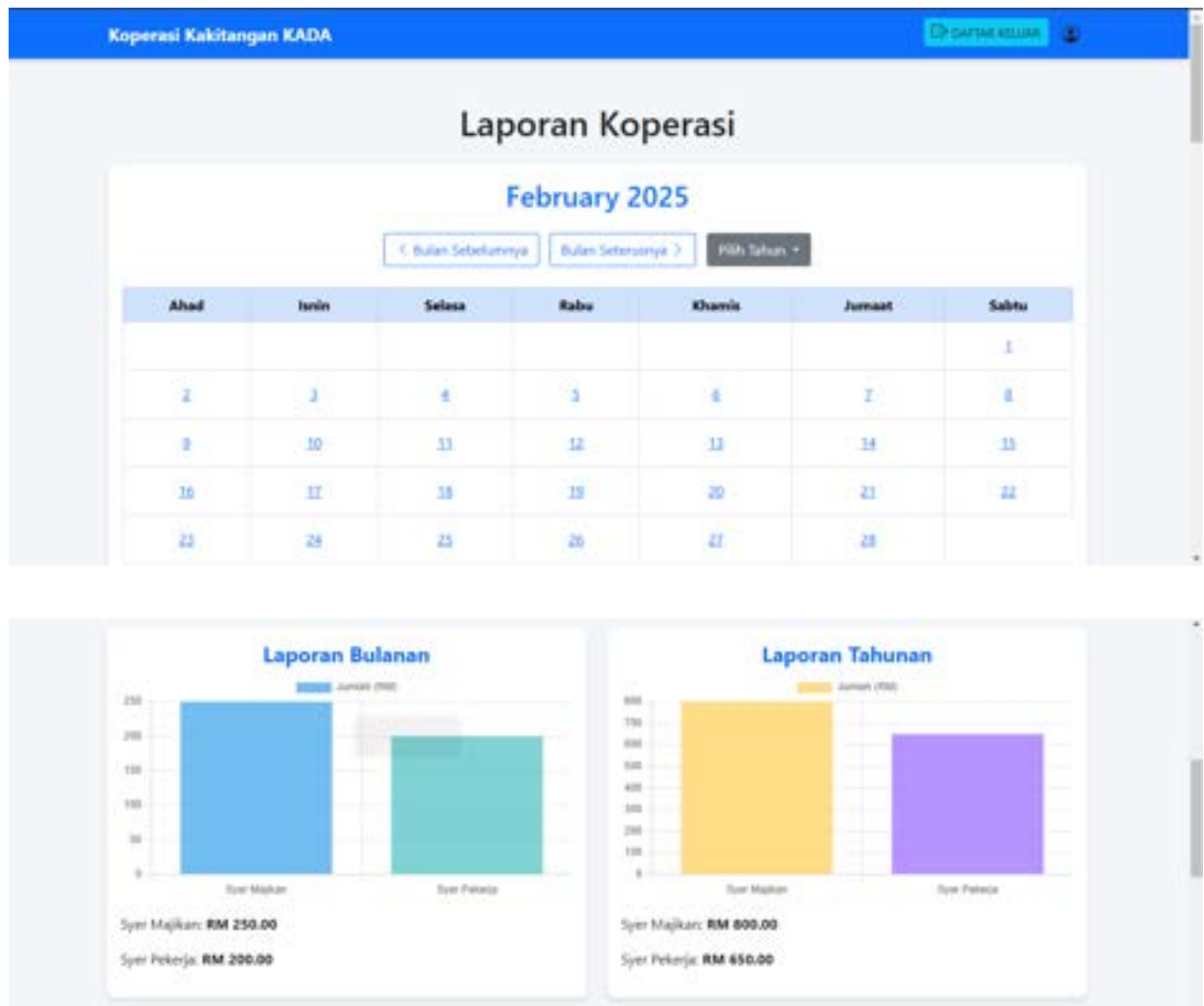
ID Pinjaman	Nama	No IC	Dokumen Permohonan Pinjaman	Status Permohonan	Tindakan
5	Rahman	0382992	Dokumen Permohonan Berhenti Ahli	Belum selesai	Kemas Kini
6	Rahman	929292	Dokumen Permohonan Berhenti Ahli	Belum selesai	Kemas Kini

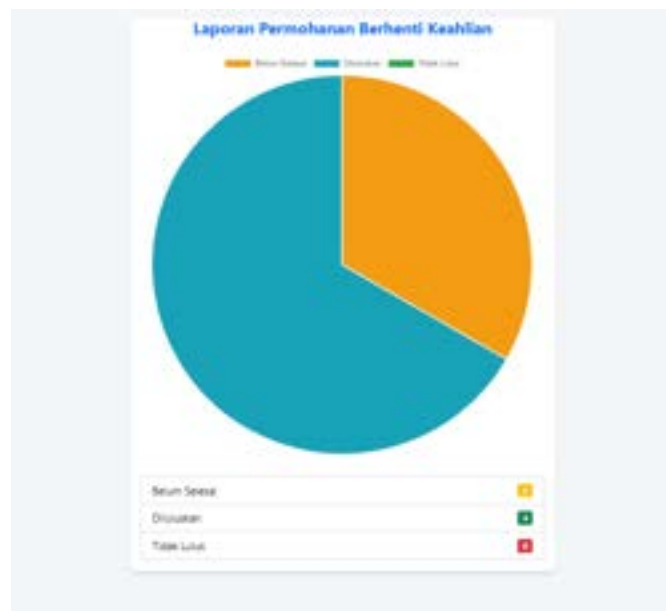
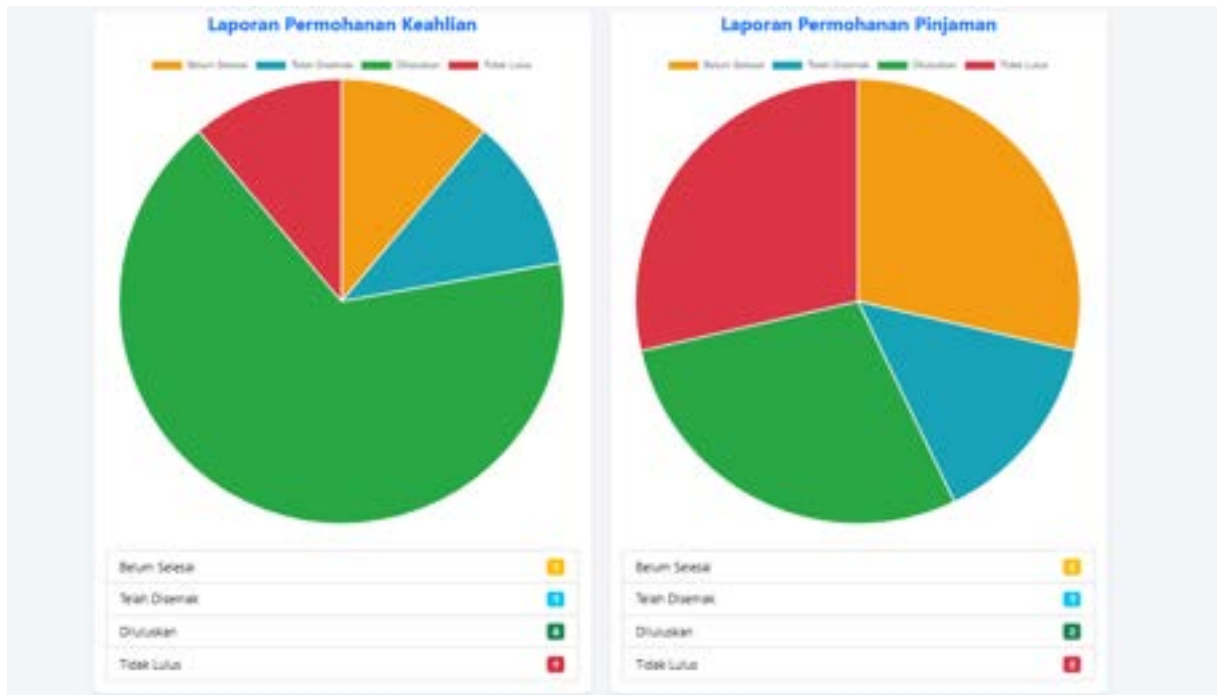
[Kembali Semula](#)

Approve Member Application decline it:

1. From the member termination application list page, press on the drop down button in the 'Status Permohonan' column and choose 'Lulus' to approve the loan application or 'Tidak Lulus' to decline the application.
2. Press the green button with 'Kemas Kini' to update the new status of the application.

3.5 View System Report for Admin

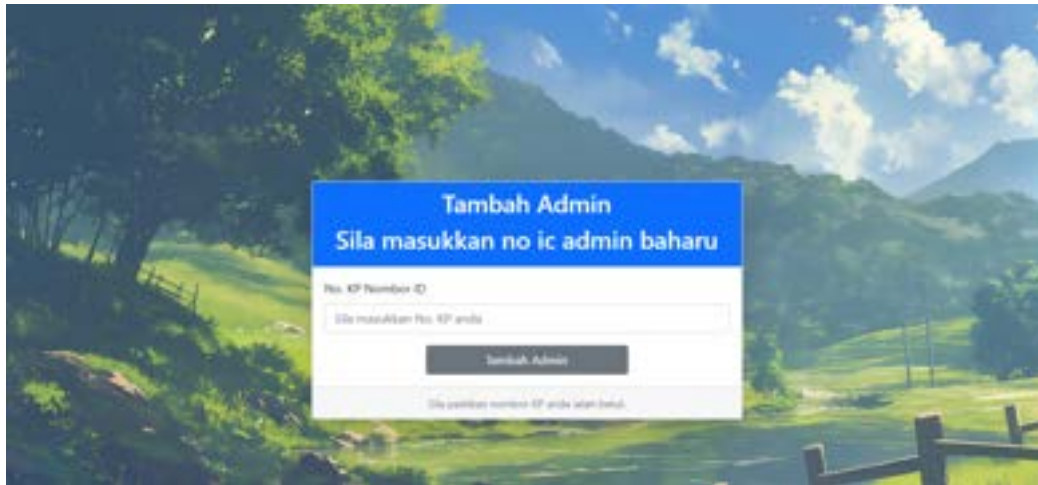




View koperasi report:

1. From the mainpage, choose 'Laporan Koperasi' to view the page for the report.
2. Enter the year, month and day in the calendar to show the graph report of the organisation until the date entered. If there is no date entered, the system will show the report until the current date.

3.6 Add New Admin and Ahli Lembaga Koperasi (ALK) to the system



Add Admin:

1. From the main page, choose 'Tambah Admin' to view the page to add a new admin.
2. Insert the new admin ic number and press 'Tambah Admin' to proceed. The new admin details can be checked at the guest page.



Add ALK:

1. From the main page, choose 'Tambah Ahli Lembaga Koperasi' to view the page to add a new ALK.

2. Insert the new ALK ic number and press 'Tambah ALK' to proceed. The new ALK details can be checked at the guest page.

3.7 Log Out



Logging Out :

1. To log out, Admins have to click on the 'Daftar Keluar' button on the navigation bar located at the top right of the page.

4.0 ALK User

4.1 ALK Log In Page



Logging In:

1. ALKs will have to click on the 'Daftar Masuk' button on the homepage to login to the system
2. A button page with three user views will be displayed, which are 'Ahli Koperasi', 'Admin Koperasi' and 'Ahli Lembaga Koperasi'. ALK have to click on the 'Ahli Lembaga Koperasi' button
3. ALKs key-in their ALK identification number and password to login to the system.

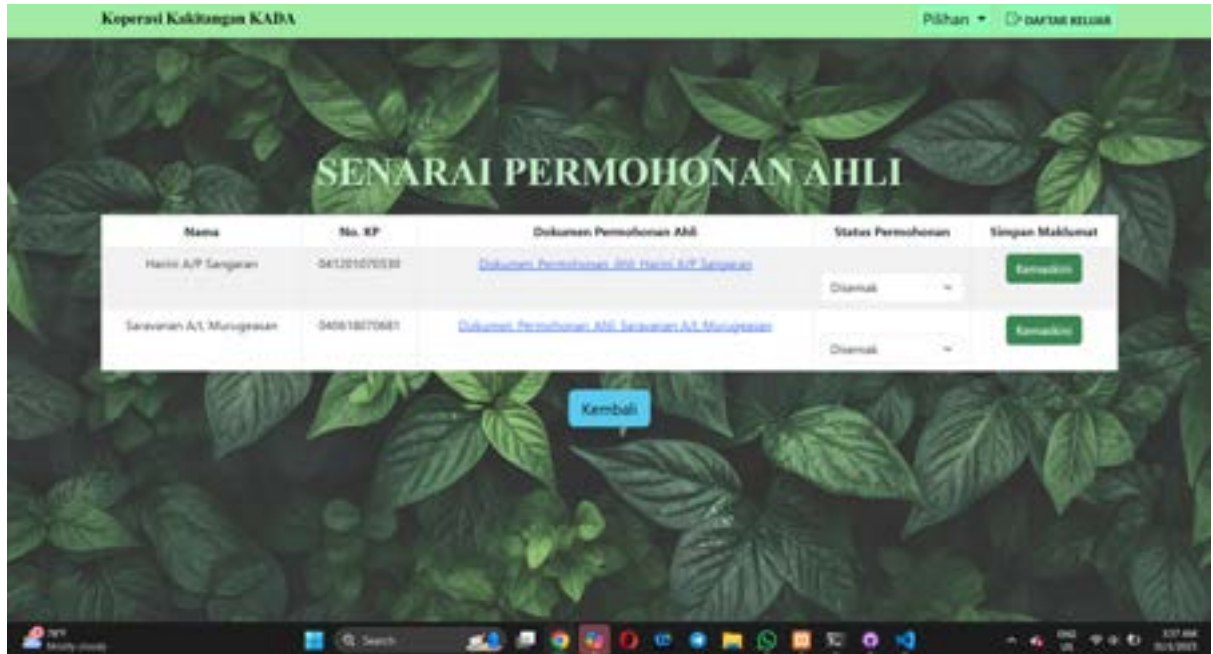
4.2 ALK Homepage



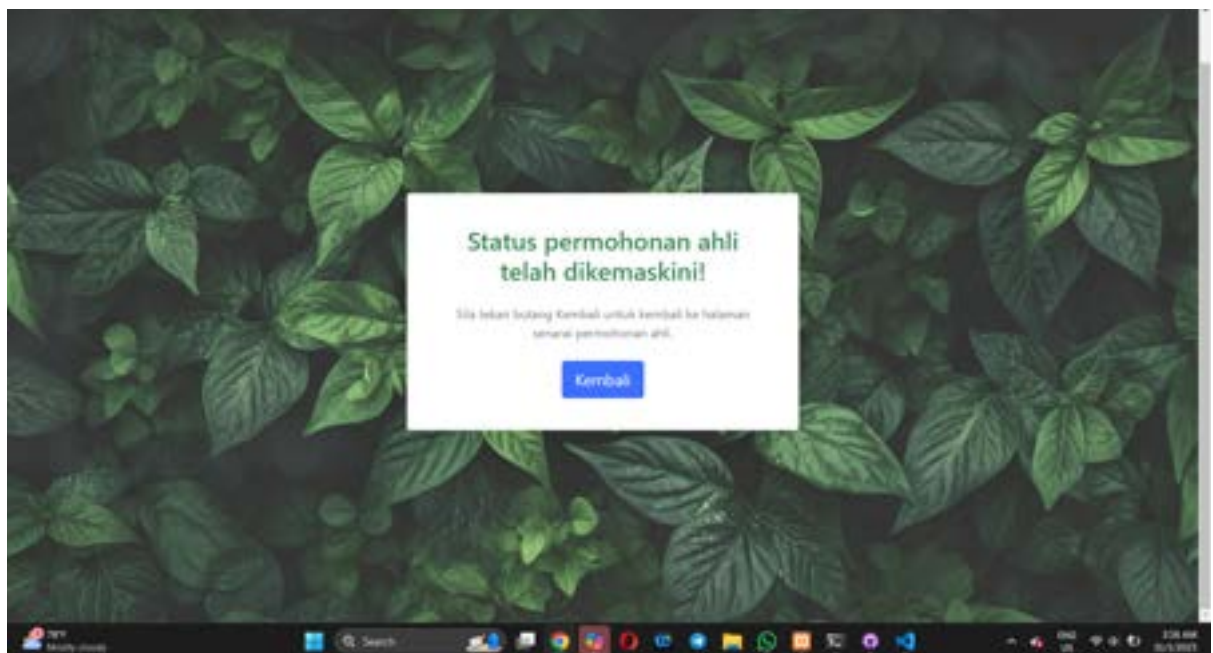
Go to Homepage:

1. Once they have logged in, they'll view a page with 3 buttons, each representing a feature for ALKs.
2. Display 'Pengesahan Permohonan Pinjaman' button to approve the loan applications, 'Pengesahan Permohonan Ahli' button to approve the membership applications and 'Laporan Tahunan' button to view Koperasi KADA's annual & monthly report.

4.3 Approve Membership Application



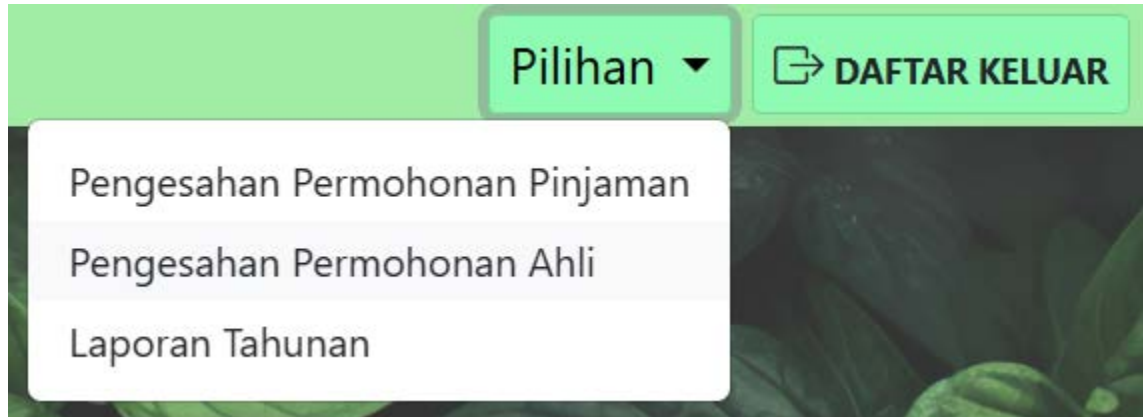
4.4 Successful Membership Application Status Update



View and Approving the Membership Application:

1. If ALKs click on the 'Pengesahan Permohonan Ahli' button, a list of membership applications with 'Disemak' status will be displayed.
2. ALKs can view the verification documents submitted by applicants
3. ALKs can update the status to 'Disahkan' if the documents fit the requirements, or update the status to 'Ditolak' if it doesn't fit the requirements
4. Once the update has been successfully done, a message will be displayed to notify the successful update.

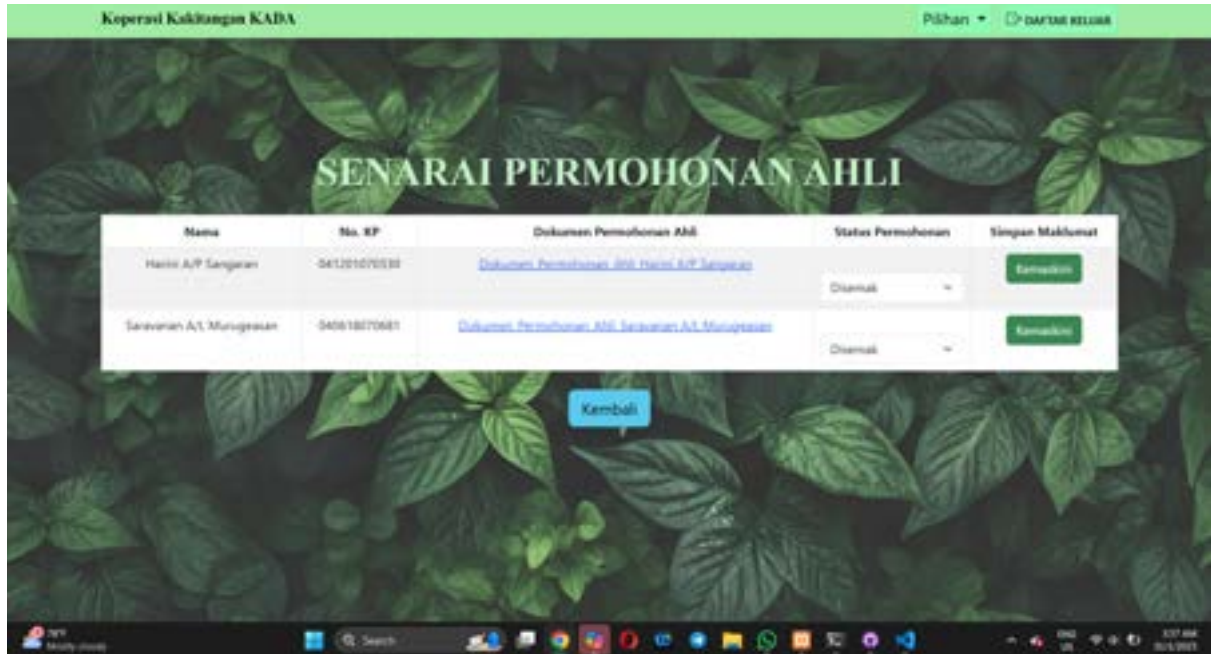
4.5 Navigation Bar



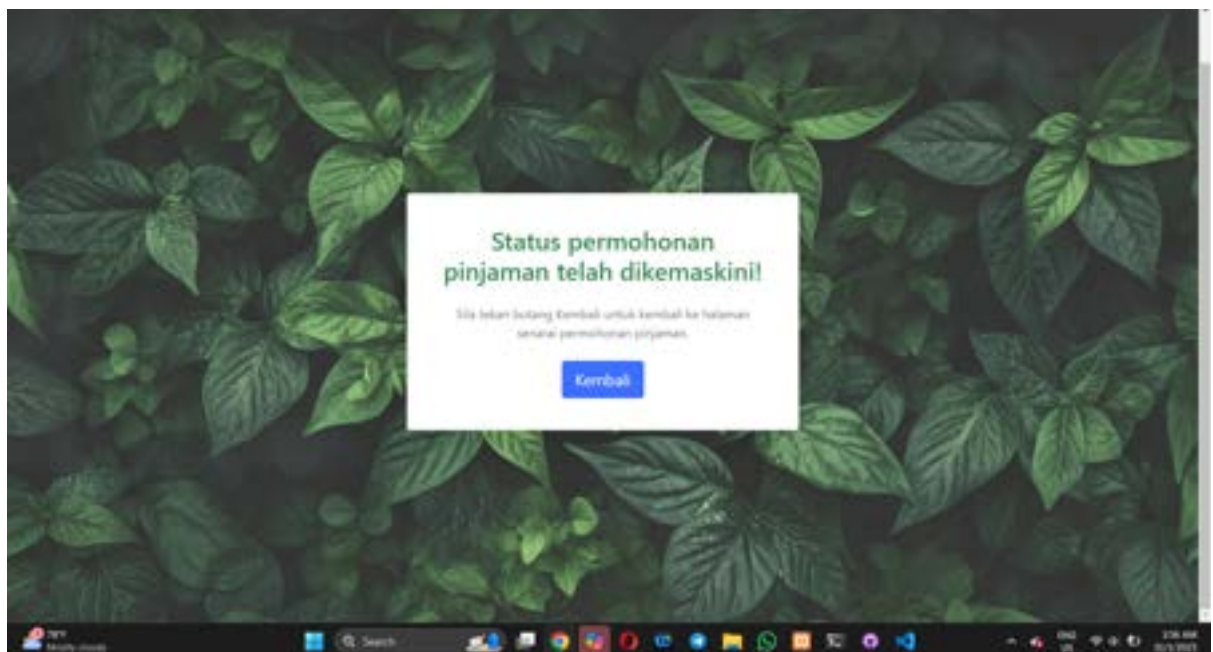
Navigate To The Page:

1. If the members click on the 'Pilihan' button, they'll view other features so they can click on them to jump pages.
2. ALKs can use the navigation bar and click to the next page which is 'Pengesahan Permohonan Pinjaman'

4.6 Approve Loan Application



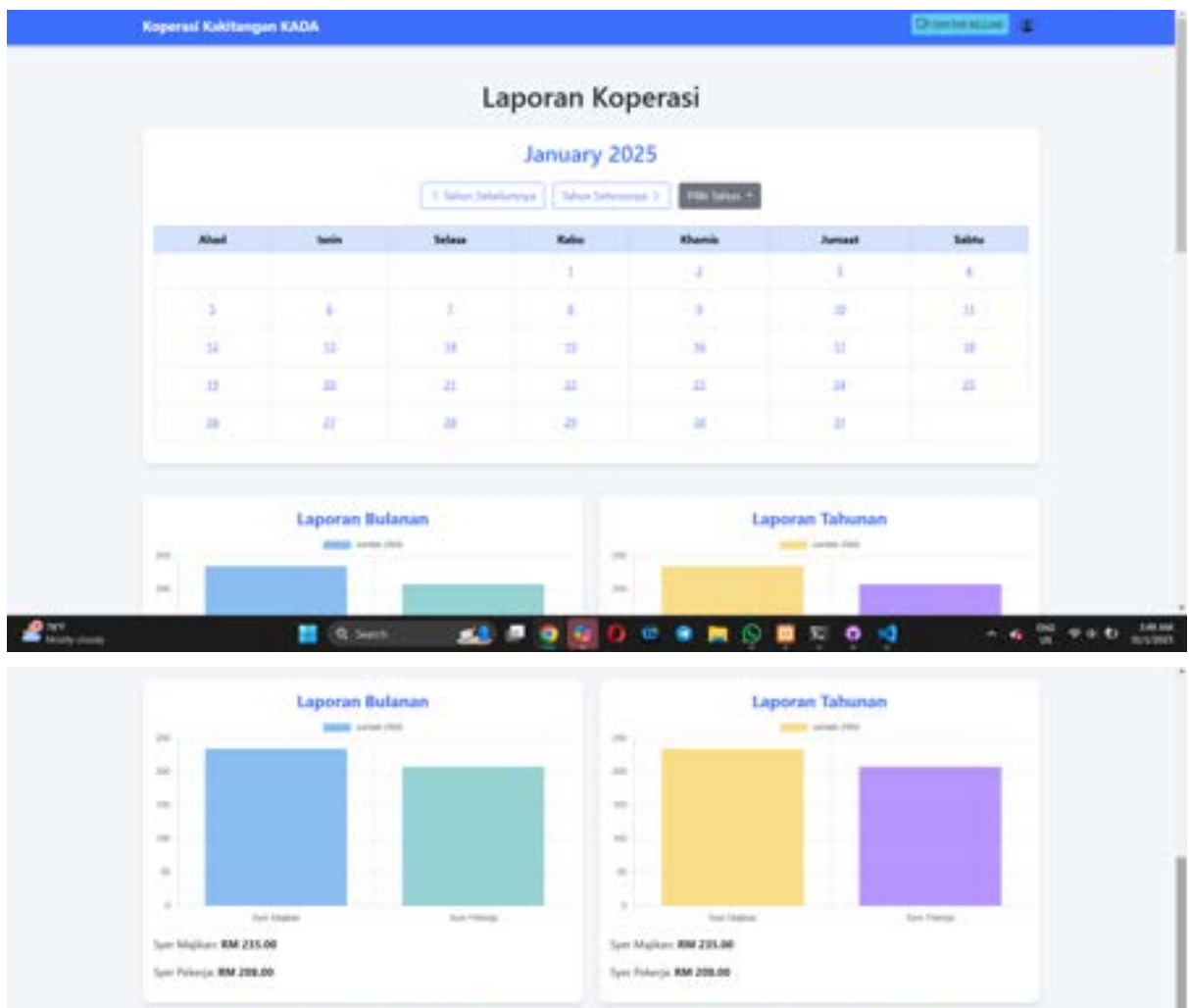
4.7 Successful Loan Application List



View and Approving the Loan Application:

5. If ALKs click on the 'Pengesahan Permohonan Pinjaman' button, a list of Loan applications with 'Disemak' status will be displayed.
6. ALKs can view the verification documents submitted by applicants
7. ALKs can update the status to 'Disahkan' if the documents fit the requirements, or update the status to 'Ditolak' if it doesn't fit the requirements
8. Once the update has been successfully done, a message will be displayed to notify the successful update.

4.8 Annual and Monthly Report

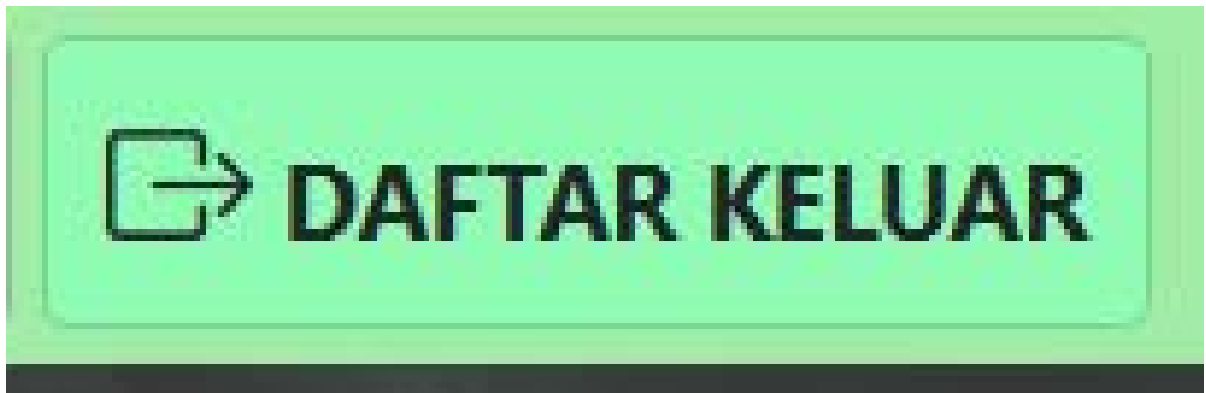




View Monthly and Annual Report:

1. If ALKs click on the 'Laporan Tahunan' button, Koperasi KADA's report will be displayed
2. The report has a monthly and annual summary for employee share displayed as 'Syer Pekerja' and employer share displayed as 'Syer Majikan' and has monthly report for membership and loan applications list.
3. If ALKs click on any specific date on the calendar displayed on the report page, the report will change up to the specific date only

4.9 Log Out ALK System



Logging Out :

1. To log out, Members, Admins and ALKs have to click on the 'Daftar Keluar' button on the top right of the page.

Presentation Slide

https://www.canva.com/design/DAGfMqHqfZM/T_45tNIQ2W8dZteKLP3xqg/edit?utm_content=DAGfMqHqfZM&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

Individual Contribution



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1

2024/2025

SECP3723 – SYSTEM DEVELOPMENT TECHNOLOGY

SECTION 02

TITLE: INDIVIDUAL CONTRIBUTION

LECTURER: DR. AHMAD NAJMI BIN AMERHAIDER NUAR

NAME	MATRIC NUMBER
NURUL IKA SYAFINY BINTI AZHAR	A23CS0164
NURAI SYAH BINTI MOHD ZIKRE	A23CS0160
HARINI A/P SANGARAN	A23CS0081
DAMIYA AINA BINTI BASIR ABD SHAMMAD	A23CS0220

Table of Content

Nurul Ika Syafiny Binti Azhar	3
Nuraisyah Binti Mohd Zikre	4
Harini A/P Sangaran	5
Damiya Aina Binti Basir Abd Shammad	7

Nurul Ika Syafiny Binti Azhar

As the leader of my development team, I played a crucial role in both the planning and execution phases of the KADA Corporation System to ensure its smooth operation and functionality. Before the development began, I took the initiative to roughly plan how the system would work, outlining its core processes and interactions. This initial blueprint served as a guide for structuring the database, modules, and system features efficiently. Additionally, I planned the database tables and their integration into different modules, ensuring that data was stored and retrieved effectively across various functionalities.

A major part of my contribution was in the coding and development of essential system functions. I was responsible for implementing the loan application process in multiple modules, including the non-member module, ALK module, and admin module. This involved coding the entire flow from application submission to approval, ensuring a seamless user experience. Furthermore, I developed the login process for all user types, including members, ALK, and admin users, ensuring secure authentication and access control. Another significant feature I worked on was the member termination application process, which included implementing both the application process in the member module and the approval process in the admin module.

In addition, I developed the functionality for viewing member savings, shares, and transactions, enabling users to track their financial activities within the system. I also contributed to the process of adding new ALK and admin users in the admin module, ensuring that the system could accommodate role-based user management effectively. Beyond functionality development, I took on the responsibility of hosting the system with hosting agents, ensuring that the system was deployed and accessible for users. This involved setting up the necessary hosting environment, configuring the system, and ensuring smooth integration with the hosting platform.

As the team leader, I was responsible for arranging meeting times, assigning tasks, and ensuring that each member had clear objectives. Recognizing the different skill levels within the team, I carefully delegated tasks to members who lacked certain technical skills, allowing them to contribute in ways that best suited their abilities. Additionally, I took the time to review my team members' work, identify problems, and fix any issues that arose during the development process. By overseeing the team's progress, providing guidance, and troubleshooting technical challenges, I ensured that the system development stayed on track and met the required standards.

Overall, as the leader of my team, my contributions to the KADA Corporation System encompassed a wide range of responsibilities, from planning and coding to leadership, problem-solving, and system deployment. Through my efforts, I helped shape the system's structure, improve its functionality, foster teamwork, and ensure successful deployment, making the system accessible and operational for users.

Nuraisyah Binti Mohd Zikre

As a part of the team, my contribution to this group does not just into the development of the system but also the documentation and preparation of the system to the industry. In terms of the development of the system, my team leader has assigned me a few functions of the system from different modules. Firstly, in the non-member module, I contribute to the making of membership applications to be applied by non-member. Next, I need to make the automated report generation for the admin and ALK module where the report needs to be accurate and updated timely from time to time. I work together with one of my team members, Harini. This is said because I continue the report generation that has been done by Harini as she completes the report generation for monthly and annual shares. Moreover, I need to make sure the report is generated monthly and annually for each application applied (membership application, loan application, membership termination application).

Other than backend, I also contribute to the frontend of the system where I need to design the interface for the admin module for all pages of the module. Besides, I also help Ika “add-on” some code for different modules to ensure the system proposed meets the client's requirement. Not just that, I also help Ika in system testing after the system is completely put into the hosting where we both check the integration of the system for every module. My contribution to the development of the system is not much but I’ve tried my best to learn from both Ika and Harini as they did so well at their part in completing the system.

In terms of documentation, I actively contribute to the task assigned by the team leader. Besides, I always prepare the link template of every documentation so that my team can complete and check all documentation together. This way everyone can check each other's tasks to avoid mistakes and so on. On the other hand, I also prepare the Lucidchart link for everyone to have one same platform to draw any related diagram to the documentation. For the same reasons, everyone can check each other's tasks to avoid mistakes. Not just that, I always ensure the documentation is in the correct format where I will check alignment and arrangement for the topic in the table of content of every documentation.

Other than documentation, I also prepare the slide to be used for Industry Day. At first, our team decided to demo the system. However, due to some confusion, I had to prepare the slide version of the demo of the system to present to the industry (Koperasi KADA). Thankfully on the day of the presentation we do not really use too much time as we use the presentation slide that I have prepared. At the same time, all of us successfully present our demo slide by taking turns to present it according to different actors/users of our system. Surprisingly our team also received a good comment from the industry even though I still think that there is a lot for me to fix.

Harini A/P Sangaran

As a team member, I contributed to the project by coding for the system development. We have a total of four user views, which are non-member, member, admin and Ahli Lembaga Koperasi (ALK). I coded features across all the modules. In the non-member module, I created the default homepage that all users would see when they access the system. The interface displays Koperasi KADA's information and has a navigation bar that allows non-members to apply for Koperasi KADA membership, all users can check their membership, admin or ALK status and log in to the system. Apart from the default homepage, I also developed the feature that enables all users to check their membership, admin or ALK status. The users will have to key in their Identification Card (IC) number and their id with default password will be displayed if their membership, admin or ALK existence is valid. If their membership is pending, being reviewed, is disapproved or does not exist, appropriate messages will be displayed. I developed the feature to change passwords on every login page, enabling all users to change their password by entering their IC number and new password.

As for the member module, I also developed the menu page for member's which displays the key features and has a navigation bar. There are six features for members, which are to apply for a loan, apply for a membership termination, check loan application status, check savings, loan balance and invoice. In these features, I developed the feature to check loan application status and loan balance. For loan application status, the system will display the loan information if the loan application is approved. If the loan is pending, being reviewed or is disapproved, appropriate messages will be displayed. The system will display messages for all loans of the specific user. For loan balance, the system displays the balance loan amount to be paid in bold and in red to emphasise the main purpose of the feature which is to show the loan balance. Other than that, the system also displays the loan type and monthly installment along with the previous transactions made to pay off the loan. This enables the members to keep track of their transactions and keep track of the loan balance. I also developed the feature that enables members to edit their personal information along with Ika.

As for the admin module, there are six features, which are to view membership applications list, loan applications list, membership terminations list, Koperasi KADA report and add new admin and ALK to the system. In this module, I developed the feature to view and validate membership applications list and did the Koperasi KADA report feature with Aisyah. For membership applications list, the name, IC number and verification documents of the applicants will be displayed. Admins can view the verification documents and update the application status to 'Reviewed' if the verification documents fits the requirements and to 'Disapproved' if not. For the Koperasi KADA report, I generated the calendar on the feature that filters the report to a certain date if any specific date is clicked on by the admin. I also did the functionality to

calculate monthly and annual reports for employer and employee shares whereas Aisyah worked on the bar and pie graphs.

As for the ALK module, there are three features, which are to approve membership applications, approve loan applications and view the Koperasi KADA report. In this module, I did the feature for ALK to approve the loan applications and the Koperasi KADA report. Approving membership applications is similar to validating the applications whereby membership applications with 'Reviewed' status is filtered and displayed to ALK. Then, ALK can approve or disapprove based on the verification documents. For the Koperasi KADA report, it is same as the admin's Koperasi KADA report.

Apart from this, I also helped my other team members to edit or correct a few features whenever they needed my help. The features mentioned above are the tasks given to me by our group leader, Ika.

Damiya Aina Binti Basir Abd Shammad

As a team member, I played a role in the development and documentation of the ALK module, ensuring that all required reports related to this module were completed accurately and on time. My responsibilities extended beyond reporting, as I also contributed to the designing of the ALK and Member interface system. Due to hardware limitations that prevented me from developing system functions in VS Code, our group leader, Ika, assigned me to focus on UI/UX design instead.

In addition to my work on the ALK module, I collaborated with Harini to finalize and refine the documentation for the SE course, ensuring that our project reports met the required standards. For the SDT course, after our team divided the tasks, I was assigned to complete crucial sections of the final report, including the Company/Client Background, System Design, Proposed Solution, and Interface for All Users. Moreover, the SDT course required the development of a user manual and slide presentation for the Industry Day on 16 February 2025. I took charge of completing the ALK module user manual and improving the slide presentation, ensuring clarity and professionalism in our delivery.

For the DB course, I worked alongside my team members to complete my assigned sections of the P1, P2, and P3 reports, contributing to the team's collective success in meeting project deadlines. Finally, on Industry Day, our group had the responsibility of presenting each module. With only six minutes per group, I successfully presented the ALK module, ensuring that key aspects were covered concisely and effectively.